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# THE ORIGINS OF INTELLIGENCE IN CHILDREN

## BY JEAN PIAGET

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# The Origins of Intelligence in Children

JEAN PIAGET, the noted Swiss child psychologist, was educated at Neuchâtel, Zurich, and the University of Paris. He has been professor of child psychology and history of scientific thought at the University of Geneva since 1929, and director of the International Bureau of Education and of the Institut J. J. Rousseau. Among his other books are *The Language and Thought of the Child, Judgment with Reasoning in the Child, The Child's Conception of the World, The Child's Conception of Physical Causality, The Moral Judgment of the Child, The Reconstruction of Reality in the Child, The Growth of Logical Thinking from Childhood to Adolescence,* and *The Child's Conception of Geometry.*

# THE ORIGINS OF INTELLIGENCE IN CHILDREN

JEAN PIAGET

*Translated by*
MARGARET COOK



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# FOREWORD TO THE SECOND EDITION

This work, a second edition of which has very kindly been requested, was followed by *La Construction du réel chez l'enfant* and was to have been completed by a study of the genesis of imitation in the child. The latter piece of research, whose publication we have postponed because it is so closely connected with the analysis of play and representational symbolism, appeared in 1945, inserted in a third work, *La formation du symbole chez l'enfant*. Together these three works form one entity dedicated to the beginnings of intelligence, that is to say, to the various manifestations of sensorimotor intelligence and to the most elementary forms of expression.

The theses developed in this volume, which concern in particular the formation of the sensorimotor schemata and the mechanism of mental assimilation, have given rise to much discussion which pleases us and prompts us to thank both our opponents and our sympathizers for their kind interest in our work. It is impossible to name here all the authors on whose observations we would like to comment, but we should single out for mention the remarkable studies made by H. Wallon and P. Guillaume.

In his fine work *De l'acte à la pensée,* H. Wallon did us the honor of discussing our work at length; we have already commented on this in *La formation du symbole chez l'enfant.* Wallon's main idea is the distinction which he makes between the realm of the sensorimotor (characterized by the "understanding of situations") and that of expression (verbal intelligence). His remarkable study on *Les origines de la pensée chez l'enfant,* published since, places the origins of thought at the age of four, as if nothing essential transpired between the attainments of the sensorimotor intelligence and the beginnings of conceptual expression. It is apparent how antithetical to everything we main-

tain in this book this radical thesis is, and we can answer it today by invoking two kinds of arguments.

In the first place, meticulous study of a definite area, that of development of spatial perceptions, has led us with B. Inhelder to discover an even greater correlation than there seemed to be between the sensorimotor and the perceptual. Doubtless nothing is directly transmitted from one of these planes to the other, and all that the sensorimotor intelligence has constructed must first be reconstructed by the growing perceptual intelligence before this overruns the boundaries of that which constitutes its substructure. But the function of this substructure is no less apparent. It is because the baby begins by constructing, in coördinating his actions, schemata such as those of the unchanging object, the fitting in of two or three dimensions, rotations, transpositions, and superpositions that he finally succeeds in organizing his "mental space" and, between preverbal intelligence and the beginnings of Euclidean spatial intuition, a series of "topological" intuitions are intercalated as manifested in drawing, stereognosis, the construction and assembling of objects, etc.; that is to say, in the areas of transition between the sensorimotor and the perceptual.

In the second place, it is primarily preverbal sensorimotor activity that is responsible for the construction of a series of perceptual schemata the importance of which in the subsequent structuring of thought cannot, without oversimplification, be denied. Thus the perceptual constants of form and size are connected with the sensorimotor construction of the permanent object: For how could the four-year-old child think without having reference to objects having form and invariable dimensions, and how would he adapt his belief without a long preliminary development by the sensorimotor?

Probably the sensorimotor schemata are not concepts, and the functional relationship which we stress in this book does not exclude the structural opposition of these extremes, despite the continuity of the transitions. But, without preliminary schemata, nascent thought would be reduced to mere verbalism, which would make one suspicious of many of the acts mentioned by Wallon in his latest work. But it is precisely on the concrete

plane of action that infancy makes its intelligence most manifest until the age of seven or eight, when coördinated actions are converted into operations, admitting of the logical construction of verbal thought and its application to a coherent structure.

In short, Wallon's thesis disregards the progressive construction of performance and that is why it goes to extremes in stressing the verbal at the expense of the sensorimotor whereas the sensorimotor substructure is necessary to the conceptual for the formation of the operational schemata which are destined to function finally in a formal manner and thus to make language consistent with thought.

As far as P. Guillaume's[1] very interesting study is concerned, it, on the other hand, agrees in the main with our conclusions, except in one essential point. In accordance with his interpretations influenced by "the theory of form," P. Guillaume presents a fundamental distinction between the perceptual mechanisms and the intellectual processes which explains the second in terms of the first (the reverse of Wallon). This controversy is too lengthy to consider in detail in a preface. Let us limit ourselves to answering that the systematic study of the child's perceptions, in which we have since collaborated with Lambertier[2] has, on the contrary, led us to doubt the permanence of perceptual constants in which P. Guillaume believes (the invariability of size, etc.) and to introduce a distinction between instantaneous perceptions which are always passive and a "perceptual activity" connecting them with each other in space and time, according to certain remarkable laws (in particular a mobility and reversibility increasing with age). This perceptual activity, which the theory of form partially disregards, is but one manifestation of the sensorimotor activities of which preverbal intelligence is the expression. In the development of the sensorimotor schema in the first year of life, there is undoubtedly a close interaction between perception and intelligence in their most elementary states.

[1] P. Guillaume, L'intelligence sensori-motrice d'après J. Piaget, *Journal de psychologie*, April–June 1940–41 (years XXXVII–XXXVIII, pp. 264–280).

[2] See Recherches sur le développement des perceptions (I–VIII), *Archives de psychologie*, 1942–1947.

# INTRODUCTION*

## The Biological Problem of Intelligence

The question of the relationships between mind and biological organization is one which inevitably arises at the beginning of a study of the origins of intelligence. True, a discussion of that sort cannot lead to any really definite conclusion at this time, but, rather than to submit to the implications of one of the various possible solutions to this problem, it is better to make a clear choice in order to separate the hypotheses which form the point of departure for our inquiry.

Verbal or cogitative intelligence is based on practical or sensorimotor intelligence which in turn depends on acquired and recombined habits and associations. These presuppose, furthermore, the system of reflexes whose connection with the organism's anatomical and morphological structure is apparent. A certain continuity exists, therefore, between intelligence and the purely biological processes of morphogenesis and adaptation to the environment. What does this mean?

It is obvious, in the first place, that certain hereditary factors condition intellectual development. But that can be interpreted in two ways so different in their biological meaning that confusing the one with the other is probably what has obfuscated the classic controversy over innate ideas and epistemological *a priorism.*

The hereditary factors of the first group are structural and are connected with the constitution of our nervous system and of our sensory organs. Thus we perceive certain physical radia-

* Another translation of this chapter was published in *Organization and Pathology of Thought,* by David Rapaport (New York: Columbia University Press, 1951). The footnote commentary to that translation provides an introduction to Piaget's thinking, and may serve as an introduction to the investigations and thinking contained in this volume.

tions, but not all of them, and matter only of a certain size, etc. Now these known structural factors influence the building up of our most fundamental concepts. For instance, our intuition of space is certainly conditioned by them, even if, by means of thought, we succeed in working out transintuitive and purely deductive types of space.

These characteristics of the first type, while supplying the intelligence with useful structures, are thus essentially limiting, in contradistinction to the factors of the second group. Our perceptions are but what they are, amidst all those which could possibly be conceived. Euclidean space which is linked to our organs is only one of the kinds of space which are adapted to physical experience. In contrast, the deductive and organizing activity of the mind is unlimited and leads, in the realm of space, precisely to generalizations which surpass intuition. To the extent that this activity of the mind is hereditary, it is so in quite a different sense from the former group. In this second type it is probably a question of a hereditary transmission of the function itself and not of the transmission of a certain structure. It is in this second sense that H. Poincaré was able to consider the spatial concept of "group" as being *a priori* because of its connection with the very activity of intelligence.

We find the same distinction with regard to the inheritance of intelligence. On the one hand, we find a question of structure: The "specific heredity" of mankind and of its particular "offspring" admits of certain levels of intelligence superior to that of monkeys, etc. But, on the other hand, the functional activity of reason (the *ipse intellectus* which does not come from experience) is obviously connected with the "general heredity" of the living organism itself. Just as the organism would not know how to adapt itself to environmental variations if it were not already organized, so also intelligence would not be able to apprehend any external data without certain functions of cohérence (of which the ultimate expression is the principle of noncontradiction), and functions making relationships, etc., which are common to all intellectual organization.

Now this second type of hereditary psychological reality is of primary importance for the development of intelligence. If

there truly in fact exists a functional nucleus of the intellectual organization which comes from the biological organization in its most general aspect, it is apparent that this invariant will orient the whole of the successive structures which the mind will then work out in its contact with reality. It will thus play the role that philosophers assigned to the *a priori;* that is to say, it will impose on the structures certain necessary and irreducible conditions. Only the mistake has sometimes been made of regarding the *a priori* as consisting in structures existing ready-made from the beginning of development, whereas if the functional invariant of thought is at work in the most primitive stages, it is only little by little that it impresses itself on consciousness due to the elaboration of structures which are increasingly adapted to the function itself. This *a priori* only appears in the form of essential structures at the end of the evolution of concepts and not at their beginning: Although it is hereditary, this *a priori* is thus the very opposite of what were formerly called "innate ideas."

The structures of the first type are more reminiscent of classic innate ideas and it has been possible to revive the theory of innateness with regard to space and the "well-structured" perceptions of Gestalt psychology. But, in contrast to the functional invariants, these structures have nothing essential from the point of view of the mind: They are only internal data, limited and delimiting, and external experience and, above all, intellectual activity will unremittingly transcend them. If they are in a sense innate, they are not *a priori* in the epistemological sense of the term.

Let us analyze first the functional invariants, and then (in §3) we shall discuss the question raised by the existence of special hereditary structures (those of the first type).

§1. THE FUNCTIONAL INVARIANTS OF INTELLIGENCE AND BIOLOGICAL ORGANIZATION.—Intelligence is an adaptation. In order to grasp its relation to life in general it is therefore necessary to state precisely the relations that exist between the organism and the environment. Life is a continuous creation of increasingly complex forms and a progressive balancing of these forms with the environment. To say that in-

telligence is a particular instance of biological adaptation is thus to suppose that it is essentially an organization and that its function is to structure the universe just as the organism structures its immediate environment. In order to describe the functional mechanism of thought in true biological terms it will suffice to determine the invariants common to all structuring of which life is capable. What we must translate into terms of adaptation are not the particular goals pursued by the practical intelligence in its beginnings (these goals will subsequently enlarge to include all knowledge), but it is the fundamental relationship peculiar to consciousness itself: the relationship of thought to things. The organism adapts itself by materially constructing new forms to fit them into those of the universe, whereas intelligence extends this creation by constructing mentally structures which can be applied to those of the environment. In one sense and at the beginning of mental evolution, intellectual adaptation is thus more restricted than biological adaptation, but in extending the latter, the former goes infinitely beyond it. If, from the biological point of view, intelligence is a particular instance of organic activity and if things perceived or known are a limited part of the environment to which the organism tends to adapt, a reversal of these relationships subsequently takes place. But this is in no way incompatible with the search for functional invariants.

In fact there exists, in mental development, elements which are variable and others which are invariant. Thence stem the misunderstandings resulting from psychological terminology some of which lead to attributing higher qualities to the lower stages and others which lead to the annihilation of stages and operations. It is therefore fitting simultaneously to avoid both the preformism of intellectualistic psychology and the hypothesis of mental heterogeneities. The solution to this difficulty is precisely to be found in the distinction between variable structures and invariant functions. Just as the main functions of the living being are identical in all organisms but correspond to organs which are very different in different groups, so also between the child and the adult a continuous creation of varied structures may be observed although the main functions of thought remain constant.

These invariant operations exist within the framework of

the two most general biological functions: *organization* and *adaptation*. Let us begin with the latter, for if everyone recognizes that everything in intellectual development consists of adaptation, the vagueness of this concept can only be deplored.

Certain biologists define *adaptation* simply as preservation and survival, that is to say, the equilibrium between the organism and the environment. But then the concept loses all interest because it becomes confused with that of life itself. There are degrees of survival, and adaptation involves the greatest and the least. It is therefore necessary to distinguish between the state of adaptation and the process of adaptation. In the state, nothing is clear. In following the process, things are cleared up. There is adaptation when the organism is transformed by the environment and when this variation results in an increase in the interchanges between the environment and itself which are favorable to its preservation.

Let us try to be precise and state this in a formal way. The organism is a cycle of physicochemical and kinetic processes which, in constant relation to the environment, are engendered by each other. Let *a, b, c,* etc., be the elements of this organized totality and *x, y, z,* etc., the corresponding elements of the surrounding environment. The schema of organization is therefore the following:

$$(1)\ a + x \longrightarrow b;$$
$$(2)\ b + y \longrightarrow c;$$
$$(3)\ c + z \longrightarrow a, \text{ etc.}$$

The processes (1), (2), etc., may consist either of chemical reactions (when the organism ingests substances *x* which it will transform into substance *b* comprising part of its structure), or of any physical transformations whatsoever, or finally, in particular, of sensorimotor behavior (when a cycle of bodily movements *a* combined with external movements *x* result in *b* which itself enters the cycle of organization). The relationship which unites the organized elements *a, b, c,* etc., with the environmental elements *x, y, z,* etc., is therefore a relationship of *assimilation,* that is to say, the functioning of the organism does not destroy it but conserves the cycle of organization and coördinates the given

data of the environment in such a way as to incorporate them in that cycle. Let us therefore suppose that, in the environment, a variation is produced which transforms $x$ into $x'$. Either the organism does not adapt and the cycle ruptures, or else adaptation takes place, which means that the organized cycle has been modified by closing up on itself:

$$(1)\ a + x' \longrightarrow b';$$
$$(2)\ b' + y \longrightarrow c;$$
$$(3)\ c + z \longrightarrow a.$$

If we call this result of the pressures exerted by the environment *accommodation* (transformation of $b$ into $b'$), we can accordingly say that *adaptation is an equilibrium between assimilation and accommodation.*

This definition applies to intelligence as well. Intelligence is *assimilation* to the extent that it incorporates all the given data of experience within its framework. Whether it is a question of thought which, due to judgment, brings the new into the known and thus reduces the universe to its own terms or whether it is a question of sensorimotor intelligence which also structures things perceived by bringing them into its schemata, in every case intellectual adaptation involves an element of assimilation, that is to say, of structuring through incorporation of external reality into forms due to the subject's activity. Whatever the differences in nature may be which separate organic life (which materially elaborates forms and assimilates to them the substances and energies of the environment) from practical or sensorimotor intelligence (which organizes acts and assimilates to the schemata of motor behavior the various situations offered by the environment) and separate them also from reflective or gnostic intelligence (which is satisfied with thinking of forms or constructing them internally in order to assimilate to them the contents of experience)—all of these adapt by assimilating objects to the subject.

There can be no doubt either, that mental life is also *accommodation* to the environment. Assimilation can never be pure because by incorporating new elements into its earlier schemata the intelligence constantly modifies the latter in order

to adjust them to new elements. Conversely, things are never known by themselves, since this work of accommodation is only possible as a function of the inverse process of assimilation. We shall thus see how the very concept of the object is far from being innate and necessitates a construction which is simultaneously assimilatory and accommodating.

In short, intellectual adaptation, like every other kind, consists of putting an assimilatory mechanism and a complementary accommodation into progressive equilibrium. The mind can only be adapted to a reality if perfect accommodation exists, that is to say, if nothing, in that reality, intervenes to modify the subject's schemata. But, inversely, adaptation does not exist if the new reality has imposed motor or mental attitudes contrary to those which were adopted on contact with other earlier given data: adaptation only exists if there is coherence, hence assimilation. Of course, on the motor level, coherence presents quite a different structure than on the reflective or organic level, and every systematization is possible. But always and everywhere adaptation is only accomplished when it results in a stable system, that is to say, when there is equilibrium between accommodation and assimilation.

This leads us to the function of *organization*. From the biological point of view, organization is inseparable from adaptation: They are two complementary processes of a single mechanism, the first being the internal aspect of the cycle of which adaptation constitutes the external aspect. With regard to intelligence, in its reflective as well as in its practical form, this dual phenomenon of functional totality and interdependence between organization and adaptation is again found. Concerning the relationships between the parts and the whole which determine the organization, it is sufficiently well known that every intellectual operation is always related to all the others and that its own elements are controlled by the same law. Every schema is thus coördinated with all the other schemata and itself constitutes a totality with differentiated parts. Every act of intelligence presupposes a system of mutual implications and interconnected meanings. The relationships between this organization and adaptation are consequently the same as on the organic level.

The principal "categories" which intelligence uses to adapt to the external world—space and time, causality and substance, classification and number, etc.—each of these corresponds to an aspect of reality, just as each organ of the body is related to a special quality of the environment but, besides their adaptation to things, they are involved in each other to such a degree that it is impossible to isolate them logically. The "accord of thought with things" and the "accord of thought with itself" express this dual functional invariant of adaptation and organization. These two aspects of thought are indissociable: It is by adapting to things that thought organizes itself and it is by organizing itself that it structures things.

§2. FUNCTIONAL INVARIANTS AND THE CATEGORIES OF REASON.—The problem now is to ascertain how these functional invariants will determine the categories of reason, in other words, the main forms of intellectual activity which are found at all stages of mental development and whose first structural crystallizations in the sensorimotor intelligence we shall now try to describe.

It is not a matter of reducing the higher to the lower. The history of science shows that every attempt at deduction to establish continuity between one discipline and another results not in a reduction of the higher to the lower but in creating a reciprocal relationship between the two terms which does not at all destroy the originality of the higher term. So it is that the functional relations which can exist between intellect and biological organization can in no way diminish the value of reason but on the contrary lead to extending the concept of vital adaptation. It is self-evident that if the categories of reason are in a sense preformed in biological functioning, they are not contained in it either in the form of conscious or even unconscious structures. If biological adaptation is a sort of material understanding of the environments, a series of later structures would be necessary in order that conscious and gnostic image may emerge from this purely active mechanism. As we have already said, it is therefore at the end and not at the point of departure of intellectual evolution that one must expect to encounter rational concepts

really expressing functioning as such, in contrast to the initial structures which remain on the surface of the organism and of the environment and only express the superficial relationships of these two terms to each other. But in order to facilitate analysis of the lower stages which we shall attempt in this work it can be shown how the biological invariants just mentioned, once they have been reflected upon and elaborated by consciousness during the great stages of mental development, give rise to a sort of functional *a priori* of reason.

Here, it seems to us, is the picture thus obtained:

| *Biological Functions* | *Intellectual Functions* | *Categories* |
|---|---|---|
| Organization | Regulating function . . . | A. Totality x Relationship (reciprocity) |
| | | B. Ideal (goal) x Value (means) |
| Adaptation . . . | Assimilation . . . Implicative function | A. Quality x Class |
| | | B. Quantitative rapport[1] x number |
| | Accommodation . . . Explicative function | A. Object x Space |
| | | B. Causality x Time |

The categories related to the function of organization constitute what Hoeffding calls the "fundamental" or regulative

[1] In this diagram we distinguish between "relationships" in the most general sense of the word and "quantitative rapport" which corresponds to what is called, on the level of thought, the "logic of relationships." The relations which the latter envisages in contradistinction to the logic of classes are always quantitative, regardless of whether they interpret "more" or "less" as comparisons (for example, "more or less dark," etc.), or whether they simply imply ideas of category or of series (for example, family relationships such as "brother of" etc.), which presuppose quantity. On the contrary, the relationships on a par with the idea of totality surpass the quantitative and only imply a general relativity in the widest sense of the term (reciprocity between the elements of a totality).

"categories," that is to say, they combine with all the others and are found again in every psychic operation. It seems to us that these categories can be defined, from the static point of view, by the concepts of *totality* and *relationship* and, from the dynamic point of view, by those of *ideal* and *value*.

The concept of *totality* expresses the interdependence inherent in every organization, intelligent as well as biological. Even though behavior patterns and consciousness seem to arise in the most uncoördinated manner in the first weeks of existence, they extend a physiological organization which antedates them and they crystallize from the outset into systems whose coherence becomes clarified little by little. For example, what is the concept of "displacement groups," which is essential to the formation of space, if not the idea of organized totality making itself manifest in movements? So also are the schemata belonging to sensorimotor intelligence controlled from the very beginning by the law of totality, within themselves and in their interrelationships. So too, every causal relation transforms an incoherent datum into an organized environment, etc.

The correlative of the idea of totality is, as Hoeffding has shown, the idea of *relationship*. Relationship is also a fundamental category, inasmuch as it is immanent in all psychic activity and combines with all the other concepts. This is because every totality is a system of relationships just as every relationship is a segment of totality. In this capacity the relationship manifests itself from the advent of the purely physiological activities and is again found at all levels. The most elementary perceptions (as shown by Köhler with regard to the color perception of chickens) are simultaneously related to each other and structured into organized totalities. It is useless to emphasize analogous facts that one finds on the level of reflective thought.

The categories of *ideal* and of *value* express the same function, but in its dynamic aspect. We shall call "ideal" every system of values which constitutes a whole, hence every final goal of actions; and we shall call "values" the particular values related to this whole or the means making it possible to attain this goal. The relations of ideal and value are therefore the same as those of totality and relation. These ideals or value of every category

are only totalities in process of formation, value only being the expression of desirability at all levels. Desirability is the indication of a rupture in equilibrium or of an uncompleted totality to whose formation some element is lacking and which tends toward this element in order to realize its equilibrium. The relations between ideal and value are therefore of the same category as those of totality and of relations which is self-evident, since the ideal is only the as yet incomplete form of equilibrium between real totalities and values are none other than the relations of means to ends subordinated to this system. Finality is thus to be conceived not as a special category, but as the subjective translation of a process of putting into equilibrium which itself does not imply finality but simply the general distinction between real equilibria and the idea equilibrium. A good example is that of the norms of coherence and unity of logical thought which translate this perpetual effort of intellectual totalities toward equilibrium, and which therefore define the ideal equilibrium never attained by intelligence and regulate the particular values of judgment. This is why we call the operations relating to totality and to values "regulative function," in contradistinction to the explicative and implicative functions.[2]

How are we to consider the categories connected with adaptation, that is to say, with assimilation and accommodation? Among the categories of thought there are some, as Hoeffding says, which are more "real" (those which, besides the activity of reason, imply a *hic* and a *nunc* inherent in experience such as causality, substance or object, space and time, each of which operates an indissoluble synthesis of "datum" and deduction) and some which are more "formal" (those which, without being less adapted, can nevertheless give rise to an unlimited deductive elaboration, such as logical and mathematical relations). Hence it is the former which express more the centrifugal process of explication and accommodation and the latter which make pos-

2 In *The Language and Thought of the Child,* London, Routledge, 1932, p. 236, we called "mixed function" this synthesis of implication and explication which at the present time we connect with the idea of organization. But this amounts to the same thing since the latter presupposes a synthesis of assimilation and accommodation.

sible the assimilation of things to intellectual organization and the construction of implications.

The implicative function comprises two functional invariants which are found again at all stages, the one corresponding to the synthesis of *qualities,* that is to say, *classes* (concepts or schemata), the other to that of *quantitative relations* or *numbers.* Ever since the formation of the sensorimotor schemata the elementary instruments of intelligence reveal their mutual dependence. With regard to the explicative function, it concerns the ensemble of operations which makes it possible to deduce reality, in other words to confer a certain permanence upon it while supplying the reason for its transformations. From this point of view two complementary aspects can be distinguished in every explication, one relating to the elaboration of *objects,* the other relating to *causality;* the former is simultaneously the product of the latter and conditions its development. Whence the circle object x *space* and causality x *time* in which the interdependence of functions is complicated by a reciprocal relation of matter to form.

We see the extent to which the functional categories of knowledge constitute a real whole which is modeled on the system of the functions of intelligence. This correlation becomes still more clear on analysis of the interrelations of organization and adaptation, on the one hand, and assimilation and accommodation, on the other.

As we have seen, organization is the internal aspect of adaptation, when the interdependence of already adapted elements and not the adaptational process in action is under consideration. Moreover, adaptation is only organization grappling with the actions of the environment. Now this mutual dependence is found again, on the level of intelligence, not only in the interaction of rational activity (organization) and of experience (adaptation) which the whole history of scientific thought reveals are inseparable but also in the correlation of the functional categories: Any objective or causal spatial-temporal structure is only possible with logical-mathematical deduction, these two kinds of reality thus forming mutually interconnected systems of totalities and relations. With regard to the circle of accommoda-

tion and assimilation—that is to say, of explication and implication—the question raised by Hume concerning causality illustrates it clearly. How can the concept of cause be simultaneously rational and experimental? If one puts causality in a purely formal category reality escapes it (as E. Meyerson has admirably shown) and if one reduces it to the level of a simple empirical sequence, necessity vanishes. Whence the Kantian solution taken up by Brunschvicg according to which it is an "analogy of experience," an irreducible interaction between the relation of implication and the spatial-temporal known data. The same can be said of the other "real" categories: They all presuppose implication although constituting accommodations to external known data. Inversely, classes and numbers could not be constructed without connection with the spatial-temporal series inherent in objects and their causal relations.

Finally, it remains for us to note that, if every organ of a living body is organized, so also every element of an intellectual organization also constitutes an organization. Consequently the functional categories of intelligence, while developing along the major lines of the essential mechanisms of organization, assimilation and accommodation, themselves comprise aspects corresponding to those three functions, the more so since the latter are certainly vicarious and so constantly change in point of application. The manner in which the functions which thus characterize the chief categories of the mind create their own organs and crystallize into structures is another question which we shall not take up in this introduction since this whole work is devoted to study of the beginnings of this construction. To prepare for this analysis it is simply fitting to say a few more words about the hereditary structures which make this mental structuring possible.

§3. HEREDITARY STRUCTURES AND THEORIES OF ADAPTATION.—Two kinds of hereditary realities exist, as we have seen, which affect the development of human reason: the functional invariants connected with the general heredity of the living substance, and certain structural organs or qualities, connected with man's particular heredity and serving as elementary instruments for intellectual adaptation. It is therefore fitting to

examine how the hereditary structures prepare the latter and how biological theories of adaptation are able to cast light on the theory of intelligence.

The reflexes and the very morphology of the organs with which they are connected constitute a sort of anticipatory knowledge of the external environment, an unconscious and entirely material knowledge but essential to the later development of real knowledge. How is such an adaptation of hereditary structures possible?

This biological problem is insoluble at present, but a brief summary of the discussions to which it has given and still gives rise seems useful to us, for the different solutions supplied are parallel to the various theories of intelligence and can thus illuminate the latter by setting off the generality of their mechanism. Five principal points of view exist concerning adaptation and each one corresponds, *mutatis mutandis,* to one of the interpretations of intelligence as such. Of course this does not mean that if a certain author chooses one of the five characteristic doctrines that can be discerned in biology he is forced by this to adopt the corresponding attitude in psychology; but whatever the possible combinations with regard to the opinions of the writers themselves may be, "common mechanisms" undeniably exist between biological and psychological explanations of general and intellectual adaptation.

The first solution is that of Lamarckism according to which the organism is fashioned from the outside by the environment which, by its constraints, trains the formation of individual habits or accommodations which, becoming hereditarily fixed, fashion the organs. There corresponds to this biological hypothesis of the primacy of habit *associationism* in psychology according to which knowledge also results from acquired habits without there being any internal activity which would constitute intelligence as such to condition those acquisitions.

*Vitalism,* on the other hand, interprets adaptation by attributing to the living being a special power to construct useful organs. So also *intellectualism* explains intelligence by itself by endowing it with an innate faculty for knowing and by consider-

ing its activity as a primary fact whence everything on the psychic plane derives.

With regard to *preformism,* the structures have a purely endogenous origin, virtual variations being made up-to-date simply by contact with the environment which thus only plays a role of "detector." It is through the same sort of reasoning that various epistemological and psychological doctrines that can be labeled *apriorism* consider mental structures as being anterior to experience which simply gives them the opportunity to manifest themselves without explaining them in any respect. Whether structures are considered to be psychologically innate, as they are thought to be by the classic proponents of innate ideas, or merely as logically eternal, "subsisting" in an intelligible world in which reason participates, is of little importance. They are preformed in the subject and not elaborated by him as a function of his experience. The most parallel excesses in this respect were committed in biology and in logic. Just as a hypothesis was made of a preformation of all the "genes" which were made manifest in the course of evolution—including genes injurious to the species—so also Russell came to allege that all the ideas germinating in our brains have existed for all eternity, including false ideas!

A separate place could be set aside for the biological doctrine of "emergent evolution," according to which structures appear as irreducible syntheses succeeding each other in a sort of continuous creation, parallel to the theory of "shapes" or "Gestalt" in psychology. But actually only a more dynamic apriorism of intention is involved which, in its particular explanations, only amounts to apriorism properly so called to the extent that it is not frankly directed toward the fifth solution.

The fourth point of view which we shall call *mutationalism* is held by biologists who, without believing in preformation, also believe that structures appear in a purely endogenous way but then consider them as arising by chance from internal transformations and adapting to the environment due to a selection after the event. Now, if one transposes this method of interpretation to the level of nonhereditary adaptations one finds it is parallel to the schema of "trials and errors" belonging to *pragmatism* or

to *conventionalism:* according to this schema, the adjustment of behavior patterns is also explained by selection after the event of behavior arising by chance in relation to the external environment. For example, according to conventionalism Euclidean space with three dimensions seems to us more "true" than the other kinds of space because of the structure of our organs of perception, and is simply more "convenient" because it permits a better adjustment of those organs to the known data of the external world.

Finally, according to a fifth solution, the organism and the environment form an indissoluble entity, that is to say, beside chance mutations there are adaptional variations simultaneously involving a structuring of the organism and an action of the environment, the two being inseparable from each other. From the point of view of awareness, that means that the subject's activity is related to the constitution of the object, just as the latter involves the former. This is the affirmation of an irreducible interdependence between experience and reasoning. Biological *relativity* is thus extended into the doctrine of the interdependence of subject and object, of assimilation of the object by the subject and of the accommodation of the latter to the former.

The parallel between the theories of adaptation and those of intelligence having been outlined, study of the development of the latter will of course determine the choice it is fitting to make between those different possible hypotheses. However, in order to prepare for this choice and primarily in order to expand our concept of adaptation—given the continuity of the biological processes and the analogy of the solutions that an attempt has been made to supply on the different planes on which this problem is encountered—we have analyzed on the plane of the hereditary morphology of the organism a case of "kinetogenesis" suitable for illustrating the different solutions we have just catalogued.[3]

[3] For details, see our two articles: Les races lacustres de la 'Limnaea stagnalis' and Recherches sur les rapports de l'adaptation héréditaire avec le milieu, *Bulletin biologique de la France et de la Belgique,* LXII, 1929, pp. 424–455; and 2. Adaptation de la Limnaea stagnalis aux milieux lacustres de la Suisse romande, *Revue Suisse de Zoologie,* XXXVI, pp. 263–531.

There is found in almost all European and Asian marshes an aquatic mollusc, the *Limnaea stagnalis* (L.) which is typically elongated in shape. Now in the great lakes of Switzerland, Sweden, etc., this species is of a lacustrine variety, shortened and globular, whose form can easily be explained by the animal's motor accommodation, during growth, to the waves and movement of the water. After having verified this explanation experimentally, we succeeded in proving, by means of many breedings in the aquarium, that this shortened variety whose geological history can be followed from the paleolithic age to our own, became hereditary and perfectly stable (those genotypes obey in particular the laws of Mendelian segregation) in the places most exposed to the winds in the lakes of Neuchâtel and Geneva.

Thus it appears at first glance as though the Lamarckian solution fits such a case: The habits of contraction acquired under the influence of waves would have ended by transmitting themselves hereditarily in a morphologico-reflex ensemble constituting a new race. In other words, the phenotype would be imperceptibly transformed into a genotype by the lasting action of the environment. Unfortunately, in the case of the Limnaea as in all others, the laboratory experiment (breeding in an agitator producing an experimental contraction) does not show a trace of the hereditary transmission of acquired characteristics. Moreover the lakes of medium size do not have all the shortened varieties. If there is an influence of the environment in the constitution of hereditary contraction this influence is subjected to thresholds (of intensity, duration, etc.) and the organism, far from suffering it passively, reacts actively by an adaptation which transcends simple imposed habits.

Regarding the second solution, vitalism would not be able to explain the particulars of any adaptation. Why does the unconscious intelligence of the species, if it exists, not intervene everywhere it could be useful? Why did contraction take centuries to appear after the post glacial stocking of the lakes and why does it not yet exist in all the lakes?

The same objections apply to the solution of the problem in accordance with the theory of preformation.

On the other hand, the fourth solution appears to be im-

pregnable to attack. According to the theory of mutation the hereditary shortened structures would be due to chance endogenous variations (that is to say, with no relation to the environment nor with the phenotypic individual adaptations) and it would only be after the event that these forms, better preadapted than the others to the rough zones of the lakes, would multiply in the very places from which the elongated shapes would be excluded by natural selection. Chance and selection after the event would thus account for adaptation without any mysterious action of the environment on hereditary transmission, whereas the adaptation of non-hereditary individual variations would remain connected with the environmental action. But, in the case of our Limnaea, two strong objections to such an interpretation can be made. In the first place, if the elongated forms of the species could not endure as such in the parts of the lakes where the water is roughest, on the other hand the shortened genotypes can live in all the environments in which the species is represented, and we have introduced some to a new climate years ago, in a stagnant pond in the Swiss Plateau. If it were, therefore, a question of chance mutations, those genotypes should be scattered everywhere; but, in fact, they only appeared in lacustrine environments and moreover in those most exposed to the wind, precisely where the individual or phenotypic adaptation to the waves is most evident! In the second place, selection after the event is, in the case of the Limnaea, useless and impossible, for the elongated forms can themselves give rise to shortened variations which are not or not yet hereditary. One cannot therefore speak of chance mutations or of selection after the event to explain such adaptation.

Therefore only a fifth and last solution remains: This is to admit the possibility of hereditary adaptations simultaneously presupposing an action of the environment and a reaction of the organism other than the simple fixation of habits. As early as the morphologico-reflex level there exist interactions between the environment and the organism which are such that the latter, without passively enduring the constraint of the former, nor limiting itself on contact with it to manifesting already preformed structures, reacts by an active differentiation of reflexes (in the

particular case by a development of the reflexes of pedal adherence and of contraction) and by a correlative morphogenesis. In other words, the hereditary fixation of phenotypes or individual adaptations is not due to the simple repetition of habits which gave rise to them but to a mechanism *sui generis* which, through recurrence or anticipation, leads to the same result on the morphologico-reflex level.

Concerning the problem of intelligence, the lessons furnished by such an example seem to us to be the following. From its beginnings, due to the hereditary adaptations of the organism, intelligence finds itself entangled in a network of relations between the organism and the environment. Intelligence does not therefore appear as a power of reflection independent of the particular position which the organism occupies in the universe but is linked, from the very outset, by biological apriorities. It is not at all an independent absolute, but is a relationship among others, between the organism and things. If intelligence thus extends an organic adaptation which is anterior to it, the progress of reason doubtless consists in an increasingly advanced acquisition of awareness of the organizing activity inherent in life itself, and the primitive stages of psychological development only constitute the most superficial acquisitions of awareness of this work of organization. *A fortiori* the morphologico-reflex structures manifested by the living body, and the biological assimilation which is at the point of departure of the elementary forms of psychic assimilation would be nothing other than the most external and material outline of the adaptation whose profound nature the higher forms of intellectual activity would express increasingly well. One can therefore believe that intellectual activity, departing from a relation of interdependence between organism and environment, or lack of differentiation between subject and object, progresses simultaneously in the conquest of things and reflection on itself, these two processes of inverse direction being correlative. From this point of view, physiological and anatomical organization gradually appears to consciousness as being external to it and intelligent activity is revealed for that reason as being the very essence of the existence of our subjects. Whence the reversal which is at work in perspectives as

mental development progresses and which explains why the power of reason, while extending the most central biological mechanisms, ends by surpassing them at the same time in complementary externalization and internalization.

# *PART I*

## Elementary Sensorimotor Adaptations

Intelligence does not by any means appear at once derived from mental development, like a higher mechanism, and radically distinct from those which have preceded it. Intelligence presents, on the contrary, a remarkable continuity with the acquired or even inborn processes on which it depends and at the same times makes use of. Thus, it is appropriate, before analyzing intelligence as such, to find out how the formation of habits and even the exercise of the reflex prepare its appearance. This is what we are going to do in the first part, dedicating one chapter to the reflex and to the psychological questions that it raises, and a second chapter to the various acquired associations or elementary habits.

CHAPTER ONE

# THE FIRST STAGE:

## The Use of Reflexes

If, in order to analyze the first mental acts, we refer to hereditary organic reactions, we must study them not for their own sake but merely so that we may describe *in toto* the way in which they affect the individual's behavior. We should begin, therefore, by trying to differentiate between the psychological problem of the reflexes and the strictly biological problem.

Behavior observable during the first weeks of life is very complicated, biologically speaking. At first there are very different types of reflexes involving the medulla, the bulb, the optic commissures, the ectoderm itself; moreover, from reflex to instinct is only a difference of degree. Next to the reflexes of the central nervous system are those of the autonomic nervous system and all the reactions due to "protopathic" sensibility. There is, above all, the whole group of postural reflexes whose importance for the beginnings of the evolution of the mind has been demonstrated by H. Wallon. It is hard to envisage the organization of the foregoing mechanisms without giving the endocrine processes their just due as indicated by so many learned or spontaneous reactions. Physiological psychology is confronted at the present time by a host of problems which consist of determining the effects on the individual's behavior of each of these separate mechanisms. H. Wallon analyzes one of the most important of these questions in his excellent book on the disturbed child (*l'Enfant turbulent*): "Is there an emotional stage, or a stage of postural and extrapyramidal reactions prior to the sensorimotor or cortical stage?" Nothing better reveals the complexity of ele-

mentary behavior and the need to differentiate between the successive stages of concurrent physiological systems than Wallon's scholarly study of their genesis in which a wealth of pathologic material always substantiates his analysis.

Notwithstanding the fascinating conclusions thus reached, it seems to us difficult at the present time to go beyond a general description when it comes to grasping the continuity between the earliest behavior of the nursling and the future intellectual behavior. That is why, although in complete sympathy with Wallon's attempt to identify psychic mechanisms with those of life itself, we believe we should limit ourselves to emphasizing functional identity, from the point of view of simple external behavior.

In this respect the problem which arises in connection with reactions in the first weeks is only this: How do the sensorimotor, postural, and other reactions, inherent in the hereditary equipment of the newborn child, prepare him to adapt himself to his external environment and to acquire subsequent behavior distinguished by the progressive use of experience?

The psychological problem begins to pose itself as soon as the reflexes, postures, etc., are considered no longer in connection with the internal mechanism of the living organism, but rather in their relationships to the external environment as it is subjected to the individual's activity. Let us examine, from this point of view, the various fundamental reactions in the first weeks: sucking and grasping reflexes, crying and vocalization,[1] movements and positions of the arms, the head or the trunk, etc.

What is striking about this is that such activities from the start of their most primitive functioning, each in itself and some in relation to others, give rise to a systematization which exceeds their automatization. Almost since birth, therefore, there is "behavior" in the sense of the individual's total reaction and not only a setting in motion of particular or local automatizations only interrelated from within. In other words, the sequential manifestations of a reflex such as sucking are not comparable to the periodic starting up of a motor used intermittently, but constitute

[1] We shall return to the subject of prehension, vision and vocalization in the course of Chapter II.

an historical development so that each episode depends on preceding episodes and conditions those that follow in a truly organic evolution. In fact, whatever the intensive mechanism of this historical process may be, one can follow the changes from the outside and describe things as though each particular reaction determined the others without intermediates. This comprises total reaction, that is to say, the beginning of psychology.

§1. SUCKING REFLEXES.—Let us take as an example the sucking reflexes or the instinctive act of sucking; these reflexes are complicated, involving a large number of afferent fibers of the trigeminal and the glossopharyngeal nerves as well as the efferent fibers of the facial, the hypoglossal and the masseteric nerves, all of which have as a center the bulb of the spinal cord. First here are some facts:

*Observation 1.*—From birth sucking-like movements may be observed: impulsive movement and protrusion of the lips accompanied by displacements of the tongue, while the arms engage in unruly and more or less rhythmical gestures and the head moves laterally, etc.

As soon as the hands rub the lips the sucking reflex is released. The child sucks his fingers for a moment but of course does not know either how to keep them in his mouth or pursue them with his lips. Lucienne and Laurent, a quarter of an hour and a half hour after birth, respectively, had already sucked their hand like this: Lucienne, whose hand had been immobilized due to its position, sucked her fingers for more than ten minutes.

A few hours after birth, first nippleful of collostrum. It is known how greatly children differ from each other with respect to adaptation to this first meal. For some children like Lucienne and Laurent, contact of the lips and probably the tongue with the nipple suffices to produce sucking and swallowing. Other children, such as Jacqueline, have slower coördination: the child lets go of the breast every moment without taking it back again by himself or applying himself to it as vigorously when the nipple is replaced in his mouth. There are some children, finally, who need real forcing: holding their head, forcibly putting the nipple between the lips and in contact with the tongue, etc.

*Observation 2.*—The day after birth Laurent seized the nipple with his lips without having to have it held in his mouth. He immediately seeks the breast when it escapes him as the result of some movement.

During the second day also Laurent again begins to make sucking-like movements between meals while thus repeating the impulsive

movements of the first day: His lips open and close as if to receive a real nippleful, but without having an object. This behavior subsequently became more frequent and we shall not take it up again.

The same day the beginning of a sort of reflex searching may be observed in Laurent, which will develop on the following days and which probably constitutes the functional equivalent of the gropings characteristic of the later stages (acquisition of habits and empirical intelligence). Laurent is lying on his back with his mouth open, his lips and tongue moving slightly in imitation of the mechanism of sucking, and his head moving from left to right and back again, as though seeking an object. These gestures are either silent or interrupted by grunts with an expression of impatience and of hunger.

*Observation 3.*—The third day Laurent makes new progress in his adjustment to the breast. All he needs in order to grope with open mouth toward final success is to have touched the breast or the surrounding teguments with his lips. But he hunts on the wrong side as well as on the right side, that is to say, the side where contact has been made.

*Observation 4.*—Laurent at 0;0 (9) is lying in bed and seeks to suck, moving his head to the left and to the right. Several times he rubs his lips with his hand which he immediately sucks. He knocks against a quilt and a wool coverlet; each time he sucks the object only to relinquish it after a moment and begins to cry again. When he sucks his hand he does not turn away from it as he seems to do with the woolens, but the hand itself escapes him through lack of coördination; he then immediately begins to hunt again.

*Observation 5.*—As soon as his cheek comes in contact with the breast, Laurent at 0;0 (12) applies himself to seeking until he finds drink. His search takes its bearings: immediately from the correct side, that is to say, the side where he experienced contact.

At 0;0 (20) he bites the breast which is given him, 5 cm. from the nipple. For a moment he sucks the skin which he then lets go in order to move his mouth about 2 cm. As soon as he begins sucking again he stops. In one of his attempts he touches the nipple with the outside of his lips and he does not recognize it. But, when his search subsequently leads him accidentally to touch the nipple with the mucosa of the upper lip (his mouth being wide open), he at once adjusts his lips and begins to suck.

The same day, same experiment: after having sucked the skin for several seconds, he withdraws and begins to cry. Then he begins again, withdraws again, but without crying, and takes it again 1 cm. away; he keeps this up until he discovers the nipple.

*Observation 6.*—The same day I hold out my crooked index finger to Laurent, who is crying from hunger (but intermittently and without

violence). He immediately sucks it but rejects it after a few seconds and begins to cry. Second attempt: same reaction. Third attempt: he sucks it, this time for a long time and thoroughly, and it is I who retract it after a few minutes.

*Observation 7.*—Laurent at 0;0 (21) is lying on his right side, his arms tight against his body, his hands clasped, and he sucks his right thumb at length while remaining completely immobile. The nurse made the same observation on the previous day. I take his right hand away and he at once begins to search for it, turning his head from left to right. As his hands remained immobile due to his position, Laurent found his thumb after three attempts: prolonged sucking begins each time. But, once he has been placed on his back, he does not know how to coördinate the movement of the arms with that of the mouth and his hands draw back even when his lips are seeking them.

At 0;0 (24) when Laurent sucks his thumb, he remains completely immobile (as though having a nippleful: complete sucking, pantings, etc.). When his hand alone grazes his mouth, no coördination.

*Observation 8.*—At 0;0 (21): Several times I place the back of my index finger against his cheeks. Each time he turns to the correct side while opening his mouth. Same reactions with the nipple.

Then I repeat the same experiments as those in observation 5. At 0;0 (21) Laurent begins by sucking the teguments with which he comes in contact. He relinquishes them after a moment but searches with open mouth, while almost rubbing the skin with his lips. He seizes the nipple as soon as he brushes against it with the mucosa of his lower lip.

That evening, the same experiment, but made during a nursing which has been interrupted for this purpose. Laurent is already half asleep; his arms hang down and his hands are open (at the beginning of the meal his arms are folded against his chest and his hands are clasped). His mouth is placed against the skin of the breast about 5 cm. from the nipple. He immediately sucks without reopening his eyes but, after a few moments, failure awakens him. His eyes are wide open, his arms flexed again and he sucks with rapidity. Then he gives up, in order to search a little further away, on the left side which happens by chance to be the correct side. Again finding nothing, he continues to change places on the left side, but the rotatory movement which he thus gives his head results in making him let go the breast and go off on a tangent. In the course of this tangential movement he knocks against the nipple with the left commissure of his lips and everything that happens would seem to indicate that he recognizes it at once. Instead of groping at random, he only searches in the immediate neighborhood of the nipple. But as the lateral movements of his head made him describe a tangential curve opposite and not parallel to the curve of

the breast, he oscillates in space guided only by light, haphazard contacts with the breast. It takes a short time for these increasingly localized attempts to be successful. This last phase of groping has been noteworthy for the speed with which each approach to it has been followed by an attempt at insertion of the nipple, while the lips open and close with maximum vigor; and noteworthy also for the progressive adjusting of the tangential movements around the points of contact.

At 0;0 (23) a new experiment. Laurent is 10 cm. from the breast, searching for it on the left and on the right. While he searches on the left the nipple touches his right cheek. He immediately turns and searches on the right. He is then moved 5 cm. away. He continues to search on the correct side. He is brought nearer as soon as he grasps the skin; he gropes and finds the nipple.

Same experiment and same result that evening. But, after several swallows, he is removed. He remains oriented to the correct side.

At 0;0 (24) Laurent, during the same experiments, seems much faster. To localize his search it suffices for the nipple to be brushed by the outside of his lips and no longer only by the mucosa. Besides, as soon as he has noticed the nipple, his head's lateral movements become more rapid and precise (less extensive). Finally, it seems that he is henceforth capable not only of lateral movements but also of raising his head when his upper lip touches the nipple.

*Observation 9.*—At 0;0 (22) Laurent is awakened an hour after his meal, and only cries faintly and intermittently. I place his right hand against his mouth but remove it before he begins to suck. Then, seven times in succession he does a complete imitation of sucking, opening and closing his mouth, moving his tongue, etc.

*Observation 10.*—Here are two facts revealing the differences in adaptation according to whether the need for nourishment is strong or weak. At 0;0 (25) Laurent is lying on his back, not very hungry (he has not cried since his last meal) and his right cheek is touched by the nipple. He turns to the correct side but the breast is removed to a distance of 5 to 10 cm. For a few seconds he reaches in the right direction and then gives up. He is still lying on his back, facing the ceiling; after a moment his mouth begins to move slightly, then his head moves from side to side, finally settling on the wrong side. A brief search in this direction, then crying (with commissures of the lip lowered, etc.), and another pause. After a moment, another search in the wrong direction. No reaction when the middle of his right cheek is touched. Only when the nipple touches his skin about 1 cm. from his lips does he turn and grasp it.

On reading this description it would seem as though all the practice of the last weeks were in vain. It would seem, above all, that the ex-

citation zone of the reflex stops about 1 cm. from the lips, and that the cheek itself is insensitive. But on the next day the same experiment yields opposite results, as we shall see.

At 0;0 (26) Laurent is lying on his back, very hungry. I touch the middle of his cheek with my index finger bent first to the right, then to the left; each time he immediately turns to the correct side. Then he feels the nipple in the middle of his right cheek. But, as he tries to grasp it, it is withdrawn 10 cm. He then turns his head in the right direction and searches. He rests a moment, facing the ceiling, then his mouth begins to search again and his head immediately turns to the right side. This time he goes on to touch the nipple, first with his nose and then with the region between his nostrils and lips. Then he twice very distinctly repeats the movement observed at 0;0 (24) (see Obs. 8): He raises his head in order to grasp the nipple. The first time he just catches the nipple with the corner of his lips and lets it go. A second or two later, he vigorously lifts his head and achieves his purpose.

The way in which he discerns the nipple should be noted; at 0;0 (29) he explores its circumference with open and motionless lips before grasping it.

The theoretical importance of such observations seems to us to be as great as their triteness.[2] They make it possible for us to understand how a system of pure reflexes can comprise psychological behavior, as early as the systematization of their functioning. Let us try to analyze this process in its progressive adaptational and organization aspects.

§2. THE USE OF REFLEXES.—Concerning its *adaptation,* it is interesting to note that the reflex, no matter how well endowed with hereditary physiological mechanism, and no matter how stable its automatization, nevertheless needs to be used in order truly to adapt itself, and that it is capable of gradual accommodation to external reality.

Let us first stress this element of *accommodation.* The sucking reflex is hereditary and functions from birth, influenced either by diffuse impulsive movements or by an external excitant (Obs. 1); this is the point of departure. In order that a useful

[2] We are particularly happy to mention their agreement with those of R. Ripin and H. Hetzer: Frühestes Lernen des Säuglings in der Ernährungssituation, *Zeitschr. f. Psychol.,* 118, 1930, pp. 82–127. Observations of our children, made several years ago, were independent of the latter which makes their convergence a real one.

function may result, that is to say, swallowing, it often suffices to put the nipple in the mouth of the newborn child, but, as we know (Obs. 1), it sometimes happens that the child does not adapt at the first attempt. Only practice will lead to normal functioning. That is the first aspect of accommodation: contact with the object modifies, in a way, the activity of the reflex, and, even if this activity were oriented hereditarily to such contact, the latter is no less necessary to the consolidation of the former. This is how certain instincts are lost or certain reflexes cease to function normally, due to the lack of a suitable environment.[3] Moreover, contact with the environment not only results in developing the reflexes, but also in coördinating them in some way. Observations 2, 3, 5 and 8 show how the child, who first does not know how to suck the nipple when it is put in his mouth, grows increasingly able to grasp and even to find it, first after direct touch, then after contact with any neighboring region.[4]

How can such accommodations be explained? It seems to us difficult to invoke from birth the mechanism of acquired associations, in the limited sense of the term, or of "conditioned reflexes," both of which imply systematic training. On the contrary, the examining of these behavior patterns reveals at once the respects in which they differ from acquired associations: Whereas with regard to the latter, including conditioned reflexes, association is established between a certain perception,

[3] Thus Larguier des Brancels (*Introduction à la Psychologie,* 1921, p. 178), after recalling Spalding's famous experiments concerning the decline of instincts in newly hatched chickens, adds: "The sucking instinct is transitory. A calf which has been separated from its mother and fed by hand for a day or two and then is taken to another cow, more often than not refuses to nurse. The child behaves somewhat similarly. If he is first spoon-fed, he subsequently has great difficulty in taking the breast again."

[4] See Preyer (*L'Ame de l'Enfant,* translated by Variguy, 1887, pp. 213–217), in particular the following lines: "To be sure, sucking is not as fruitful the first as the second day and I have often observed in normal newborn children (1869) that attempts at sucking were completely vain in the first hours of life: when I made the experiment of putting an ivory pencil in their mouth, they were still uncoördinated" (p. 215). Also: "It is well known that newborn children, when put to the breast do not find the nipple without help; they only find it by themselves a few days later (in one case only on the eighth day), that is to say, later than animals" (pp. 215–216). And: "When the child is put to the breast the nipple often does not enter his mouth and he sucks the neighboring skin; this is still evident in the third week . . ." (p. 216).

foreign to the realm of the reflex, and the reflex itself (for example, between a sound, a visual perception, etc., and the salivary reflex), according to our observations, it is simply the reflex's own sensibility (contact of the lips with a foreign body) which is generalized, that is to say, brings with it the action of the reflex in increasingly numerous situations. In the case of Observations 2, 3, 5 and 8, for example, accommodation consists essentially of progress in the continuity of the searching. In the beginning (Obs. 2 and 3) contact with any part of the breast whatever sets in motion momentary sucking of this region, immediately followed by crying or a desultory search, whereas after several days (Obs. 5), the same contact sets in motion a groping during which the child is headed toward success. It is very interesting, in the second case, to see how the reflex, excited by each contact with the breast, stops functioning as soon as the child perceives that sucking is not followed by any satisfaction, as is the taking of nourishment (see Obs. 5 and 8), and to see how the search goes on until swallowing begins. In this regard, Observations 2 to 8 confirm that there is a great variety of kinds of accommodation. Sucking of the eider-down quilt, of the coverlet, etc., leads to rejection, that of the breast to acceptance; sucking of the skin (the child's hand, etc.) leads to acceptance if it is only a matter of sucking for the sake of sucking, but it leads to rejection (for example when it involves an area of the breast other than the nipple) if there is great hunger; the paternal index finger (Obs. 6) is rejected when the child is held against the breast, but is accepted as a pacifier, etc. In all behavior patterns it seems evident to us that learning is a function of the environment.

Surely all these facts admit of a physiological explanation which does not at all take us out of the realm of the reflex. The "irradiations," the "prolonged shocks," the "summations" of excitations and the intercoördination of reflexes probably explains why the child's searching becomes increasingly systematic, why contact which does not suffice to set the next operation in motion, does suffice in doing so a few days later, etc. Those are not necessarily mechanisms which are superposed on the reflex such as habit or intelligent understanding will be, later. But it remains no less true that the environment is indispensable to this opera-

tion, in other words, that reflex adaptation is partly accommodation. Without previous contact with the nipple and the experience of imbibing milk, it is very likely that the eider-down quilt, the wool coverlet, or the paternal index finger, after setting in motion the sucking reflex, would not have been so briskly rejected by Laurent.[5]

But if, in reflex adaptation, allowances must be made for accommodation, accommodation cannot be dissociated from progressive *assimilation,* inherent in the very use of the reflex. In a general way, one can say that the reflex is consolidated and strengthened by virtue of its own functioning. Such a fact is the most direct expression of the mechanism of assimilation. Assimilation is revealed, in the first place, by a growing need for repetition which characterizes the use of the reflex (functional assimilation) and, in the second place, by this sort of entirely practical or sensorimotor recognition which enables the child to adapt himself to the different objects with which his lips come in contact (recognitory and generalizing assimilations).

The need for repetition is in itself alone very significant; in effect, it is a question of a behavior pattern which shows a history and which proceeds to complicate the simple stimuli connected with the state of the organism considered at a given moment in time. A first stimulus capable of bringing the reflex into play is contact with an external object. Preyer thus set in motion the sucking movements of a newborn child by touching his lips, and Observation 1 shows us that children suck their hand a quarter of an hour or half an hour after birth. In the second place, there are internal stimuli, connected with the somato-affective states: diffuse impulsive movements (Obs. 1) or excitations due to hunger. But to these definite excitations, connected with particular moments in the life of the organism, there is added, it seems to us, the essential circumstance that the very repetition of the reflex movements constitutes a cynamogeny for them. Why, for instance, does Lucienne suck her fingers soon after birth for ten minutes in succession? This could not be

[5] In animals every slightly complicated reflex mechanism occasions reactions of the same kind. The beginnings of copulation in the mollusks, for example, give way to very strange gropings before the act is adapted.

because of hunger, since the umbilical cord had just been cut. There certainly is an external excitant from the moment the lips touch the hand. But why does the excitation last, in such a case, since it does not lead to any result except, precisely, to the use of the reflex? It therefore seems that, from the start of this primitive mechanism, a sort of circular process accompanies the function, the activity of the reflex having augmented due to its own use. If this interpretation remains doubtful, in so far as the point of departure is concerned, it obtains increasingly, on the other hand, with regard to subsequent behavior patterns. After the first feedings one observes, in Laurent (Obs. 2), sucking-like movements, in which it is difficult not to see a sort of autoexcitation. Besides, the progress in the search for the breast in Observations 2–5 and 8 seems also to show how much the function itself strengthened the tendency to suck. The counterproof of this is, as we have seen, the progressive decay of reflex mechanisms which are not used. How to interpret these facts? It is self-evident that "circular reaction," in Baldwin's sense of the term, could not yet be involved, that is to say, the repetition of a behavior pattern acquired or in the process of being acquired, and of behavior directed by the object to which it tends. Here it is only a matter of reflex and not acquired movements, and of sensibility connected with the reflex itself and not with the external objective. Nevertheless the mechanism is comparable to it from the purely functional point of view. It is thus very clear, in Observation 9, that the slightest excitation can set in motion not only a reflex reaction but a succession of seven reactions. Without forming any hypothesis on the way of conserving this excitation, or *a fortiori,* without wanting to transform this repetition into intentional or mnemonic behavior, one is compelled to state that, in such a case, there is a tendency toward repetition, or, in objective terms, cumulative repetition.

This need for repetition is only one aspect of a more general process which we can qualify as assimilation. The tendency of the reflex being to reproduce itself, it incorporates into itself every object capable of fulfilling the function of excitant. Two distinct phenomena must be mentioned here, both equally significant from this particular point of view.

The first is what we may call "generalizing assimilation," that is to say, the incorporation of increasingly varied objects into the reflex schema. When, for example, the child is hungry but not sufficiently so to give way to rage and to crying, and his lips have been excited by some accidental contact, we witness the formation of this kind of behavior pattern, so important due to its own future developments and the innumerable analogous cases which we shall observe in connection with other schemata. Thus, according to chance contacts, the child, from the first two weeks of life, sucks his fingers, the fingers extended to him, his pillow, quilt, bedclothes, etc.; consequently he assimilates these objects to the activity of the reflex.

To be sure, we do not claim, when speaking of "generalizing" assimilation, that the newborn child begins by distinguishing a particular object (the mother's breast) and subsequently applies to other objects the discoveries he has made about this first one. In other words, we do not ascribe to the nursling conscious and intentional generalization with regard to transition from the particular to the general, especially as generalization, in itself intelligent, never begins by such a transition but always proceeds from the undifferentiated schema to the individual and to the general, combined and complementary. We simply maintain that, without any awareness of individual objects or of general laws, the newborn child at once incorporates into the global schema of sucking a number of increasingly varied objects, whence the generalizing aspect of this process of assimilation. But is it not playing on words to translate a fact so simple into the language of assimilation? Would it not suffice to say "the setting in motion of a reflex by a class of analogous excitants?" And, if one sticks to the term assimilation, must the conclusion then be reached that the nonhabitual excitants of any reflex (for example the aggregate of objects capable of setting in motion the palpebral reflex when they approach the eye) give rise to an identical phenomenon of generalizing assimilation? There is nothing to it. What does present a particular and truly psychological problem, in the case of the sucking reflex, is that the assimilation of objects to its activity will gradually be generalized until, at the stage of acquired circular reactions and even at the

stage of intentional movements, it gives birth to a very complex and strong schema. From the end of the second month the child will suck his thumb systematically (with acquired coördination and not by chance), then at nearly five months his hands will carry all objects to his mouth and he will end by using these behavior patterns to recognize bodies and even to compose the first form of space (Stern's "buccal space"). It is thus certain that the first assimilations relating to sucking, even if they reveal a lack of differentiation between contact with the breast and contact with other objects, are not simple confusion destined to disappear with progress in nutrition, but constitute the point of departure for increasingly complex assimilations.

How to interpret this generalizing assimilation? The sucking reflex can be conceived as a global schema of coördinated movements which, if it is accompanied by awareness, certainly does not give rise to perception of objects or even of definite sensorial pictures but simply to an awareness of attitudes with at most some sensorimotor integration connected with the sensibility of the lips and mouth. Now this schema, due to the fact that it lends itself to repetitions and to cumulative use, is not limited to functioning under compulsion by a fixed excitant, external or internal, but functions in a way for itself. In other words, the child does not only suck in order to eat but also to elude hunger, to prolong the excitation of the meal, etc., and lastly, he sucks for the sake of sucking. It is in this sense that the object incorporated into the sucking schema is actually assimilated to the activity of this schema. The object sucked is to be conceived, not as nourishment for the organism in general, but, so to speak, as aliment for the very activity of sucking, according to its various forms. From the point of view of awareness, if there is awareness, such assimilation is at first lack of differentiation and not at first true generalization, but from the point of view of action, it is a generalizing extension of the schema which foretells (as has just been seen) later and much more important generalizations.

But, apart from this generalizing assimilation, another assimilation must be noted from the two first weeks of life, which we can call "recognitory assimilation." This second form seems inconsistent with the preceding one; actually it only reveals

progress over the other, however slight. What we have just said regarding the lack of differentiation which characterizes generalizing assimilation is, in effect, true only with respect to states of slight hunger or of satiety. But it is enough that the child be very hungry for him to try to eat and thus to distinguish the nipple from the rest. This search and this selectivity seem to us to imply the beginning of differentiation in the global schema of sucking, and consequently a beginning of recognition, a completely practical and motor recognition, needless to say, but sufficient to be called recognitory assimilation. Let us examine, from this point of view, the way in which the child rediscovers the nipple. Ever since the third day (Obs. 3), Laurent seems to distinguish the nipple from the surrounding teguments; he tries to nurse and not merely to suck. From the tenth day (Obs. 4), we observe the alacrity with which he rejects the eider-down quilt or the coverlet which he began to suck, in order to search for something more substantial. Furthermore, his reaction to his father's index finger (Obs. 6) could not be more definite: disappointment and crying. Lastly, the gropings on the breast itself (Obs. 5 and 8) also reveal selectivity. How is this kind of recognition to be explained?

Of course there could be no question, either here or in connection with generalizing assimilation, of the recognition of an "object" for the obvious reason that there is nothing in the states of consciousness of a newborn child which could enable him to contrast an external universe with an internal universe. Supposing that there are given simultaneously visual sensations (simple vision of lights without forms or depth), acoustic sensations and a tactile-gustatory and kinesthetic sensibility connected with the sucking reflex, it is evident that such a complexus would in no way be sufficient to constitute awareness of objects: the latter implies, as we shall see,[6] characteristically intellectual operations, necessary to secure the permanence of form and substance. Neither could there be a question of purely perceptive recognition or recognition of sensorial images presented by the external world, although such recognition considerably precedes the elaboration of objects (recognizing a person, a toy or a linen

[6] Volume II, *La Construction du Réel chez l'Enfant.*

cloth simply on "presentation" and before having a permanent concept of it). If, to the observer, the breast which the nursling is about to take is external to the child and constitutes an image separate from him, to the newborn child, on the contrary, there can only exist awareness of attitudes, of emotions, or sensations of hunger and of satisfaction. Neither sight nor hearing yet gives rise to perceptions independent of these general reactions. As H. Wallon has effectively demonstrated, external influences only have meaning in connection with the attitudes they arouse. When the nursling differentiates between the nipple and the rest of the breast, fingers, or other objects, he does not recognize either an object or a sensorial picture but simply rediscovers a sensorimotor and particular postural complex (sucking and swallowing combined) among several analogous complexes which constitute his universe and reveal a total lack of differentiation between subject and object. In other words, this elementary recognition consists, in the strictest sense of the word, of "assimilation" of the whole of the data present in a definite organization which has already functioned and only gives rise to real discrimination due to its past functioning. But this suffices to explain in which respect repetition of the reflex leads by itself to recognitory assimilation which, albeit entirely practical, constitutes the beginning of knowledge.[7] More precisely, repetition of the reflex leads to a general and generalizing assimilation of objects to its activity, but, due to the varieties which gradually enter this activity (sucking for its own sake, to stave off hunger, to eat, etc.), the schema of assimilation becomes differentiated and, in the most important differentiated cases, assimilation becomes recognitory.

In conclusion, assimilation belonging to the adaptation reflex appears in three forms: cumulative repetition, generalization of the activity with incorporation of new objects to it, and

[7] Let us repeat that we do not claim to specify the states of consciousness which accompany this assimilation. Whether these states are purely emotional or affective, connected with the postures accompanying sucking, or whether there exists at first conscious sensorial and kinesthetic discrimination, we could not decide by studying behavior of the first two or three weeks. What this behavior simply reveals is the groping and the discernment which characterizes the use of the reflex, and these are the two fundamental facts which authorize us to speak of psychological assimilation at this primitive stage.

finally, motor recognition. But, in the last analysis, these three forms are but one: The reflex must be conceived as an organized totality whose nature it is to preserve itself by functioning and consequently to function sooner or later for its own sake (repetition) while incorporating into itself objects propitious to this functioning (generalized assimilation) and discerning situations necessary to certain special modes of its activity (motor recognition). We shall see—and this is the sole purpose of this analysis—that these processes are again found, with the unwedging accounted for by the progressive complexity of the structures, in the stages of acquired circular reactions, of the first voluntary schemata and of truly intelligent behavior patterns.

The progressive adaptation of the reflex schemata, therefore, presupposes their *organization.* In physiology this truth is trite. Not only does the reflex arc as such presuppose an organization but, in the animal not undergoing laboratory experimentation, every reflex system constitutes in itself an organized totality. According to Graham Brown's theories, the simple reflex is, in effect, to be considered as a product of differentiation. From the psychological point of view, on the other hand, there is too great a tendency to consider a reflex, or even a complex instinctive act such as sucking, to be a summation of movements with, eventually, a succession of conscious states juxtaposed, and not as a real totality. But two essential circumstances induce us to consider the sucking act as already constituting psychic organization: The fact that sooner or later this act reveals a meaning, and the fact that it is accompanied by directed searching.

Concerning the meanings, we have seen how much sucking acts vary according to whether the newborn child is hungry and tries to nurse, or sucks in order to calm himself, or whether in a way he plays at sucking. It seems as though they have a meaning for the nursling himself. The increasing calm which succeeds a storm of crying and weeping as soon as the child is in position to take nourishment and to seek the nipple is sufficient evidence that, if awareness exists at all, such awareness is from the beginning awareness of meaning. But one meaning is necessarily relative to other meanings, even on the elementary plane of simple motor recognitions.

Furthermore, that organization exists is substantiated by the fact that there is directed search. The precocious searching of the child in contact with the breast, in spite of being commonplace, is a remarkable thing. Such searching, which is the beginning of accommodation and assimilation, must be conceived, from the point of view of organization, as the first manifestation of a duality of desire and satisfaction, consequently of value and reality, of complete totality and incomplete totality, a duality which is to reappear on all planes of future activity and which the entire evolution of the mind will try to abate, even though it is destined to be emphasized unceasingly.

Such are, from the dual point of view of adaptation and organization, the first expressions of psychological life connected with hereditary physiological mechanisms. This survey, though schematic, we believe suffices to show how the psyche prolongs purely reflex organization while depending on it. The physiology of the organism furnishes a hereditary mechanism which is already completely organized and virtually adapted but has never functioned. Psychology begins with the use of this mechanism. This use does not in any way change the mechanism itself, contrary to what may be observed in the later stages (acquisition of habits, of understanding, etc.). It is limited to strengthening it and to making it function without integrating it to new organizations which go beyond it. But within the limits of this functioning there is room for a historical development which marks precisely the beginning of psychological life. This development undoubtedly admits of a physiological explanation: if the reflex mechanism is strengthened by use or decays through lack of use, this is surely because coördinations are made or unmade by virtue of the laws of reflex activity. But a physiological explanation of this kind does not exclude the psychological point of view which we have taken. In effect, if, as is probable, states of awareness accompany a reflex mechanism as complicated as that of the sucking instinct, these states of awareness have an internal history. The same state of awareness could not twice reproduce itself identically. If it reproduces itself it is by acquiring in addition some new quality of what has already been seen, etc., consequently some meaning. But if, by chance, no state of aware-

ness yet occurred, one could nevertheless speak of behavior or of behavior patterns, given, on the one hand, the *sui generis* character of their development and, on the other, their continuity with those of subsequent stages. We shall state this in precise terms in our conclusion.

The true character of these behavior patterns involves the individual utilization of experience. In so far as the reflex is a hereditary mechanism it perhaps constitutes a racial utilization of experience. That is a biological problem of which we have already spoken (Introduction, §3) and which, while of highest interest to the psychologist, cannot be solved by his particular methods. But, inasmuch as it is a mechanism giving rise to use, and consequently a sort of experimental trial, the sucking reflex presupposes, in addition to heredity, an individual utilization of experience. This is the crucial fact which permits the incorporation of such a behavior pattern into the realm of psychology, whereas a simple reflex, unsubordinated to the need for use or experimental trial as a function of the environment (sneezing for example) is of no interest to us. Of what does this experimental trial consist? An attempt can be made to define it without subordinating this analysis to any hypothesis concerning the kinds of states of consciousness which eventually accompany such a process. Learning connected with the reflex or instinctive mechanism is distinguished from the attainments due to habits or intelligence by the fact that it retains nothing external to the mechanism itself. A habit, such as that of a 2- or 3-month-old baby who opens his mouth on seeing an object, presupposes a mnemonic fixation related to this object. A tactile-motor schema is formed according to the variations of the object and this schema alone explains the uniformity of the reaction. In the same way the acquisition of an intellectual operation (counting, for instance) implies memory of the objects themselves or of experiments made with the objects. In both cases, therefore, something external to the initial mechanism of the act in question is retained. On the other hand, the baby who learns to suck retains nothing external to the act of sucking; he undoubtedly bears no trace either of the objects or the sensorial pictures on which later attempts have supervened. He merely records the se-

ries of attempts as simple acts which condition each other. When he recognizes the nipple, this does not involve recognition of a thing or of an image but rather the assimilation of one sensorimotor and postural complex to another. If the experimental trial involved in sucking presupposes environment and experience, since no functional use is possible without contact with the environment, this is a matter of a very special kind of experimental trial, of an autoapprenticeship to some extent and not of an actual acquisition. This is why, if these first psychological behavior patterns transcend pure physiology—just as the individual use of a hereditary mechanism transcends heredity—they still depend on them to the highest degree.

But the great psychological lesson of these beginnings of behavior is that, within the limits we have just defined, the experimental trial of a reflex mechanism already entails the most complicated accommodations, assimilations and individual organizations. Accommodation exists because, even without retaining anything from the environment as such, the reflex mechanism needs the environment. Assimilation exists because, through its very use, it incorporates to itself every object capable of supplying it with what it needs and discriminates even these objects thanks to the identity of the differential attitudes they elicit. Finally, organization exists, inasmuch as organization is the internal aspect of this progressive adaptation. The sequential uses of the reflex mechanism constitute organized totalities and the gropings and searchings apparent from the beginnings of this period of experimental trial are oriented by the very structure of these totalities.

But if these behavior patterns transcend pure physiology only to the very slight extent in which individual use has a history independent of the machine predetermined by heredity (to the point where it could seem almost metaphorical to characterize them as "behavior patterns" as we have done here), they nevertheless seem to us to be of essential importance to the rest of mental development. In effect, the functions of accommodation, of assimilation and of organization which we have just described in connection with the use of a reflex mechanism will be found once more in the course of subsequent stages and will

acquire increasing importance. In a certain sense, we shall even see that the more complicated and refined intellectual structures become, the more this functional nucleus will constitute the essence of these very structures.

§3. ASSIMILATION: BASIC FACT OF PSYCHIC LIFE.— In studying the use of reflexes we have ascertained the existence of a fundamental tendency whose manifestations we shall rediscover at each new stage of intellectual development: the tendency toward repetition of behavior patterns and toward the utilization of external objects in the framework of such repetition. This assimilation—simultaneously reproductive, generalizing, and recognitory—constitutes the basis of the functional use which we have described with respect to sucking. Assimilation is therefore indispensable to reflex accommodation. Moreover, it is the dynamic expression of the static fact of organization. From this double point of view it emerges as a basic fact, the psychological analysis of which must yield genetic conclusions.

Three circumstances induce us to consider assimilation the fundamental fact of psychic development. The first is that assimilation constitutes a process common to organized life and mental activity and is therefore an idea common to physiology and psychology. In effect, whatever the secret mechanism of biological assimilation may be, it is an empirical fact that an organ develops while functioning (by means of a certain useful exercise and fatigue). But when the organ in question affects the external behavior of the subject, this phenomenon of functional assimilation presents a physiological aspect inseparable from the psychological aspect; its parts are physiological whereas the reaction of the whole may be called psychic. Let us take for example the eye which develops under the influence of the use of vision (perception of lights, forms, etc.). From the physiological point of view it can be stated that light is nourishment for the eye (in particular in primitive cases of cutaneous sensibility in the lower invertebrates, in whom the eye amounts to an accumulation of pigment dependent on environing sources of light). Light is absorbed and assimilated by sensitive tissues and this action brings with it a correlative development of the organs affected. Such a

process undoubtedly presupposes an aggregate of mechanisms whose start may be very complex. But, if we adhere to a global description—that of behavior and consequently of psychology—the things seen constitute nourishment essential to the eye since it is they which impose the continuous use to which the organs owe their development. The eye needs light images just as the whole body needs chemical nourishment, energy, etc. Among the aggregate of external realities assimilated by the organism there are some which are incorporated into the parts of the physico-chemical mechanisms, while others simply serve as functional and general nourishment. In the first case, there is physiological assimilation, whereas the second may be called psychological assimilation. But the phenomenon is the same in both cases: the universe is embodied in the activity of the subject.

In the second place, assimilation reveals the primitive fact generally conceded to be the most elementary one of psychic life: repetition. How can we explain why the individual, on however high a level of behavior, tries to reproduce every experience he has lived? The thing is only comprehensible if the behavior which is repeated presents a functional meaning, that is to say, assumes a value for the subject himself. But whence comes this value? From functioning as such. Here again, functional assimilation is manifest as the basic fact.

In the third place, the concept of assimilation from the very first embodies in the mechanism of repetition the essential element which distinguishes activity from passive habit: the coördination of the new with the old which foretells the process of judgment. In effect, the reproduction characteristic of the act of assimilation always implies the incorporation of an actual fact into a given schema, this schema being constituted by the repetition itself. In this way assimilation is the greatest of all intellectual mechanisms and once more constitutes, in relation to them, the truly basic fact.

But could not this description be simplified by eliminating a concept which is so fraught with meaning that it might seem equivocal? In his remarkable essays on functional psychology Claparède[8] chooses without adding anything as a point of de-

8 See *l'Education fonctionnelle,* Delachaux and Niestlé, 1931.

parture of all mental activity the very fact of need. How can it be explained that certain behavior patterns give rise to spontaneous repetition? How does it happen that useful acts reproduce themselves? Because, says Claparède, they answer a need. Needs thus mark the transition between organic life, from which they emanate, and psychic life, of which they constitute the motive power.

The great advantage of this phraseology is that it is much simpler than that of assimilation. Besides, on the basis of what Claparède maintains, it is very difficult not to agree with him. Since need is the concrete expression of what we have called the process of assimilation, we could not raise doubts concerning the ground for this conception to which we personally owe much. But the question is to know whether, precisely because of its simplicity, it does not bring up initial problems which the concept of assimilation permits us to refer to biological study. There seems to us to be two difficulties.

In the first place, if need as such is the motive power for all activity, how does it direct the movements necessary to its satisfaction? With admirable analytical acuteness, Claparède himself has raised the question. Not only, he says, does one not understand why the pursuit of a goal coördinates useful actions, but furthermore, one does not see how, when one means fails, others are attempted. It transpires, in effect, especially when acquired associations are superimposed on the reflex, that an identical need releases a succession of different behavior patterns, but always directed toward the same end. What is the instrument of this selection and of this coördination of advantageous reactions?

It is self-evident that it would be useless to try to resolve these fundamental problems now. But does not the question arise because one begins by dissociating the need from the act in its totality? The basic needs do not exist, in effect, prior to the motivating cycles which permit them to be gratified. They appear during functioning. One could not say, therefore, that they precede repetition: they result from it as well, in an endless circle. For example empty sucking or any similar practice constitutes training which augments need as well as the reverse. From the psychological point of view, need must not be conceived as be-

ing independent of global functioning of which it is only an indication. From the physiological point of view, moreover, need presupposes an organization in "mobile balance" of which it simply indicates a transitory imbalance. In both kinds of terminology, need is thus the expression of a totality momentarily incomplete and tending toward reconstituting itself, that is to say, precisely what we call a cycle or a schema of assimilation: Need manifests the necessity of the organism or an organ to use an external datum in connection with its functioning. The basic fact is therefore not need, but the schemata of assimilation of which it is the subjective aspect. Henceforth it is perhaps a pseudo question to ask how need directs useful movements. It is because these movements are already directed that need sets them in motion. In other words, organized movements, ready for repetition, and need itself constitute only one whole. True, this conception, very clear with regard to the reflex or any innate organization, ceases to seem so with respect to acquired associations. But perhaps the difficulty comes from taking literally the term "associations," whereas the fact of assimilation makes it possible to explain how every new schema results from a differentiation and a complication of earlier schemata and not of an association between elements given in an isolated state. This hypothesis even leads to an understanding of how a sole need can set in motion a series of successive efforts. On the one hand, all assimilation is generalizing and, on the other hand, the schemata are capable of intercoördination through reciprocal assimilation as well as being able to function alone. (See stages IV–VI concerning this.)

A second difficulty seems to us to appear when one considers need as the basic fact of psychic life. Needs are supposed, in such a case, to insure the transition between organism and psyche; they constitute in some way the physiological motive power for mental activity. But if certain corporeal needs play this role in a large number of lower behavior patterns (such as the search for food in animal psychology), in the young child the principal needs are of a functional category. The functioning of the organs engenders, through its very existence, a psychic need *sui generis*, or rather a series of vicarious needs whose complexity transcends, from the very beginning, simple organic satisfaction. Further-

more, the more the intelligence develops and strengthens, the more the assimilation of reality to functioning itself is transformed into real comprehension, the principal motive power of intellectual activity thus becoming the need to incorporate things into the subject's schemata. This vicariousness of needs, which unceasingly transcend themselves to go beyond the purely organic plane, seems to show us anew that the basic fact is not need as such but rather the act of assimilation, which embodies in one whole functional need, repetition and that coördination between subject and object which foretells discrepancy and judgment.

To be sure, invoking the concept of assimilation does not constitute an explanation of assimilation itself. Psychology can only begin with the description of a basic fact without being able to explain it. The ideal of absolute deduction could only lead to verbal explanation. To renounce this temptation is to choose as a principle an elementary fact amenable to biological treatment simultaneously with psychological analysis. Assimilation answers this. Explanation of this fact is in the realm of biology. The existence of an organized totality which is preserved while assimilating the external world raises, in effect, the whole problem of life itself. But, as the higher cannot be reduced to the lower without adding something, biology will not succeed in clarifying the question of assimilation without taking into account its psychological aspect. At a certain level life organization and mental organization only constitute, in effect, one and the same thing.

CHAPTER II

# THE SECOND STAGE:

## The First Acquired Adaptations and the Primary Circular Reaction

The hereditary adaptations are doubled, at a given moment, by adaptations which are not innate to which they are subordinated little by little. In other words, the reflex processes are progressively integrated into cortical activity. These new adaptations constitute what are ordinarily called "acquired associations," habits or even conditioned reflexes, to say nothing of intentional movements characteristic of a third stage. Intent, which is doubtless imminent to the more primitive levels of psychological assimilation, could not, in effect, be aware of itself, and thus differentiate behavior, before assimilation through "secondary" schemata, that is to say, before the behavior patterns born of the exercise of prehension and contemporaneous with the first actions brought to bear on things. We can therefore ascribe to the present stage intentional movements as the higher limit and the first nonhereditary adaptations as the lower limit.

In truth it is extremely precarious to specify when acquired adaptation actually begins in contradistinction to hereditary adaptation. From a theoretical point of view, the following criterion can be adopted: in every behavior pattern of which the adaptation is determined by heredity, assimilation and accommodation form one entity and remain undifferentiated, whereas with acquired adaptation they begin to dissociate themselves. In other words, hereditary adaptation does not admit of any apprenticeship outside its own use, whereas acquired adaptation implies an apprenticeship related to the new conditions of the external environment simultaneously with an incorporation of

the objects to the schemata thus differentiated. But if one proceeds from theory to the interpretation of particular facts, great difficulties arise in distinguishing real acquisition from simple preformed coördination.

In effect, how is it possible to have a clear idea of the moment whence there is retention of some other condition external to the reflex mechanism itself? In the use of the reflex, as we have seen, there is only fixation of the mechanism as such, and it is in this respect that accommodation of a hereditary schema, while presupposing experience and contact with the environment, forms only one entity with assimilation, that is to say, with the functional use of this schema. At a given moment, on the other hand, the child's activity retains something external to itself, that is to say, it is transformed into a function of experience; in this respect there is acquired accommodation. For instance, when the child systematically sucks his thumb, no longer due to chance contacts but through coördination between hand and mouth, this may be called acquired accommodation. Neither the reflexes of the mouth nor of the hand can be provided such coördination by heredity (there is no instinct to suck the thumb!) and experience alone explains its formation. But if this is clear with regard to that kind of behavior pattern, in how many others is it impossible to draw a clear boundary between the pure reflex and the utilization of experience? The multiple aspects of visual accommodation, for example, comprise an inseparable mixture of reflex use and true acquisition.

There is the same difficulty from the point of view of assimilation. Psychological assimilation characteristic of the reflex consists, as we have seen, in a cumulative repetition with progressive incorporation of the objects into the cycle which has thus been reproduced. But nothing, in such a behavior pattern, yet implies that it is directed by the new results to which it leads. To be sure, in the sucking act, there is from the beginning directed searching and, in case of hunger, success alone gives meaning to the series of gropings. But the result sought yields nothing new in relation to the primitive sensorimotor field of the reflex itself. On the contrary, in the realm of acquired adaptation toward a new result (new either through the character of the sen-

sorial pictures which define it, or through the procedures set in motion to obtain it), this directs repetition. Whereas, in the reflex, assimilation only formed one entity with accommodation, henceforth the reproduction of the new act, or the assimilation of objects to the schema of this act, constitutes a process distinct from accommodation itself. Such a process can be very slightly differentiated when the acquired adaptation merely prolongs the reflex adaptation, but it is the more distinct from accommodation as the new act is more complex. Thus it is that, in the acquisition of prehension, it is one thing to repeat indefinitely a maneuver which has been successful and quite another thing to attempt to grasp an object in a new situation. The repetition of the cycle which has been actually acquired or is in the process of being acquired is what J. M. Baldwin has called the "circular reaction": this behavior pattern will constitute for us the principle of assimilation *sui generis* characteristic of this second stage. But if such a distinction between the simple repetition of the reflex and the "circular reaction" is theoretically clear, it goes without saying that here again the greatest difficulties confront concrete analysis.

Now let us proceed to examining the facts, first grouping them according to separate and distinct realms of activity.

§1. ACQUIRED SUCKING HABITS.—Superimposed on the reflex behavior patterns, which we have described in the first chapter, are, from the second or third month, certain forms of sucking which are unquestionably new. We shall begin by describing the two principal circular reactions—the systematic protrusion of the tongue (later with the action of saliva, of the lips, etc.), and the sucking of the thumb. These two activities will provide us with the type of that which is spontaneous acquired habit, with active assimilation and accommodation. Thereupon we shall discuss some facts concerning accommodation, commonly called "association transfers" or "sensorimotor associations" (setting in motion of sucking by various signals: position, noises, optical signals, etc.) and we shall see that these partial accommodations, however mechanical and passive they may appear to be, in reality constitute simple, isolated and abstract links of the

cycles inherent in circular reaction. Finally we shall speak of certain coördinations between sucking and vision.

Here are examples of the first group of facts (circular reactions):

*Observation 11.*—Laurent at 0;0 (30) stays awake without crying, gazing ahead with wide open eyes. He makes sucking-like movements almost continually, opening and closing his mouth in slow rhythm, his tongue constantly moving. At certain moments his tongue, instead of remaining inside his lips, licks the lower lip; the sucking recommences with renewed ardor.

Two interpretations are possible. Either at such times there is searching for food and then the protrusion of the tongue is merely a reflex inherent in the mechanism of sucking and swallowing, or else this marks the beginning of circular reaction. It seems, for the time being, that both are present. Sometimes protrusion of the tongue is accompanied by disordered movements of the arms and leads to impatience and anger. In such a case there is obviously a seeking to suck, and disappointment. Sometimes, on the other hand, protrusion of the tongue is accompanied by slow, rhythmical movements of the arms and an expression of contentment. In this case the tongue comes into play through circular reaction.

*Observation 12.*—At 0;1 (3) Laurent puts out his tongue several times in succession. He is wide awake, motionless, hardly moves his arms and makes no sucking-like movements; his mouth is partly open and he keeps passing his tongue over the lower lip.—At 0;1 (5) Laurent begins sucking-like movements and then the sucking is gradually replaced by the preceding behavior.—At 0;1 (6) he plays with his tongue, sometimes by licking his lower lip, sometimes by sliding his tongue between his lips and gums.—The following days this behavior is frequently repeated and always with the same expression of satisfaction.

*Observation 13.*—At 0;1 (24) Lucienne plays with her tongue, passing it over her lower lip and licking her lips unceasingly. Observation is made of the existence of a habit acquired a certain number of days previous. The behavior is extended to sucking the thumb and beyond.

*Observation 14.*—During the second half of the second month, that is to say, after having learned to suck his thumb, Laurent continues to play with his tongue and to suck, but intermittently. On the other hand, his skill increases. Thus, at 0;1 (20) I notice he grimaces while placing his tongue between gums and lips and in bulging his lips, as well as making a clapping sound when quickly closing his mouth after these exercises.

*Observation 15.*—During the third month he adds to the protrusion of his tongue and finger sucking new circular reactions involving the mouth. Thus from 0;2 (18) Laurent plays with his saliva, letting it accumulate within his half-open lips and then abruptly swallowing it. About the same period he makes sucking-like movements, with or without putting out his tongue, changing in various ways the position of his lips; he bends and contracts his lower lip, etc.—These exercises subsequently become increasingly varied and do not deserve more detailed examination from the point of view we have taken in this study.

Finger sucking also gives rise to evident acquisition.

*Observation 16.*—At 0;1 (1) Laurent is held by his nurse in an almost vertical position, shortly before the meal. He is very hungry and tries to nurse with his mouth open and continuous rotations of the head. His arms describe big rapid movements and constantly knock against his face. Twice, when his hand was laid on his right cheek, Laurent turned his head and tried to grasp his fingers with his mouth. The first time he failed and succeeded the second. But the movements of his arms are not coördinated with those of his head; the hand escapes while the mouth tries to maintain contact. Subsequently, however, he catches his thumb; his whole body is then immobilized, his right hand happens to grasp his left arm and his left hand presses against his mouth. Then a long pause ensues during which Laurent sucks his left thumb in the same way in which he nurses, with greed and passion (pantings, etc.).

There is therefore a complete analogy with Observation 7 of Chapter I. But it is more firmly established that nothing external forces the child to keep his hand in his mouth; the arms are not immobilized by the reclining position of the subject but by a spontaneous attitude. Nevertheless the fact observed lends itself to two interpretations: either, as may be the case from the first consecutive days after birth, sucking immobilizes the whole body and consequently the hands—the arms remain tight against the torso while the newborn child nurses, and it is conceivable that it may be the same when he sucks his thumb which he has found by chance—or else there is direct coördination between sucking and the arm movements. Subsequent observations seem to show that actual behavior foretells this coördination.

*Observation 17.*—At 0;1 (2) Laurent in his crib cries with hunger. He is lifted to an almost vertical position. His behavior then goes through four sequential phases quite distinct from one another. He begins by calming himself and tries to suck while turning his head from left to right and back again while his arms flourish without direction. Then (second phase) the arms, instead of describing movements of maximum breadth, seem to approach his mouth. Several times each hand brushes

his lips; the right hand presses against the child's cheek and clasps it for a few seconds. Meanwhile the mouth is wide open and unceasingly attempts to grasp something. The left thumb is then caught and the two arms become rigid, the right arm against the chest under the left arm which is held by the mouth. During a third phase, the arms again wave about in space without direction, the left thumb leaving the mouth after a few minutes. During this time the child becomes angry, his head thrown back and his cries alternating with attempts to suck. Finally a fourth phase begins during which the hands again approach the mouth which tries to seize the fingers which touch it. These last attempts meet with no success and crying ensues.

Can coördination be mentioned this time? Each of these phases finds its parallel in the behavior of the preceding weeks: from the first days of life babies are seen slashing their faces with their fingers while the mouth seems to try to grasp something. Nevertheless the sequence of the four phases seems to indicate a beginning of a connection between the movements of the arms and the sucking attempts.

*Observation 18.*—At 0;1 (3) Laurent (same position) does not seem to reveal any coördination between hands and mouth before nursing. On the other hand, after a meal, when he was still wide awake and trying to suck, his arms, instead of gesticulating aimlessly, constantly move toward his mouth. To be more precise, it has occurred to me several times that the chance contact of hand and mouth set in motion the directing of the latter toward the former and that then (but only then), the hand tries to return to the mouth. Laurent succeeded in sucking his fingers four times, his hand and arms immediately becoming immobilized. But that has never lasted more than a few seconds.—The evening of the same day Laurent, after nursing, remained wide awake and continued to try to suck, interspersing his attempts with vigorous cries. I then grasped his right arm and held it until his mouth began to suck his hand. As soon as the lips were in contact with the hand, the arms stopped resisting and remained still for several moments. This phenomenon has been confirmed since I made the experiment—since 0;0 (15)—but as a rule the position is not maintained. Only when the thumb is sucked does immobility result (see Obs. 7 and 16). This time, on the contrary, the arm remained immobile for a moment, although the back of the hand only was in contact with the lips; the latter obviously tried to explore the whole hand. After a moment, the hand lost the contact but rediscovered it. It is no longer the mouth that seeks the hand, but the hand which reaches for the mouth. Thirteen times in succession I have been able to observe the hand go back into the mouth. There is no longer any doubt that coördination exists. The mouth may be seen opening and the hand directing itself toward it simultaneously. Even the failures are significant. It thus happens that

the fingers are planted on the cheek while the open mouth is ready to receive them.

*Observation 19.*—At 0;1 (4) after the 6 P.M. meal Laurent is wide awake (as was not the case at the preceding meals) and not completely satisfied. First he makes vigorous sucking-like movements, then his right hand may be seen approaching his mouth, touching his lower lip and finally being grasped. But as only the index finger was grasped, the hand fell out again. Shortly afterward it returned. This time the thumb was in the mouth while the index finger was placed between the gums and the upper lip. The hand then moves 5 cm. away from the mouth only to reënter it; now the thumb is grasped and the other fingers remain outside. Laurent then is motionless and sucks vigorously, drooling so much that after a few moments he is removed. A fourth time the hand approaches and three fingers enter the mouth. The hand leaves again and reënters a fifth time. As the thumb has again been grasped, sucking is resumed. I then remove the hand and place it near his waist. Laurent seems to give up sucking and gazes ahead, contented and satisfied. But after a few minutes the lips move and the hand approaches them again. This time there is a series of setbacks; the fingers are placed on the chin and lower lip. The index finger enters the mouth twice (consequently the sixth and seventh time this has succeeded). The eighth time the hand enters the mouth, the thumb alone is retained and sucking continues. I again remove the hand. Again lip movements cease, new attempts ensue, success results for the ninth and tenth time, after which the experiment is interrupted.

*Observation 20.*—At 0;1 (5) and 0;1 (6) Laurent tries to catch his thumb as soon as he awakes but is unsuccessful while he is lying on his back. His hand taps his face without finding his mouth. When he is vertical, however (held by the waist, his arms and torso free), he quickly finds his lips.—At 0;1 (7) on the other hand, I find him sucking his thumb while he is lying down. But it keeps escaping him because it does not go far into his mouth but between the upper lip and the gum. Progress ensues, however, because the thumb, after leaving the mouth, returns to it several times in succession. Unfortunately, between these successful attempts, Laurent taps his nose, cheeks and eyes. Finally he becomes angry as the result of an unsuccessful attempt.—The following days, coordination is accomplished. At 0;1 (9), for example, Laurent sucks his thumb while lying on his back. I take it out of his mouth and, several times in succession, he puts it back into his mouth again almost immediately (having at most groped between nose and mouth) and only grasping the thumb, his other fingers remaining outside the mouth.

*Observation 21.*—At the end of the second month Laurent sucked his left thumb as well as his right. At 0;1 (21), for example, while lying on

his left side, he tries to suck his left thumb. After failure due to his position, he raises his right arm. Unable to grasp the thumb he then turns to the right, manages to lie on his back and continues searching. He almost reaches his right thumb but, happening to fail, he returns to his left hand and directs it toward his mouth. Failing once more, he again turns to the right and this time succeeds in seizing the right thumb.—This example reveals that Laurent is equally adept at sucking both thumbs. Subsequently, however, he became more accustomed to sucking the left thumb so that he injured it slightly and it had to be bandaged with his hand attached. After some anger and groping he then resumed sucking his right thumb (0; (7) and the days following).

*Observation 22.*—During the third month thumb sucking grew less important to Laurent due to the pressure of new interests, visual, auditory, etc. From 0;2 (15) I note that Laurent now sucks his thumb only to assuage his hunger and chiefly to put himself to sleep. This is an interesting example of specialization of the habit, also observable in Jacqueline. When Laurent cries his thumb goes to the rescue. At 0;2 (19) I note that he even closes his eyes and turns on his right side to go to sleep at the moment his thumb touched his lips.—During the third month the thumb is opposite the fingers at the moment sucking takes place. At the end of the second month Laurent began by sucking the back of his hand and of his fingers, or several fingers together, or the thumb and index finger, before finding the thumb alone. During the third month, on the contrary, the thumb gradually placed itself opposite the other fingers and Laurent managed to grasp it at the first attempt and suck it alone.

*Observation 23.*—In the case of Lucienne who did not undergo the sort of training to which I subjected Laurent, the coördination between arm movements and sucking was only definitely established at 0;2 (2). At 0;1 (25) and 0;1 (26) the hands touch the mouth constantly but I still observe Lucienne's incapacity to hold her thumb between her lips for a long time and above all to find it again once it has left. On the other hand, at 0;2 (2) I was able to make the two following observations. At 6 P.M., after the meal, her hands wandered around her mouth and she alternately sucked her fingers (chiefly the index finger), the back of her hand and her wrist. When her hand escapes her mouth it approaches it again and coördination is reëstablished. At 8 P.M. Lucienne is awakened and again sucks her fingers: her hand remains still for long moments and then the mouth opens to grasp it at the same time as the hand approaches the mouth. The following day, the same observations: coördination was reëstablished during the whole morning and for several moments in the evening. I particularly noted the following: the hand groping in the right direction, then an abrupt

movement of the fingers into the mouth which was already open and motionless. The rest of the observations confirmed the fact that coordination had become permanent.

*Observation 24.*—In the case of Jacqueline the first sure signs date from 0;1 (28) and the days following. She puts her left hand in her mouth when she is very hungry, a few moments before nursing. After the meal she often puts her fingers in her mouth again, to prolong sucking. From approximately 0;4 (5) the habit becomes systematic and she must suck her thumb in order to go to sleep.

In addition it is to be noted that the objects grasped are carried to the mouth from approximately 0;3 (15).

Putting out the tongue and finger sucking thus constitute the first two examples of a behavior pattern which prolongs the functional use of the reflex (sucking-like movements), but with the acquisition of some element external to the hereditary mechanisms. The new use of the tongue seems to go beyond the simple reflex play involved in sucking. With regard to the thumb, let us repeat that no instinct to suck the fingers exists and, even if the act of bringing food to the mouth were a hereditary behavior pattern, it is evident that the late appearance of this act indicates the interdiction of acquired associations, superimposed on ultimate reflex coördination. In characterizing these acquisitions it must also be noted that they imply an active element. It is not a question of associations imposed by the environment, but rather of relationships discovered and even created in the course of the child's own searchings. It is this twofold aspect of acquisition and activity which characterizes what we shall henceforth call "circular reactions" not in the rather loose sense of the term as used by Baldwin, but in Mr. Wallon's limited sense;[1] the functional use leading to the preservation or the rediscovery of a new result.

Along with actual circular reactions sucking also gives rise to behavior patterns in which accommodation is predominant. Here are involved those acquired associations which are often called "associative transfers" when one does not wish to go so far as to speak of "conditioned reflexes." Let us first note that circular reaction brings such transfers with it. In the course of progressive coördination between sucking and hand and arm

[1] *L'enfant turbulant*, p. 85.

movements it is evident that associations are established which direct the thumb to the mouth. The contact of the fingers with the covers, with the face, lips, etc., thus serves sooner or later as a signal which directs the hand. But, outside these mnemonic acquisitions or transfers inherent in circular reaction, there are some which seem to result from single automatic training without the appearance of the element of activity characteristic of the preceding reactions. What must we think of this?

It is appropriate to recall here the fine observations of two of Charlotte Bühler's collaborators, M. Hetzer and R. Ripin,[2] on the nursling's training as a function of feeding conditions (*Ernährungssituation*). According to these authors, three stages in the child's behavior may be distinguished. The first stage comprises the first week: the nursling attempts to suck only when his lips are in contact with the breast or the bottle. This we have seen in Chapter I (§§ 1 and 2). The second stage extends from the second to the eighth or ninth week: the nursling seeks the breast as soon as he finds himself in situations which regularly precede the meal (dressing, diaper changing, a stretched-out position, etc.). Finally the third stage begins between 0;3 and 0;4 and can be recognized by the appearance of visual signals. It is enough that the child sees the bottle or the objects which remind him of the meal for him to open his mouth and cry. Let us examine separately the second and third of these behavior patterns; both of them are in the category of acquired associations, but under different headings.

The behavior patterns characteristic of these stages seem to constitute the prototype of passive association (*Signalwirkung*). Contrary to the transfers characteristic of active circular reaction, the former seem due to the pressure of external circumstances subject to repetition. But, as we shall see, this is only a probability and such accommodations presuppose an element of activity. Concerning the reality of the facts observed we obviously agree with Charlotte Bühler and her collaborators. It is certain that, at a given moment in development, relationships are established between the position of the child, tactile and

[2] H. Hetzer, and R. Ripin, *op. cit.*; and Ch. Bühler, *Kindheit und Jugend*, 3rd edition, Jena: Fischer, 1931, p. 14 f.

acoustic signals, etc., and the release of sucking movements. On the other hand, the date of the appearance of these behavior patterns as well as their interpretation both seem to us to be subjects for discussion. First, here are two observations which will clarify the meaning of our remarks:

*Observation 25.*—I tried to determine with respect to Laurent when there began to be association between the position of the baby and the seeking of the breast. But it seemed to me impossible to affirm the existence of the association before the second month. At 0;0 (6) and the days following, Laurent, it is true, sought to nurse as soon as he was put on the scale, the dressing table, or his mother's bed, whereas previously he sought nothing and cried in his crib. At 0;0 (9) Laurent is half asleep in his crib; he sought nothing as long as he was being carried, but as soon as he was placed on the bed he opened his mouth and turned his head from side to side with more rapid arm movements and tension of the whole body. At 0;0 (10) he no longer seeks while in his crib but as soon as he is in the nurse's arms, etc. This was his behavior until the end of the first month. But is it a matter of pure coincidence or of an actual association betweeen position and sucking? It is impossible for us to decide this question, because the facts can be interpreted quite independently from the existence of an associative transfer. It is sufficient to state, as we have done in Chapter I, how precocious sucking-like movements and the groping characteristic of the reflex are, to understand that the child will try to nurse as soon as he is neither crying, nor asleep, nor distracted by movement. In his crib he does not seek because nothing distracts him from his cries of hunger, and these cries engender others through this sort of reflex repetition of which we have already spoken; so long as he is carried he seeks nothing because the rocking motion absorbs him; but as soon as he is placed on the scale, on the dressing table where his diapers are changed, or in his nurse's or his mother's arms, he tries to suck before recommencing to cry because neither his weeping nor the excitements of motion prevent him any longer from sucking. Does this mean there is a connection between *Trinklage* and sucking? Nothing authorizes us to deny it, or to affirm it either as yet. Besides, when one knows the difficulty of establishing a conditioned reflex in animals and especially the necessity to "strengthen" it all the time in order to preserve it, one can only be prudent in invoking such a mechanism in so far as the behavior patterns of the first weeks are concerned.[3]

[3] We do not mean to deny that certain conditioned reflexes may be constituted at birth, as D. P. Marquis succeeded in proving this with babies from 3 to 10 days old by associating certain sounds with sucking reflexes

On the other hand, from the moment Laurent knows how to find his thumb (beginning of the second month), the seeking of the breast may be differentiated from the other tendencies and one thus succeeds in establishing a connection between the *Trinklage* and this seeking. Before the meal the child is only inclined to suck his fingers in the crib when he is not crying or is not too sleepy; but, as soon as he is in position to eat (in his mother's arms or on the bed, etc.) his hands lose interest, leave his mouth, and it becomes obvious that the child no longer seeks anything but the breast, that is to say, contact with food. At 0;1 (4) for instance, no experiment involving finger sucking was possible before the meal as Laurent turned his head from side to side as soon as he was in position to eat.

During the second month coördination between position and seeking the breast has made considerable progress. Thus at the end of the month Laurent only tries to nurse when he is in his mother's arms and no longer when on the dressing table.

*Observation 26.*—In correlation with this progressive accommodation to the situation as a whole, it seemed to us that accommodation to the breast itself made some progress during the second month and went beyond the reflex accommodation of the first weeks. We noted in Jacqueline from 0;1 (14) and in Lucienne from 0;1 (27) the natural disposition to turn the head to the correct side when the breast was changed; whereas their body's rotation should have directed the head to the outside, they themselves turned it in the direction of the breast. Such behavior does not of course imply in any way correct orientation in space; it only indicates that henceforth the child knows how to utilize the contacts with his mother's arms as signals enabling him to mark the location of the food. Now if this is the case, there is obviously acquired association, that is to say, accommodation which transcends simple reflex accommodation.

From the second month we again find the correlations observed by Charlotte Bühler and her collaborators. But do these correlations between the situation as a whole and sucking necessarily presuppose the hypothesis of the "associative transfer" (*"Signalwirkung"*)?

That is a general problem to which we shall return in §5.

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(*Journ. of Genet. Psychol.*, XXXIX, 1931, p. 479) and W. S. Ray was even able to provoke conditioned reflexes in the fetus (*Child Development*, III, 1932, p. 175). We only claim that, granted the difficulties of the question of conditioning which increase daily, caution compels us whenever possible to have recourse to more satisfactory explanations than those which one sometimes hopes to draw from the existence of the conditioned reflex.

Let us limit ourselves to emphasizing from now on the fact that the association acquired between the signals characteristic of the *Trinklage* and the sucking reflex was not imposed on the child in a wholly mechanical way. There is not only passive recording. Through the constant seeking which characterizes the sucking instinct it is always in connection with the efforts and gropings of the subject himself that the association is acquired. Here again let us beware of too simple a comparison with the conditioned reflex. As we understand it, if association is established between *Trinklage* and sucking, it is not through mere training, otherwise one would not see why optic signals would not also give rise to training of the same kind from the second month. It is simply that the sucking schema—that is to say, the organized totality of the movements and attitudes peculiar to sucking—comprises certain postures which extend beyond the buccal sphere. Now these attitudes are not entirely passive and sooner or later involve the compliance of the whole body: the limbs become rigid, the hands clenched, etc., as soon as the nursling adopts the position characteristic of nursing. Thenceforth the simple recall of these attitudes sets in motion the whole cycle of the sucking act because the kinesthetic sensations and postural sensibility thus released are immediately assimilated to the schema of this act. Therefore association between an independent signal and a given sensorimotor schema does not exist, nor coördination between two groups of independent schemata (as will be the case between vision and sucking, etc.), but rather the constitution and progressive enlargement of a single schema of accommodation and assimilation combined. At most can it be said, in such a case, that accommodation prevails over assimilation.

Let us now come to the most complex acquisitions pertaining to sucking (the third of the stages of Hetzer and Ripin)—the associations between sucking and vision. From the third and fourth month, according to Hetzer and Ripin, the child may be observed getting ready to eat as soon as he sees the bottle or any object connected with food. In such a behavior pattern there is no longer simple, more or less passive, association between a

signal and the act but recognition of an external image and of meanings attributed to this image.

We have been able to make similar observations:

*Observation 27.*—Jacqueline, at 0;4 (27) and the days following, opens her mouth as soon as she is shown the bottle. She only began mixed feeding at 0;4 (12). At 0;7 (13) I note that she opens her mouth differently according to whether she is offered a bottle or a spoon.

Lucienne at 0;3 (12) stops crying when she sees her mother unfastening her dress for the meal.

Laurent too, between 0;3 (15) and 0;4 reacts to visual signals. When, after being dressed as usual just before the meal, he is put in my arms in position for nursing, he looks at me and then searches all around, looks at me again, etc.—but he does not attempt to nurse. When I place him in his mother's arms without his touching the breast, he looks at her and immediately opens his mouth wide, cries, moves about, in short reacts in a completely significant way. It is therefore sight and no longer only the position which henceforth is the signal.

Such behavior patterns are surely superior to those which are governed only by coördination between position and sucking. They imply, in effect, actual recognition of visual images and the attribution of a meaning to these images through reference to the sucking schema. Is this tantamount to saying that the bottle, etc., already constitute "objects" for the child, as Ch. Bühler maintains?[4] We would not dare to go so far.[5] Sensorial images can be recognized and endowed with meaning without by the same token acquiring the characteristics of the substantial and spatial permanence inherent in the object. But we recognize that such images are evidently perceived by the child as "external"; that is to say, they are projected in a coherent whole of images and relationships. In effect, through the very fact that for the nursling the bottle belongs to two series of schemata capable of giving rise to adaptations and functions independent of each other (vision and sucking) and through the fact that it realizes the coördination of these two schemata, it is necessarily endowed with a certain externality. On the other hand, thumb sucking does not realize this condition. Even though this sucking presupposes for the observer coördination between the move-

[4] *Op. cit.*, p. 18.

[5] We shall see why in the course of Volume II.

ments of the hand and those of the mouth, the thumb is at first only known by the child to the extent that it is sucked and there is no coördination between two independent schemata for the subject himself. We shall speak, therefore, in the case of the release of sucking through visual signals, of recognition as function of the coördination of two schemata of assimilation (sucking and vision).

In conclusion, the acquisitions which characterize the sucking mechanism past the stage of purely hereditary adaptations, are three in number. In the first place there is actual "circular reaction"—playing with the tongue, systematic thumb sucking, etc. This reaction constitutes an essentially active behavior pattern which prolongs the reflex use described in the first chapter but with, in addition, an acquired element of accommodation to the facts of experience. Passivity increases, on the other hand, in the accommodations which are constituted more or less automatically as a function of the external environment, but these accommodations, too, presuppose, at their point of departure, activity of the subject. Finally, the behavior is complicated by the coördination of heterogeneous schemata at the time of the recognition of the visual signals for sucking.

Without wanting to anticipate the theoretical conclusions which we shall try to draw from the facts in §5, it is possible at the beginning to ask ourselves what these three types of conduct represent from the point of view of the mechanisms of adaptation. Circular reaction is surely to be conceived as an active synthesis of assimilation and accommodation. It is assimilation to the extent that it constitutes a functional use prolonging the assimilation reflex described in the first chapter: to suck thumb or tongue is to assimilate these objects to the very activity of sucking. But circular reaction is accommodation to the extent that it realizes a new coördination, not given in the hereditary reflex mechanism. With regard to the so-called associative transfer, it is chiefly accommodation in so far as it presupposes associations suggested by the external environment. But it implies an element of assimilation of earlier circular reactions, to the extent that it proceeds by differentiation. Between its own accommodation and that of the circular reaction there is only a

difference of degree: the latter is more active and the former more passive. Finally, the coördination of the schemata in which the recognition of the visual signals for sucking consists, is only a complication of these same mechanisms: it is assimilation to the second degree inasmuch as it is coördination of two schemata of assimilation (vision and sucking) and it is accommodation to the second degree inasmuch as it prolongs the chain of acquired associations.

§2. VISION.—We are here not at all going to study perceptions and visual accommodations in themselves but only attempt, in accordance with the aim of this work, to distinguish in the behavior patterns pertaining to vision the different aspects applying to the development of intelligence. We shall resume consideration of the particulars of certain visual accommodations connected with the formation of the idea of space.

As with sucking, we shall distinguish in the behavior patterns controlled by vision a certain number of types proceeding from the pure reflex to the circular reaction and from there to the acquired coördinations between the visual schemata and those of other activities.

The reflexes should have been dealt with in the first chapter. But, as they are far from interesting inasmuch as the sucking reflexes, we can limit ourselves to mentioning them here as a memorandum. Perception of light exists from birth and consequently the reflexes which insure the adaptation of this perception (the pupillary and palpebral reflexes, both to light). All the rest (perception of forms, sizes, positions, distances, prominence, etc.) is acquired through the combination of reflex activity with higher activities. But the behavior patterns connected with the perception of light imply—as they do with sucking, but to a much lesser degree—a sort of reflex apprenticeship and actual searching. I noted, for example, from the end of the first week how much Laurent's expression changed when he was near luminous objects and how he sought them, as soon as they were moved, without of course being able to follow them with his glance. His head alone followed their movement for an instant, but without continuous coördination. Preyer (*op. cit.*, p. 3) notes during the first days the child's expression of satisfaction at soft

light; from the sixth day, his son turned his head toward the window when he was moved away from it. It seems as though such behavior patterns are explainable in the same way as the reflex behavior pertaining to sucking. Light is an excitant (consequently a functional aliment) for visual activity, whence a tendency to preserve perception of light (assimilation) and a groping to rediscover it when it vanishes (accommodation). But nothing acquired is doubtless yet superimposed on this reflex adaptation and, if it is already possible to speak of activity at this level, since there is searching, this activity does not necessarily imply apprenticeship as a function of the external environment.

On the other hand, toward the end of the first month the situation changes, as the result of progress in directing the glance. It is known that there is surface participation as early as the motor accommodation of the eye to the moving of objects. From the point of view of psychological accommodation, the stage thus surmounted during the fourth week is extremely significant. As Preyer says, the child begins "really to look, instead of contemplating vaguely" and the face assumes "a definitely intelligent expression" (*op. cit.*, p. 35) this is the time when the baby stops crying in order to look before him for long minutes in succession without even making sucking-like movements. Here are a few examples:

*Observation 28.*—Jacqueline at 0;0 (16) does not follow with her eyes the flame of a match 20 cm. away. Only her expression changes at the sight of it and then she moves her head as though to find the light again. She does not succeed despite the dim light in the room. At 0;0 (24), on the other hand, she follows the match perfectly under the same conditions. The subsequent days her eyes follow the movements of my hand, a moving handkerchief, etc. From this date she can remain awake without crying, gazing ahead.

*Observation 29.*—Lucienne also has directed her glance since the fourth week and is able to rediscover the object when it has escaped her sight and it follows its previous movement. She also finds the object by fits and starts, moving her eyes slightly, then losing sight of the object, then readjusting her head, then following the object with her eyes only, etc.

*Observation 30.*—Laurent, until 0;0 (21) has only been capable of badly coördinated movements of the head previously reported in con-

nection with the perception of light and simply revelatory of an attempt to make the excitement last. At 0;0 (21) on the other hand, for the first time his eyes follow a match in a dimly lit room, 20 cm. from his eyes.—At 0;0 (23) he is lying down, his head resting on his right cheek; I show him my fingers 20 cm. away and he follows them so that he turns all the way to the left.—At 0;0 (25) same experiment with a handkerchief: I make his head describe an angle of 180° moving backward and forward, so attentively does he follow the object.

*Observation 31.*—Laurent at 0;0 (24) watches the back of my hand which is motionless, with such attention and so marked protrusion of the lips that I expect him to suck it. But it is only visual interest.—At 0;0 (25) he spends nearly an hour in his cradle without crying, his eyes wide open. At 0;0 (30) same observation. He stares at a piece of fringe on his cradle with continuous little readaptive movements as though his head had difficulty in not changing position and his gaze brought it to the right place. So long as he gazes thus his arms are still, whereas when sucking-like movements are paramount, his arms swing to and fro again.—At 0;1 (6) Laurent stops crying when I put my handkerchief 10 cm. away from his eyes. He looks at it attentively, then follows it; but when he loses sight of it, he does not succeed in catching sight of it again.

*Observation 32.*—Laurent at 0;1 (7) begins to look at immobile objects with direction, naturally without much coördination. But for this it is essential that a previous movement excite his curiosity. He is, for example, lying in his bassinet, looking at a certain place in the hood. I pull down the hood to the other end of the bassinet so that instead of having over his head the usual material, he finds an empty space, limited by the edge of the hood. Laurent immediately looks at this, seeking from side to side. Thus he follows, roughly, the line of a white fringe which edges the hood and he finally fixes his gaze on a particularly visible point of this fringe. At 0;1 (8) same experiment and same result. But when he looks at the fringe he sees my motionless face (I stood there in order to observe his eyes). He then gazes alternately at the fringe and my head, directing his gaze himself, without having any external movement distracting his attention.

How can such behavior patterns be classified? There is not involved, it goes without saying, any interest of the child in the objects themselves that he tries to watch. These sensorial images have no meaning, being coördinated neither with sucking, grasping or anything which could constitute a need for the subject. Moreover, such images have neither depth nor prominence (the first accommodations to distance are exactly contemporaneous to

the beginnings of the directing of the glance). They therefore only constitute spots which appear, move, and disappear without solidity or volume. They are, in short, neither objects, independent images, nor even images charged with extrinsic meaning. What then is the motivating force of the child's behavior? There only remains the very need to look which can play this role. Just as, from the earliest days, the newborn child reacts to light and seeks it to the extent that the reflex use concomitant with this perception makes of the latter a need, so also, as soon as the glance is able to follow a moving spot, the use of the glance suffices to confer a functional value on objects which can be followed with the eyes. In other words, if the child looks at moving objects it is simply because, at the beginning, they constitute an aliment for the activity of the glance. Later, when the various accommodations to distance, prominence, etc., enrich visual perception, the objects looked at serve as more differentiated nourishment for these multiple operations. Still later, or concurrently, the visual images acquire meanings connected with hearing, grasping, touching, with all the sensorimotor and intellectual combinations. Thus they support increasingly subtle functions. The rough initial assimilation of the object to the very activity of the glance gradually becomes recognition and organization of images, projecting in space and, to sum it up, "objective" vision. But, before reaching this state of solidification the visual perception of the nursling is only a functional exercise. The object is, in the true sense, assimilated to the subject's activity. The perseverance and searching characteristic of the beginnings of looking are therefore of the same kind as the functional exercise of sucking activity, to take an example which has already been analyzed. At first purely reflex, this exercise is doubled by an acquired exercise or "circular reaction." At the second or third month level circular reaction seems to us definitely to exist. The direction of the look itself depends on the play of reflexes but these, being cortical, can from the beginning be extended into acquired reactions—that is to say, from the very beginning there is apprenticeship as a function of the objects themselves.

Having stated this, let us now try to analyze these circular

reactions. Circular reaction is, therefore, an acquired functional exercise which prolongs the reflex exercise and has the effect of fortifying and maintaining, no more only a completely assembled mechanism, but a sensorimotor whole with new results pursued for their own sake. Inasmuch as it is adaptation, circular reaction involves, according to the rule, a pole of accommodation and a pole of assimilation.

*Accommodation* is the whole of the associations acquired at the contact of objects due to the increasingly complex play of the "reflexes of accommodation": accommodation of the crystalline lens, pupillary reflex to distance, and binocular convergence. The instruments of this accommodation are certainly reflex and are contained in the hereditary structure of the eye itself. But the instruments only achieve effective utilization in the course of exercise in which experience is a factor. In other words, it is only in exerting himself to perceive forms, prominence, depth, in measuring distances, in seeing things in perspective, in short in making his accommodation reflexes function with respect to things that the child arrives at the correct handling of these instruments. It is useless to emphasize here the particulars of these mechanisms since we shall come across some of them again when dealing with space (Vol. II). Let us limit ourselves to one remark. It is an observed fact that the child at the stage under present consideration does not yet know how to measure distances. Not only are pupillary accommodation and binocular convergence not stabilized with regard to all distances at the age of 4–5 months, but the child makes all sorts of mistakes when he wishes to grasp objects (see Vol. II, Chapter II). Does this mean that the sense of depth is entirely due to acquired experience? Obviously no, because the existence of "accommodation reflexes'" shows that, even if the subject's first evaluations are erroneous, he is necessarily led, by means of his hereditary constitution, sooner or later to attribute depth to space. Is this to say henceforth that accommodation to depth is a pure reflex exercise comparable to the exercise by means of which the newborn child learns to suck —an apprenticeship presupposing the external environment because every function is relative to the environment but owing nothing to it because retaining nothing of the things themselves?

This could be maintained if space were independent of the objects it contains. But it is apparent that depth is nothing independent of concrete evaluations of the distances of objects. To say that a certain subject possesses the sense of depth necessarily means that he perceives a particular object as being farther away or nearer than another. But it is precisely in the acquisition of these particular perceptions that experience plays a role. For the baby to discover that the handle of his bassinet is farther removed in depth than the edge of the same bassinet, it is not enough that he possess the sense of depth by heredity, but he must put things in perspective, compare his perceptions, in short, make experiments. Therefore no accommodation reflex to depth in itself exists; there are only accommodations peculiar to the different objects perceived which presuppose, in addition to hereditary adaptation, acquired "circular reactions." It is in this respect that the functional exercise of looking, of which we are now speaking in general, involves an element of acquired accommodation and not only reflex use.

But the circular reaction proper to looking also presupposes an element of *assimilation*. First, as we have already said, there is essentially reproductive assimilation. If the child looks constantly, and more each day, at the objects surrounding him, this is not, at the beginning, because he is interested in them as objects nor as signals devoid of external meaning, nor even (at the very beginning) as sensorial images capable of being recognized, but simply because these moving, luminous spots are an aliment for his glance and permit it to develop while functioning. Objects are therefore first assimilated to the very activity of looking; their only interest lies in being objects of vision.

How shall we proceed from this purely functional assimilation (through pure repetition) to objective vision—that is to say, to an assimilation which presupposes the precise adaptation of the structure of the subject to the structure of things and vice versa? Three steps must be considered here: generalizing assimilation, recognitory assimilation, and the coördination of the schemata of visual assimilation with the other schemata of mental assimilation.

We can use the term "generalizing assimilation" (in the

same sense as in the first chapter dealing with the sucking system) to designate the fact—which is as important as it is trite —that from the fourth and fifth week the child looks at an increasing number of things but by proceeding through concentric waves. In the beginning, as revealed by the above observations, the nursling limits himself to watching objects which are slowly moved at a distance of 20–30 cm. from his face (Obs. 30) or to staring in front of him (Obs. 31). Then (Obs. 32) he applies himself to directing his glance to certain objects. From now on it becomes possible to make a general appraisal of the child's spontaneous visual interests. Then one observes that the subject looks neither at what is too familiar, because he is in a way surfeited with it, nor at what is too new because this does not correspond to anything in his schemata (for instance, objects too remote for there yet to be accommodation, too small or too large to be analyzed, etc.). In short, looking in general and the different types of visual accommodation in particular are put to use progressively in increasingly varied situations. It is in this sense that the assimilation of objects to visual activity is "generalizing."

Here are a few examples:

*Observation 33.*—Having learned to direct his glance (Obs. 32), Laurent explores his universe little by little. At 0;1 (9), for example, as soon as he is held vertically in the arms of his nurse, he examines successively the various images before him. First he sees me, then raises his eyes and looks at the walls of the room, then turns toward a dormer window, etc. At 0;1 (15) he systematically explores the hood of his bassinet which I shook slightly. He begins by the edge, then little by little looks backward at the lowest part of the roof, although this had been immobile for a while. Four days later he resumes this exploration in the opposite direction. He begins with the hood itself and then examines a piece of veiling which extends beyond the edge of the roof, a part of the coverlet (in the same position), my face which he finds before him and finally empty space. Subsequently he constantly resumes examining the cradle but, during the third month, he only looks at the toys hanging from the hood or at the hood itself when an unwonted movement excites his curiosity or when he discovers a particular new point (a pleat in the material, etc.).

*Observation 34.*—His examination of people is just as marked, especially after 0;1 (15); that is to say, after his first smiles. When one

leans over him, as when dressing him, he explores the face section by section: hair, eyes, nose, mouth, everything is food for his visual curiosity. At 0;1 (10) he alternately looks at his nurse and at me and, in examining me, his eyes oscillate between my hair and face. At 0;1 (21) he watches his nurse enter and leave the room. At 0;1 (25) he looks in turn at his nurse, his mother, and myself with a change of attitude when confronted by each new face and a abrupt and spontaneous moving of his glance from one face to the other.

But, quickly enough, his interest in faces is no longer a purely visual one. Through coördination with the schemata of hearing in particular and with the global situations of eating, dressing, etc., the familiar faces become fraught with meaning. Thus we leave the realm of purely generalizing assimilation. This reappears, on the other hand, as soon as an unfamiliar feature appears, to alter his visual image of people. Thus at 0;2 (4) Laurent notices his mother wearing a pearl necklace in which he is more interested than in her face. At 0;2 (13) my béret catches his attention. At 0;2 (18) the shaving soap on my chin, then my pipe; the following days it is my tongue which I stick out at him having in mind experiments concerning imitation, etc. At 0;2 (29) he watches me eat most attentively. He successively examines the bread I hold and my face, then my glass and my face. He watches my hand when I raise it to my mouth, my mouth, etc.

*Observation 35.*—There is generalizing assimilation, not only with respect to the successive objects which the child sees, but also in connection with the successive positions which the subject assumes in order to look. The acquisition of the "alternate" glance may be cited in this connection. During the second month we have seen Laurent look in turn at various objects or different parts of the same object, as example (Obs. 34) three motionless people next to his bassinet or the hair and face of the same person. But in this case he looks at each image irregularly. On the other hand, during the third month, the emergence of the following behavior pattern may be observed: the glance compares, so to speak, two distinct objects while alternately examining them. For example, at 0;2 (11) Laurent is looking at a rattle suspended from the hood of his bassinet when I hang a handkerchief parallel to the rattle. He then looks alternately at the handkerchief and at the rattle and smiles. At 0;2 (17) he explores a part of the hood of his bassinet when I shake it slightly. Laurent then looks at a certain place in the hood, then observes the moving rattle, then returns to the hood and so on, six times in succession. I repeat the experiment a moment later and count nine more alternate glances.[6] Such behavior surely constitutes the beginning of comparison, but as yet, it seems to us, purely

[6] See also Observations 92, at 0;3 (13), the example of the case and the chain.

visual comparison. It is inconceivable that Laurent should already give a causal interpretation to the relationship he observes between the movement of the hood and that of the rattle; he simply compares two things seen.

*Observation 36.*—Here is another example of generalization due to the subject's position. At 0;2 (21), in the morning, Laurent spontaneously bends his head backward and surveys the end of his bassinet from this position. Then he smiles, returns to his normal position and then begins again. I observed this several times. As soon as Laurent awakens after the short naps to which he is accustomed, he resumes this activity. At four o'clock in the afternoon after a long sleep he has barely awakened before he bends his head backward and bursts out laughing. Such behavior reveals all the characteristics of a typical circular reaction. The days following he continues to explore and the next week his interest is almost as keen.

Thus it may be seen how the child's spontaneous looking develops through being exercised. The bassinet hood, having at first only been the object of "looking for the sake of looking," arouses growing interest through its particularities as well as through its successive modifications (the objects hanging from it). Interest in certain faces adduces interest in all others and in everything which complicates the original appearance of the former. New perspectives due to positions fortuitously discovered, arouse immediate interest through comparison with habitual perspectives, etc. In short, practice of looking brings with it the generalization of its activity.

But this growing generalization of the schema of sight is accompanied by a complementary differentiation of the global schema in particular schemata, this differentiation leading to "recognition." The purely functional assimilation which prevailed in the beginning (looking for the sake of looking) is thus transformed into an assimilation of objects to limited schemata which is tantamount to saying that sight is on the way to objectification (looking in order to see). For example, among the things which the child contemplates all the time are some which are immobile (the hood of the bassinet), some which sometimes move slightly (the fringe of the hood), some which constantly change position, appear and disappear, remain stationary for a while and suddenly disappear (human faces). Each of these cate-

gories of visual images gives rise to progressive exercises (generalization) but, at the same time, to differentiations in functioning. Each one presupposes, in effect, an exercise *sui genesis* of vision, just as the breast, the thumb, the pillow, etc., actuate sucking in different ways: so generalizing assimilation brings with it the formation of particular schemata. The child, in assimilating to these schemata the objects which appear in his field of vision, "recognizes" them through this very act. This recognition is therefore probably global in the beginning. The child does not recognize a certain face as such, but at first recognizes this face in a given situation. Only, the more generalizing assimilation permits the subject to incorporate the visual environment into his schemata, the more the latter dissociate themselves and permit precise recognition.

But if purely functional and generalizing assimilation can be observed thanks to the mere behavior of the child, how can what we have just said about recognitory assimilation be verified? From the time when the nursling is able to smile and thus to differentiate his gestures and the expression of his emotions, the analysis of recognition becomes possible without too great a risk of error. Let us try, from this point of view, to analyze the first smiles produced in the presence of visual images and to collect what they can teach us about the beginnings of recognition.

The smile is, as we know, a reflex mechanism whose association with pleasurable states makes it possible sooner or later to make a social sign assuming varied meanings but always related to contact with people. Must it be said, therefore, that the smile is a hereditary social behavior pattern which from the beginning constitutes, as Ch. Bühler maintains, a "reaction to people" or is it possible to think that the smile only becomes specialized progressively in its functions as a social sign and consists during the first months of a simple pleasurable reaction to the most varied excitants, even though it begins in the presence of the voice or movements of the human face? Ours is the second interpretation which is why the smile seems to us a good indication of the existence of recognition in general. Ch. Bühler's interpretation does not seem to us to withstand factual examination as

has been already elucidated by C. W. Valentine.[7] In a somewhat categorical note[8] Ch. Bühler has answered him by presenting statistics which contradict his observations. But an acute observation, especially when made by as good an observer as C. W. Valentine, surpasses all statistics. As for us, examining our three children has left us no doubt concerning the fact that the smile is primarily a reaction to familiar images, to what has already been seen, inasmuch as familiar objects reappear suddenly and release emotion, or again inasmuch as a certain spectacle gives rise to immediate repetition. It is only very gradually that people monopolize the smile precisely in so far as they constitute familiar objects most inclined to this kind of reappearances and repetitions. But in the beginning anything at all can give rise to the emotional recognition which elicits the smile.

*Observation 37.*—Laurent smiled for the first time at 0;1 (15) at 6 o'clock, 10 o'clock and 11:30 while looking at his nurse who is wagging her head and singing. Apparently there is a global impression involving visual recognition, perception of a rhythmic movement, and hearing. The following days the voice remains necessary to produce the smile but at 0;1 (25) merely seeing the nurse suffices. Same observation at 0;1 (30). On the other hand, it is not until 0;2 (2) that he smiles at his parents when they do not make noises. At 0;2 (3) he refuses to smile at his grandmother and an aunt despite all their advances, but he finally smiles at the latter when she removes her hat. At 0;2 (4) he smiles a lot at his mother (while she remains silent) but a few moments later refuses to smile at a woman of the same age. During this third month I do not succeed in making him smile only on seeing me if I remain immobile (without head movements) or if I appear at a distance (of 1 meter or more). On the other hand, during the fourth month these conditions are no longer inhibiting. At 0;2 (26) Laurent does not recognize me in the morning before I am groomed. He looks at me with a frightened expression and drooping mouth, then he suddenly rediscovers me and smiles. Seeing his systers does not cause him to smile as quickly as seeing his parents, but the reaction became identical after the middle of the third month. During the fourth month he even seems already to prefer children to adults when his acquaintance with both is equally slight. Thus, at 0;3 (7) Laurent is afraid of a neighbor but reveals great interest, with smiling eyes, in the man's 12-year-old son (blond with a very youthful appearance comparable to Laurent's sisters).

[7] C. W. Valentine, British Assoc. 1930, *The Foundations of Child Psychology.*

[8] Ch. Bühler, *op. cit.*

*Observation 38.*—With regard to inanimate objects, from the beginning of the third month Laurent revealed great interest in the cloth and celluloid toys hanging from the hood of his bassinet. At 0;2 (5) he looks at them as yet unsmilingly but emitting periodically the sound *aa* with an expression of enchantment. At 0;2 (11) he smiles broadly when he sees his toys move. He has not seen or heard anyone either previously or when confronted by this spectacle, for I move the toys from a distance with a stick. Besides, the toys have no human appearance; they are little balls of wool or celluloid. The sound of the toys which could have played a role in this first smile, does not do so subsequently; five times in succession on the same day Laurent smiles at these motionless toys. The evening of the same day I hung a handkerchief next to the toys. Laurent compares them (see above, Obs. 35) then smiles (he has not seen or heard me). The following days the reaction is just as definite and frequent. At 0;2 (15) I notice seven smiles at things (the motionless toys and hood of the bassinet, the movements of the bassinet when it is carried without the person making noise or showing himself to Laurent, etc.), and three smiles at people (his mother). At 0;2 (18) he smiles five times in succession while looking at the mosquito net (I observe this through the bassinet hood). The same day he laughs and babbles with great excitement while watching the toy. As soon as he is naked he laughs loudly, gesticulating and looking at the objects surrounding him including the brown wall of the balcony. At 0;2 (19) he did not smile at people a single time in a whole day; on the other hand, he smiled at all the familiar objects. In particular he smiles for the first time (five times during the day) at his left hand which he looks at since about fifteen days before (see Obs. 62). At 0;2 (21) he even smiles beforehand while drawing his hand toward his face. The same day he learns to look backward (as seen in Obs. 36) and almost infallibly smiles at this new perspective. From 0;2 (25) he smiles during his experiments with grasping; in shaking a toy, etc. At 0;3 (6 and 7) for example, he manifests a certain astonishment and even anxiety in the presence of new objects which he would like to grasp (shiny paper, tinfoil, medical tubes, etc.) but smiles (or smiles only with his eyes) while taking familiar objects (cloth and celluloid toys, package of tobacco, etc.).

*Observation 39.*—Lucienne likewise expresses with smiles certain definite recognitions of things and people. She too begins by smiling at a person—at 0;1 (24)—as the result of head movements and sounds. Then she smiles at her mother merely at the sight of her, at 0;1 (27) before smiling at her father. Then from 0;2 (2) she smiles at familiar objects attached to the bassinet or its hood. At 0;2 (13) for example, she smiles at the hood. She looks attentively at a particular place, then smiles while wriggling all over, then returns to this place, etc. At 0;2 (19) the ribbon which always hangs from this hood arouses her hi-

larity; she looks at it, laughs while twisting herself about, looks at it again, etc. At 0;2 (27) same reactions with, in addition, broad smiles at the toys which are swinging. At 0;3 (0) smile at the hood which is being replaced in position (without Lucienne's seeing or hearing the person).

Thus may be seen the extent to which smiles evidence subtle differences in recognition. The reactions differ with respect to different people and to the same person, to different situations (according to distances, movements, etc.). If, then, primitive recognition is "global"—that is to say, related to varied situations and to different types of looking becoming differentiated as a function of generalizing assimilation and of accommodation combined—nevertheless this recognition becomes more and more precise. The reaction is exactly the same with regard to things.

In conclusion, visual circular reaction or acquired adaptation in the realm of looking requires a component of accommodation of the function to the object and a component of assimilation of the object to the function. Such assimilation, at first simply functional and reproductive (repetition or pure circular reaction), becomes simultaneously generalizing and recognitory. It is when it attains a certain level of recognition that visual perception may be considered as perception of images distinct from one another and no longer only as a simple exercise of which the sensorial image constitutes the aliment without exciting interest in itself.

But the process is far from adequate to explain the growing objectification of visual adaptation. It is not enough that a sensorial image be recognized when it reappears for it to constitute by itself an external object. Any subjective state can be recognized without being attributed to the action of objects independent of the ego. The newborn child who nurses recognizes the nipple by the combination of sucking and swallowing reflexes without making the nipple a thing. So also a month-old child can recognize certain visual images without, however, really exteriorizing them. What is the next condition necessary for the solidification of such images? It seems to us essential that the visual schemata be coördinated with other schemata of assimilation such as those of prehension, hearing, or sucking. They

must, in other words, be organized in a universe. It is their insertion in a totality which is to confer upon them an incipient objectivity.

This leads us to the third aspect of visual circular reactions —their *organization*. It may be stated that the visual images to which the child adapts himself are, through the very fact of this adaptation, coördinated among themselves and also in relation to other kinds of schemata. The organization of visual images among themselves can itself give rise to a distinction. First there are the coördinations of distance, size, etc., which constitute visual space and which we shall not mention here because the matter deserves special study (see Vol. II). Then there are the wholly qualitative coördinations (relationships of color, light, etc., and the sensorimotor relationships), whose activity is made manifest in generalizing and recognitory assimilation. Thus it may be said that, independently of any coördination between vision and the other schemata (prehension, touch, etc.), the visual schemata are organized among themselves and constitute more or less well-coördinated totalities. But the essential thing for this immediate question is the coördination of the visual schemata, no longer among themselves, but with the other schemata. Observation shows that very early, perhaps from the very beginnings of orientation in looking, coördinations exist between vision and hearing (see Obs. 44–49). Subsequently the relationships between vision and sucking appear (see Obs. 27), then between vision and prehension, touch, kinesthetic impressions, etc. These intersensorial coördinations, this organization of heterogeneous schemata will give the visual images increasingly rich meanings and make visual assimilation no longer an end in itself but an instrument at the service of vaster assimilations. When the child seven or eight months old looks at unknown objects for the first time before swinging, rubbing, throwing and catching them, etc., he no longer tries to look for the sake of looking (pure visual assimilation in which the object is a simple aliment for looking), nor even to look for the sake of seeing (generalizing or recognitory visual assimilation in which the object is incorporated without adding anything to the already elaborated visual schemata), but he looks in order to act, that is to say, in order to assimilate the

new object to the schemata of weighing, friction, falling, etc. There is therefore no longer only organization inside the visual schemata but between those and all the others. It is this progressive organization which endows the visual images with their meanings and solidifies them in inserting them in a total universe.

From the point of view of the functional categories of thought which correspond to the biological invariants of mental development, it is interesting to note the extent to which this element of organization is, here as everywhere, the source of totalities and of values. In so far as the organization of the visual schemata forms a more or less closed totality, vision constitutes a value in itself and the assimilation of things is an assimilation to vision itself. On the other hand, in so far as the visual universe is coördinated to the other universes—that is to say, where there is reciprocal organization and adaptation between the visual and other schemata—visual assimilation becomes a simple means at the service of higher ends, and consequently a value derived in relation to principal values (the latter being constituted by the totalities pertaining to hearing, prehension and the activities proceeding from it). This is what we shall see in the following pages.

§3. PHONATION AND HEARING.—As is the case with sucking and vision, phonation and hearing give rise to acquired adaptations superimposing themselves on hereditary adaptations and, again in this case, the first acquired adaptations consist in circular reactions to the breast in which it is possible to distinguish the processes of accommodation, assimilation, and organization.

Phonation is evidenced at birth by the cry of the newborn child and his whining in the first weeks. That this reflex behavior might from the beginning be subject to some complications analogous to those we have noted in connection with vision and especially sucking, is not impossible if one considers these two observations, both unfortunately to be received with caution. The first is this sort of rhythm which appears very early in the child's cries. Laurent has hardly ever cried at night during the

first three weeks but almost every day between 4 and 6 P.M.; Lucienne cries mostly in the morning, etc. The second is the possibility of a contagious spreading of crying beginning the first week. When a baby cries in the room shared by the newborn babies in a clinic, several seem to copy him; furthermore it seemed to me that my voice (I said, "Aha, aha," etc.) made Laurent cry beginning 0;0 (4 and 5). But the rhythm in question may be due to an organic rhythm (particularly digestive) without any reflex involvement, and the supposed contagious spreading of crying may be due to coincidence or to the simple fact that the others' voices awaken the child and a newborn child cries almost immediately upon awaking. Let us therefore conclude nothing.

On the other hand, circular reaction is superimposed on reflex phonation as soon as, at one or two months, the little wail which precedes crying is kept up for its own sake and gradually gives rise to modulations. This is the point of departure for our analysis of phonation in so far as it is acquired adaptation.

With regard to hearing, an interest in sound may be observed in the first days of life. At the end of the second week, for instance, Laurent stopped crying for a moment in order to listen to a sound coming from near his pillow. But it cannot be called acquired adaptation until the second month from the time the heard sound provokes a somewhat prolonged interruption of the action in progress and an actual search.

If we study phonation and hearing simultaneously, we observe that, from the stage when circular reaction prolongs, in these two realms, hereditary adaptation, to the child hearing and voice are connected. Not only does the normal child regulate his own phonation primarily according to the acoustic effects he notices, but also the voices of others seem to react on his own voice. Is such a connection between hearing and phonation partly hereditary and consolidated by acquired adaptation, or is it only acquired? It is very difficult to decide. If from birth the cries were really imitative there would definitely exist a hereditary connection. But, as we have just seen, even if the fact of a contagious spreading of crying were established, it could be explained otherwise than by imitation. Let us therefore

not form hypotheses concerning the heredity of connections between phonation and hearing and limit ourselves to studying the behavior patterns related to these functions from the time when acquired adaptation exists.

First, here are some observations concerning phonation:

*Observation 40.*—Jacqueline, until the middle of the second month, has only used her voice for daily wails and certain more violent cries of desire and anger when hunger became persistent. Around 0;1 (14) it seems as though crying stops simply expressing hunger or physical discomfort (especially intestinal pains) to become slightly differentiated. The cries cease, for example, when the child is taken out of the crib and resume more vigorously when he is set down for a moment before the meal. Or again, real cries of rage may be observed if the feeding is interrupted. It seems evident, in these two examples, that crying is connected with behavior patterns of expectation and disappointment which imply acquired adaptation. This differentiation of mental states concomitant with phonation is soon accompanied by a differentiation in the sounds emitted by the child. Sometimes crying is imperious and enraged, sometimes plaintive and gentle. It is then that the first "circular reactions" related to phonation may definitely be observed. Sometimes, for instance, the wail which precedes or prolongs the crying is kept up for its own sake because it is an interesting sound: 0;1 (22). Sometimes the cry of rage ends in a sharp cry which distracts the child from his pain and is followed by a sort of short trill: 0;2 (2). The smile may be accompanied by indistinct sounds: 0;1 (26). Finally, the sounds thus produced as prolongations of crying or of smiles are immediately rediscovered and sustained as such: at 0;2 (12) Jacqueline prattles for a moment without smiling or wailing. At 0;2 (13) she emits a sort of trill. At 0;2 (15) the crying is transformed into playing with the voice, "aha," "ahi," etc. At 0;2 (15) she even interrupts her meal to resume her babbling. Finally, at 0;2 (18) playing with her voice becomes routine when she is awake.

It is to be noted, as we shall see concerning imitation, that these first circular reactions are almost immediately accompanied by vocal contagion and then, at 0;2, by definite imitation.

*Observation 41.*—Until 0;1 (8) I noticed nothing in Laurent resembling a vocal circular reaction. His phonation only consists of cries of hunger and pain or in wails preceding and prolonging the cries. True, at 0;0 (9) Laurent makes a sound similar to *aha,* without crying, but only once; usually this sound precedes crying. On the other hand, beginning 0;1 (8) vague voice exercises may be observed, but these could be the beginning of a wail interrupted by a visual or auditory interest. At 0;1 (9) on the other hand, the wailing is maintained for its own sake, for

several seconds before the crying. As soon as the first cry ensues I imitate Laurent's wailing; he then stops crying and begins to wail again. This first vocal imitation seems to me to substantiate the existence of circular reaction. If imitation of others exists, there also exists, in effect and *a fortiori,* imitation of oneself, that is to say "circular reaction." At 0;1 (15) I note a sort of fleeting *arr* or *rra,* and at 0;1 (20) a sound resembling *en* indicating contentment interspersed with sucking-like movements in which he indulges, alone and wide awake. The latter sound reappears intermittently at 0;1 (22) and at 0;1 (26) in the same situations, whereas the sound *aa* or *rra* which I emit in Laurent's presence in order to copy him releases analogous sounds, after a smile, at 0;1 (22). At 0;1 (28) circular reaction begins with the sounds *aha, enhen,* etc., and at the third month vocalizations are produced. At 0;2 (7) Laurent babbles in the twilight and at 0;2 (16) he does this on awakening early in the morning often for half an hour at a time.

*Observation 42.*—In certain special cases the tendency to repeat, by circular reactions, sounds discovered by pure chance, may be observed. Thus at 0;2 (12) Lucienne, after coughing, recommences several times for fun and smiles. Laurent puffs out his breath, producing an indefinite sound. At 0;2 (26) he reproduces the peals of his voice which ordinarily accompany his laughter, but without laughing and out of pure phonetic interest. At 0;2 (15) Lucienne uses her voice in similar circumstances, etc.

It is useless to continue this description since phonation does not interest us for its own sake but simply inasmuch as it is subject to adaptations of general form. In this respect it is easy to find in circular vocal reactions of which we have just spoken, the processes of accommodation, assimilation and organization to which sucking and vision have already accustomed us. Accommodation, first, because circular reaction is an effort to rediscover the new sound discovered by chance. There is thus perpetual accommodation of the vocal organs to phonic reality perceived by hearing (see for example Obs. 42) even though this reality is the product of their own activity. Very early too, vocal accommodation will consist in the imitation of new sounds made by others, but we can remit study of this question to the volume on "Imitation." The use of the voice is then assimilation in the triple sense of the term. There is assimilation through repetition, to the extent that each vocal system is consolidated while functioning. There is generalizing assimilation to the extent that circular reaction progressively diversifies the phonic material in

indefinite combinations which the authors have noted in detail. There is recognitory assimilation to the extent that circular reaction and beginning imitation entail the discrimination of one sound in relation to another. Finally phonation is organization in two complementary senses, first inasmuch as the totality of the sounds produced constitutes a system of interdependent articulations and then inasmuch as phonation is immediately coördinated with other schemata and in particular with the auditory schemata.

This leads us to hearing. The first acquired adaptations related to hearing date from the second month, from the time when two essential coördinations are established—coördination with phonation and coördination with vision. Until then the only reaction observed is the child's interest in the voice. But as this reaction is accompanied by no other visible accommodation except the smile and the coördinations of which we have just spoken, it is very difficult to fix the boundary of reflex adaptation and of acquired adaptation.

*Observation 43.*—At 0;1 (0) Jacqueline still limits herself to interrupting her crying when she hears an agreeable voice or sound but she does not try to mark the sound. At 0;1 (6 and 13), same reaction. At 0;1 (10) on the other hand, she begins to smile at the voice. From now on it is possible in a general way to distinguish the sounds which she recognizes and which make her smile (vocalizations, singing intonations, etc., resembling her own phonations) from those which astonish, worry, or interest her. The same is true of Lucienne, beginning 0;1 (13). The sound *rra* which is a copy of her own vocalizations almost invariably makes her smile for three or four weeks, beginning at 0;1 (25) and produces a vague imitation beginning 0;1 (26).—Laurent smiles at the voice alone beginning 0;1 (20), but at 0;0 (12) the voice sufficed to interrupt his crying and this interest in sound gave rise to attempts at localization from 0;1 (8). As a rule high-pitched sounds in a childish intonation make him smile; deep tones surprise and disturb him. The sound *bzz* is sure to make him smile during the third month (before he himself emits it) provided that it is sung on a sufficiently high key. At 0;1 (22) he easily recognizes the sound of the metal rattle in his celluloid balls and he immediately looks in the right place as soon as he hears them.

These facts suffice to make us state that the child behaves with respect to sounds as with respect to vision. On the one hand,

he progressively accommodates himself to them; on the other hand, he assimilates them. This assimilation is at first the simple pleasure of hearing (circular reaction to the sound or assimilation through repetition). Then, to the extent that there is discrimination of the sounds heard, there is simultaneously generalizing assimilation (that is to say, interest in increasingly varied sounds) and recognition of certain sounds (*rra, bzz,* etc.).

Let us proceed to the coördinations between sound and sight:

*Observation 44.*—At 0;2 (12) Jacqueline turns her head to the side whence the sound comes. For example, hearing a voice behind her, she turns in the right direction. At 0;2 (26) she localizes the source of the sound quite accurately. It seems she searches until she finds the person who is speaking but it is of course difficult to say whether she identifies the source of the sound and the original image or whether there is simply accommodation to the sound.

*Observation 45.*—At 0;1 (26) Lucienne, whose head is turned to the left when I call her from the right, turns her head at once and seeks by looking. At 0;1 (27) she is carried under my window whence I call her; she turns her head left and right and finally above her at an angle 45° too much to the left but revealing obvious control. In this last example it seems as though she tries to see what has produced the sound and not only to accommodate herself to the sound. At 0;2 (12) also, she turns her head when I call her and looks until she has seen me, even if I remain motionless.

*Observation 46.*—At 0;1 (8) Laurent reveals an incipient localization of sound. He lies on his back without seeing me and looks at the roof of the cradle while moving his mouth and arms. Then I call him softly, "Aha, aha." His expression immediately changes, he listens, motionless, and seems to try to locate the sound by looking. His head oscillates slightly from right to left without yet finding the right location and his glance, instead of remaining fixed as previously, also searches. The following days Laurent better directs his head toward the sound and of course he then looks in the right direction, but it is impossible to decide whether the child tries to see the source of the sound or whether his looking simply accompanies pure auditory accommodation.

*Observation 47.*—At 0;1 (15), on the other hand, it seems that on hearing my voice Laurent tries to see the face that goes with it but with two conditions which we shall try to specify. That morning Laurent smiled for the first time, three times, and, as we have seen, it is probable that the smile was started by a global impression, auditory as well as visual. That afternoon I stand at Laurent's left while he is lying in his cradle and looks to the right. I call, "Aha, aha." Laurent then slowly turns his

head to the left and suddenly sees me after I have stopped singing. He looks at me at length. Then I move to his right (without his being able to follow me with his eyes) and I call. Laurent again turns in my direction and his eyes seem to search. He sees me and looks at me but this time without an expression of understanding (I am immobile at this moment). I move back to the left, call, and he turns back again. As a counterproof I repeat the same experiment but I tap the window panes with my hand (the cradle is between the two leaves of a French window). Each time Laurent turns to the correct side and looks in the direction of my face which, however, he perceives in passing. It appears therefore that he associates the sound of the voice with the visual image of the human face and that he tries to see something else upon hearing a new sound.—But the rest of the observation shows that two conditions are still necessary for Laurent to look at a face when he has heard a voice: he must have seen this face shortly beforehand and it must be in motion. For example at 0;1 (20) I enter unobserved by Laurent and say. "Aha." He looks and searches most attentively (his arm movements stop completely) but limits himself to exploring the visual field exposed to him through his initial position (he examines the hood of the bassinet, the ceiling of the room, etc.). A moment later I appear in front of Laurent, then disappear and call him sometimes at the left sometimes at the right of the bassinet. Henceforth he searches in the right direction every time. The next day, same experiment and same result; furthermore I note that if I remain immobile he looks at me without interest and without even recognizing me, whereas if I move he looks at me and his searching ends as though he knew it was I who sang. At 0;1 (22) in the same way he searches anywhere at all although manifesting much attention to my voice; then he perceives me while I am immobile and continues searching without attributing importance to my visual image; after this I shake my head and thereafter he turns toward me whenever I call and seems satisfied as soon as he has discovered me. The following days, same phenomenon.

*Observation 48.*—From 0;1 (26) on the other hand, Laurent turns in the right direction as soon as he hears my voice (even if he has not seen me just before) and seems satisfied when he has discovered my face even when it is immobile. At 0;1 (27) he looks successively at his father, his mother, and again at his father after hearing my voice. It therefore seems that he ascribes this voice to a visually familiar face. At 0;2 (14) he observes Jacqueline at 1.90 to 2 meters, at the sound of her voice; same observation at 0;2 (21). At 0;3 (1) I squat before him while he is in his mother's arms and I make the sound *bzz* (which he likes). He looks to his left, then to his right, then ahead, then below him; then he catches sight of my hair and lowers his eyes until he sees my motionless face. Finally he smiles. This last observation may be con-

sidered as definitely indicating identification of the voice and visual image of the person.

*Observation 49.*—Regarding the noises made by things, it seems as though Laurent acquired his auditory-visual coördination around the same time as that relating to persons. At 0;1 (22) for example, he turns immediately in the direction of a celluloid ball in which there is a rattle. True, it is moving, but at 0;1 (26) he finds it again when it is immobile.—At 0;2 (6) he looks at an electric kettle as soon as I produce a sound by means of its lid. At 0;2 (11) he is sucking his thumb while looking to the right when I shake a celluloid rattle which has been attached to the hood of his bassinet for several days only (two weeks at most). He immediately lets go his thumb to look up at the right place, thus showing that he knows where the sound came from. That evening, same reaction, very rapid even though he was half asleep after a long nap. The next day and the following days: same phenomena. At 0;2 (14) Laurent observes, one meter away, my pipe which I knock lightly on wood; he stops looking at the place of contact when the sound stops and immediately finds it again when I resume. Same reactions at 0;2 (15), with a cane (at 1.50 to 2 meters), then he rediscovers the cane in various places when I change the point of contact.

It is therefore permissible to regard as certain the existence of coördination between sight and hearing from the third month on, whereas the facts observed during the second month can be due to a simple accommodation of the head to the direction of the sound. These ideas coincide with the results obtained by B. Löwenfeld.[9]

This coördination between sound and vision poses an interesting question. The coördinations which we have hitherto encountered oscillate between two extreme types. On the one hand, there is the more or less passive association imposed by the environment; thus the special position at mealtime is accompanied in the 1- or 2-month-old nursling by a search for the breast. True, such associations have seemed to us only capable of being constituted through accommodations and searchings, indicating a certain activity. But, granted this element of active accommodation, it must be recognized that it is reduced to its simplest expression and that the environment imposes the content of these accommodations before the child really assimilates

9 Berthold Löwenfeld, Systematisches Studium der Reaktionen der Säuglinge auf Klange und Gerausche, *Zeitschr. f. Psychol.* CIV, 1927, pp. 62–96.

them in detail (through recognition, etc.). At the other extreme we have the active recognition of a sign charged with meaning. It is thus that the 3- to 4-month-old nursling recognizes his bib by visual perception and knows that it announces the coming meal. With regard to the coördination between hearing and sight, we are now confronted by behavior patterns contemporaneous with coördinations of position and of sucking (first type), but behavior patterns which resemble the later coördinations of vision and of sucking (second type). How should they be interpreted? Must we state that the sound of the voice is a simple signal forcing the baby to search with his eyes for the face corresponding to this voice in the manner in which the sound of the bell sets in motion salivation in the dog by conditioned reflex, or must we think that the sound of the voice constitutes a sign charged with meaning and is recognized by the child as going with the visual perception of someone's face? If, in the coördinations of position and sucking, we admit the existence of an element of active accommodation, however small, then it is evident that a series of intermediaries will link the two extreme types (active and passive coördination) and that the coördination between sight and hearing will be located midway between these extremes. In other words, the association between a sound and a visual perception is never a purely passive association, but it is not at the outset a relationship of understanding (recognition of meanings). How then can this intermediary state and the progress of understanding be explained?

In view of all we have seen regarding assimilation we may hypothesize that every assimilatory schema tends to conquer the whole universe including the realms assimilable by means of other schemata. Only the resistances of the environment or the incompatibilities due to the conditions of the subject's activity curb this generalization. So it is that the child sucks everything that touches his mouth or face and learns to coördinate the movements of his hands with those of sucking as a function of his pleasure in sucking his thumb. When he will know how to grasp he will suck everything he will have in his hands. Concerning what he sees or hears, if the nursling does not try to suck this from the outset it is perhaps less because these realms have no

connection with sucking (it often happens that he makes sucking-like movements as soon as he hears a sound) than because it is difficult for the child to do two things at once (looking attentively and making sucking-like movements, etc.). But instead of immediate coördination between sucking and sight it is possible that there exists nevertheless excitation of the sucking cycle in the presence of especially interesting visual images. The remarkable protrusion of the lips observed in the youngest children (see Obs. 31) in states of great attention could not be other than sucking-like movements if it cannot be explained by a purely automatic or tonic postural mechanism.[10] In the same way, with regard to the visual, hearing and grasping schemata, etc., the child will try little by little to see everything, hear everything, take everything, etc. This is well put by Ch. Bühler when she says with regard to the first sensorial reactions that the response to an excitant during the first months depends more on the subject's functional needs than on the nature of this excitant.[11] Consequently it is natural that the nursling should try in the course of his first auditory adaptations to look at the same time as listen, at least from the time when he has learned to direct the movement of his eyes [at 0;1 (7) in Laurent's case (see Obs. 32)]. This beginning of coördination between hearing and sight does not necessarily presuppose a passive association but can be explained by active assimilation. It is true that, when he turns his head to accommodate himself to the sound, the child comes automatically, in the case of the human voice, to perceive an interesting visual image (the corresponding face); the element of passive association is not to be entirely excluded. But simple associations would never give rise to actual searching in the coördination between sight and hearing (looking for the face which corresponds to the voice and later for the sounds which correspond to the objects seen) if the schemata of visual and audi-

10 Preyer, *op. cit.*, p. 251–252, construes this protrusion of the lips as being a hereditary association between sucking and sight (his son evidenced it the tenth day while looking at a candle). But it goes without saying that, if association exists, it can be explained by reflex assimilation without recourse to heredity.

11 *Op. cit.*, p. 26.

tory assimilation did not succeed in reciprocally directing their respective realms by assimilating them in an active way.

More precisely, if at a given moment the child applies himself to searching systematically for the visual images which correspond to the sounds heard, this is so in the first place because he forces himself to look at everything. Without yet knowing that a sound necessarily comes from a visible object, the child is visually excited by the sound as well as through hearing it. Thus in Observation 46 the sound *aha* releases a need in Laurent to look as well as to listen; and this is true, probably, not because Laurent already knows that this sound emanates from a precise visual image but simply because the excitant arouses all his needs at once; in other words, because the child tries to integrate the new reality into all the available schemata of assimilation. In the second place, the child turns his head in the direction of the sound through an accommodation to the sound comparable to the movements of the eye following an object. It is self-evident, consequently, that the glance is directed to the same side as the head, whence the observer's impression that the baby tries to see what he hears (see end of Obs. 46), whereas he undoubtedly only tries to see at the same time he hears. In the third place, in certain cases success confirms the searching. The sound of the voice of others in this respect constitutes a privileged example; such a sound nearly always gives rise to double assimilation, auditory and visual. In other words, the human face has the almost unique property, in the universe of the child of 1 to 2 months, of lending itself to a totality of simultaneous assimilation. This face is at the same time recognizable and mobile, thus exciting visual interests to the highest degree; it is this face that the baby contemplates or rediscovers when he fixes his attention on the sound of the voice; again it is this face which is central in the most interesting moments of existence (coming out of the bassinet, dressing, meal, etc.). In the case of the appearance of other people it is possible to speak, not of association between various assimilations, but of global assimilation, and it is apparently this fact which explains why the smile occurs more frequently in the presence of persons than with respect to things. As far as coördination between hearing and sight is concerned it is thus

evident that the child early identifies someone's face inasmuch as it is a visual image of this same face, inasmuch as it is a sonorous image. It is self-evident that, to the child, another person is not yet an object conceived as cause of the voice. But it cannot be said, inversely, that sound and vision are simply associated. This is why it must be asserted that visual and auditory schemata are reciprocally assimilated. The child tries, in a sense, to listen to the face and to look at the voice. It is this reciprocal assimilation that constitutes the identification of visual and sonorous images prior to the more complex solidifications which are to give rise to the object and to the causality.[12] In other words, the human face is one entity with regard to looking, listening, etc., and once he has acquired, in this case and some other privileged examples (rattles, etc.), coördination between hearing and sight, the child will search systematically and everywhere for correlations between sounds and visual images.

Let us finally proceed to the coördination between hearing and phonation. This coördination seems much simpler since every phonation is accompanied from the outset by auditory perception and is controlled by it. It seems therefore that here there is not intersensorial coördination, but pure circular reaction; a series of movements culminating in a sensorial effect and maintained by the interest of this result. But if that is true of simple phonation, the inverse process may also be observed: the action of hearing on phonation. In effect, as we have seen (Obs. 41) vocal contagion is almost as precocious as the first circular reactions which are the basis of phonation; the wailing of another person maintains that of the child, etc. What does this mean, if not that the schemata of phonation and hearing are reciprocally assimilated and in the same way as those of hearing and sight? Just as the child comes to listen to the sound of his voice instead of merely crying and thus inaugurates acquired circular reactions, so also he listens to the voice of another and, inasmuch as the sounds heard are analogous to the sounds he himself makes, he can only perceive them by means of corresponding auditory-vocal schemata. The imitation of sounds, in the beginning, is thus only

12 This explains why attributing the voice to a face is only achieved by relatively long stages; see Obs. 47 and 48.

the confusion of one's own voice with that of another, coming from the fact that the voice of others is actively perceived, that is to say, assimilated to the schemata of phonation.

In conclusion, analysis of the schemata of phonation, of hearing and of their coördination completely confirms what we have stated with regard to sucking and vision. Each of these adaptations brings with it a measure of accommodation to the external environment—accommodation to the direction of sounds, to their gradual variety, etc. But each one also involves an element of assimilation. First it is assimilation by pure repetition—listening for the sake of listening, crying or wailing in order to hear these sounds, etc. Then it is generalizing assimilation—listening to or producing increasingly varied sounds. Finally it is recognitory assimilation—rediscovering a definite sound. These perceived or produced sounds at first only present an internal organization. Related to each other they only have meaning in relation to the system they form; it is this system that the child maintains and uses, to which he assimilates the various heard sounds and which he accommodates as much as possible to the new heard sounds. Then this internal organization is itself inserted into a wider organization which gives it new meanings; sound is coördinated with vision, etc. But this coördination involves no new process; it is constituted by reciprocal assimilation of the visual and auditory schemata, etc.

If the latter process is difficult to study at so early an age as 1 to 2 months, analyzing prehension will now afford us the opportunity to extend the description of the mechanism of the coördinations among heterogeneous schemata.

§4. PREHENSION.—With the mouth, the eye, and the ear, the hand is one of the most essential instruments of which the intelligence, once it has been established, will make use. One can even say that the definitive conquest of the mechanisms of grasping marks the beginning of the complex behavior patterns which we shall call "assimilations through secondary schemata" and which characterize the first forms of deliberate action. It is therefore important to analyze fundamentally the way in which this discovery of grasping takes place: here, more even than with the

preceding schemata, we find an indispensable connecting link between organic adaptation and intellectual adaptation.

The hand's chief activity is grasping. But it is self-evident that this function cannot entirely be dissociated from touching or the coördinations between kinesthesia and sight, etc. We shall therefore touch on these questions in passing, but only in passing. The aim of this work is not to supply an inventory of behavior patterns of the first year of life and we shall only dwell on examples of use to the analysis of intelligence.

It seems to us five stages may be discerned in the development of grasping. If, as revealed by studying our three children, these stages do not correspond to definite ages, their sequence nevertheless seems necessary (except perhaps with regard to the third stage). Let us therefore examine the facts by classifying them according to the way they succeed one another.

The *first stage* is that of *impulsive movements* and of *pure reflex*. The newborn child closes his hand when the palm is lightly touched. Lucienne, a few hours after birth, closed her fingers around my index finger without resistance of the thumb. But at first it seems as though this reflex were unaccompanied by any search or appreciable use: the child immediately relinquishes that which he grasped. It is only while nursing when his hands are tightened and almost clenched, before the general relaxation of tonicity, that the child is able to hold on to a solid object for a few minutes (pencil, etc.). But it would be rash to conclude that this is due to a pure automatism and thus to contrast the grasping reflexes to those of sucking whose use we have seen presupposed to a great extent active accommodation and assimilation. In effect when the child closes his hand around the object which touched his palm, he reveals a certain interest. Laurent, at 0;0 (12) stops crying when I put my finger in his hand and recommences shortly afterward. The grasping reflex is thus comparable to sight or hearing during the first two weeks and not at all comparable to reflexes such as sneezing, yawning, etc., which do not attract the subject's attention in any way. True, things remain thus for a long time and prehension does not from the outset lend itself to systematic use as does sucking But we may ask ourselves whether the impulsive movements of arms, hands,

and fingers, which are almost continuous during the first weeks (waving the arms, slowly opening and closing the hands, moving the fingers, etc.) do not constitute a sort of functional use of these reflexes.

The *second stage* is that of *the first circular reactions related to hand movements, prior to any actual coördination between prehension and sucking or vision.* We shall group here the whole of the circular reactions leading to prehension for its own sake (grasping and holding objects without seeing them or attempting to carry them to the mouth), tactile and kinesthetic reactions (scratching a body, moving the fingers, hands or arms, etc.), the coördinations between sucking and hand movements (finger sucking, etc.) and finally the coördination between sucking and actual prehension (grasping an object in order to carry it to the mouth), the coördination which characterizes the third stage and realizes notable progress in the way of systematic prehension and the coördinations between sight and grasping (grasping in order to look, grasping objects perceived in the visual field), which will be formed during the fourth and fifth stages and indicate definitive success in grasping.

Thus defined, the first circular reactions related to hand movements and to prehension begin by autonomous activities of hands or fingers which prolong in a continuous way the impulsive movements and reflexes of the first stage. We have stated, in effect, that from birth certain impulsive movements seem to constitute an empty use of the grasping mechanism. From the second month it becomes evident that some of these movements are so systematized that they give rise to true circular reactions, capable of gradual accommodation and assimilation.

*Observation 50.*—At 0;1 (8) Laurent's arm is stretched out and almost immobile while his hand opens, half closes and then opens again, etc. When the palm of his hand strikes the covers, he grasps them, lets them go in unceasing oscillating motion. It is difficult to describe these vague movements, but it is also difficult not to see in them grasping for the sake of grasping, or even empty grasping analogous to the phenomena described in connection with sucking, vision, etc. But there does not yet exist, in such behavior patterns, either true accommodation to the object or even any continuity.

*Observation 51.*—Until 0;1 (19) I did not observe in Laurent any accommodation, even momentary, of the hand to the object outside of reflex accommodation. It seems, on the contrary, that today the contact of my hand with his little finger or of a handkerchief with the outer surface of his finger sets in motion a certain searching. True, his hand does not remain on the spot as it will do later on. There are attempts, back and forth, and each time his hand touches my fingers or the handkerchief and seems more ready to grasp (the palm seems to be directed toward the object). But it is self-evident that the interpretation of such movements remains a very delicate one. At 0;1 (20) also, contact of his closed left hand with a rolled up handkerchief which I hold produces this result: the hand moves away while opening, then returns, open, to strike the object, grasps it feebly, then moves away again, returns to grasp it, etc. The hand seems to be stimulated by contact with the object, a beginning of accommodation. But the hand comes and goes instead of remaining immobile and really searching.

*Observation 52.*—Beginning at 0;1 (22), on the other hand, there seems to be more continuity in the grasping movements. Thus at 0;1 (22) Laurent holds in his hand four and a half minutes an unfolded handkerchief which he grasped by chance (his arm is occasionally immobile and occasionally in slow movement). At 0;1 (23) he holds about two minutes a toy which I placed on his palm. When he half lets it go he grasps it again by himself (twice). But soon complete lack of interest ensues. Same observation at 0;1 (26) and 0;1 (29). At 0;1 (25) he opens his hand and grasps my index finger when I touch the back of his fingers. This observation seems doubtful at first but seems to be confirmed on the following days. In particular, at 0;1 (30) for a few moments Laurent pulls my thumb without letting it go, having by chance knocked it with the back of his hand.

*Observation 53.*—From 0;2 (3) Laurent evidences a circular reaction which will become more definite and will constitute the beginning of systematic grasping; he scratches and tries to grasp, lets go, scratches and grasps again, etc. On 0;2 (3) and 0;2 (6) this can only be observed during the feeding. Laurent gently scratches his mother's bare shoulder. But beginning 0;2 (7) the behavior becomes marked in the cradle itself. Laurent scratches the sheet which is folded over the blankets, then grasps it and holds it a moment, then lets it go, scratches it again and recommences without interruption. At 0;2 (11) this play lasts a quarter of an hour at a time, several times during the day. At 0;2 (12) he scratches and grasps my fist which I placed against the back of his right hand. He even succeeds in discriminating my bent middle finger and grasping it separately, holding it a few moments. At 0;2 (14) and 0;2 (16) I note how definitely the spontaneous grasping of the sheet reveals the characteristics of circular reaction—groping at first, then

regular rhythmical activity (scratching, grasping, holding and letting go), and finally progressive loss of interest.

But this behavior grows simpler as it evolves in that Laurent scratches less and less, and instead really grasps after a brief tactile exploration. Thus already at 0;2 (11) Laurent grasps and holds his sheet or handkerchief for a long time, shortening the preliminary scratching stage. So also at 0;2 (14) he pulls with his right hand at a bandage which had to be applied to his left. The following days his tactile interest is entirely absorbed by reciprocal hand grasping and tactile exploration of the face to which we shall return. With regard to object grasping Laurent (whose precocity has been noted with regard to thumb sucking) begins, at the end of the third month, to grasp in order to suck. He thus passes from the second to the third stage.

*Observation 54.*—Lucienne manifested the same vague reactions as Laurent (see Obs. 50–52) until about the age of 2 and ½ months. About 0;2 (12) I note agitation of her hands when in contact with the covers—grasping and releasing, scratching the material, etc. Same reactions on the following days. At 0;2 (16) she pulls at a pillow. At 0;2 (20) she opens and shuts her hands in space, and scratches a piece of material. At 0;2 (27) she holds her cover for a few moments, then a corner of the sheet which she grasped by chance, then a small doll which I placed against her right palm. At 0;3 (3) she strikes her quilt with her right hand; she scratches it while carefully watching what she is doing, then lets it go, grasps it again, etc. Then she loses contact with it, but as soon as she feels it again, she grasps it without scratching it first. Same reaction several times in succession. There exists therefore a quite systematic circular reaction directed by touch and not by sight.

It is not difficult to find in these reactions the first behavior patterns pertaining to sight or hearing: assimilation by pure repetition (grasping for the sake of grasping) and the beginning of accommodation (orientation of hand and fingers as a function of the object when they are in contact with this object). But there are not yet subtler accommodations or recognitory or generalizing assimilations.

From the onset of these primitive behavior patterns, on the other hand, a coördination between hand movements and those of sucking may be observed. Actually, with regard to our three children, the systematic sucking of the fingers either preceded or accompanied the first acquired activities involving only the hand or the fingers. It is also possible to find other very precocious reactions of the fingers coördinated not only with sucking

but also with all the tactile sensibility of the face and discovered parts of the body:

*Observation 55.*—Jacqueline, while learning to suck her fingers [achieved at 0;1 (28)] constantly moved her hand over her face without appearing to explore it systematically but undoubtedly learning to recognize certain contacts. For instance at 0;2 (77) she puts her right hand on her nose when it is being cleaned. So also, during the third month, she rubs her eyes several times in succession so that they become irritated.

*Observation 56.*—At 0;2 (17) and the days following, Lucienne more or less systematically puts the fingers of her right hand on her right eye and goes to sleep in this position. Perhaps the irritation of the eye before her nap provoked this repeated reaction. At 0;2 (25) she scratches her eye with the back of her hand and recommences momentarily so that the whole eyelid is reddened.

*Observation 57.*—Beginning 0;2 (8) Laurent constantly pulls at his face before, during, or after sucking his fingers. This behavior slowly gains interest for its own sake and thus gives rise to two distinct habits. The first consists in holding his nose. Thus at 0;2 (17) Laurent babbles and smiles without any desire to suck, while holding his nose with his right hand. He begins this again on 0;2 (18) while sucking (he holds his nose with four fingers while sucking his thumb), then continues later. At 0;2 (19) he grasps his nose sometimes with his right, sometimes with his left hand, rubs his eye in passing but constantly returns to his nose. That evening he holds his nose with both hands. At 0;2 (22) he seems to raise his right hand to his nose when I pinch it. At 0;2 (24) and the following days he touches his nose again.

*Observation 58.*—The second habit acquired by Laurent at the same period consists in rubbing his eyes sometimes with the back of his hand, sometimes with the fingers. This may be observed when he awakens and is stretching but not only a particular reflex must be involved for stretching is present from birth but eye rubbing has just occurred and only sporadically. Furthermore and more important, Laurent rubs his eyes all the time independently of his nap as though he has made the tactile discovery of his eyes and kept returning to it through circular reaction. At 0;2 (16) I even note that he closes his eye before his right hand approaches it and while he does not yet see it. At 0;2 (18), same reaction: both of his eyes close before he scratches the right eye. At 0;2 (19) he turns his head to the left as his left hand is being directed toward his eye. Then he rubs both eyes simultaneously with both hands At 0;2 (20) he makes fists in order to rub his eyes, again closes his eyes beforehand and smiles with joy; there is no connection with stretching The following days, same reactions.

*Observation 59.*—The activity of the hands with respect to the body itself is not limited to the nose and eyes. Sometimes the whole face is covered by both hands joined together. Sometimes—in Laurent's case at 0;2 (24)—the chest receives regular blows. But it is chiefly the hands which discover and touch one another. This phenomenon was particularly important with respect to Laurent not only because it gave rise to an especially tenacious habitual schema but also because this schema subsequently set in motion very precocious behavior patterns of prehension coördinated with sucking and above all with vision. It is noteworthy in the first place that already during the acquisition of thumb sucking (Obs. 6–21) Laurent often clasped his hands while he sucked the fingers of one of them. This pattern was revealed sporadically until the end of the second month. At the beginning of the third month it gave rise to a very systematic habit. I note that at 0;2 (4) and 0;2 (10) he seems to touch his hands. At 0;2 (14) his right hand pulls a bandage on his left. At 0;2 (17) I draw away his left hand by means of a string (attached to it to prevent Laurent from sucking his left thumb), he catches this hand several times by means of his right hand. The precision with which he performs this function while his left hand tries to overcome the resistance of the string and to enter his mouth, shows that a solidly constructed schema has already been formed. At 0;2 (19) Laurent clasps his hands several times and toward evening does it almost continuously. He touches them, then sucks them together, lets them go, grasps them again, etc. The interest is primarily in grasping and only secondarily in sucking. The following days this behavior is increasingly frequent but here we must interrupt our description of it because looking intervenes and begins to modify this "schema of junction." Beginning 0;2 (24) Laurent is observed to examine his clasped hands so attentively that their movement is transformed by this, which is characteristic of the third stage. Primarily, the systematization of this habit of joining results in hastening the moment when Laurent will grasp with both hands some object in order to keep it in his mouth which is also typical of this third stage (it is even by this last characteristic that we shall arrive at defining the transition from the second to the third stage of prehension).

These coördinations between the movement of hands and face (Obs. 55–58) do not raise any particular question. They are not, like the coördination between sight and hearing, for example, reciprocal assimilations of independent schemata; they only actually constitute an extension of the primitive and purely tactile schemata of prehension (Obs 50–54). The clasping of the hands on the contrary, is in one sense a mutual assimilation, but not outside the realm of tactile prehension. Until now the

above-mentioned coördination of the thumb and of sucking (Obs. 16–24) involves a beginning of reciprocal assimilation between independent schemata; but if the mouth sucks the hand and the hand directs itself to the mouth, the hand is not yet able to grasp everything that the mouth sucks.

Let us now proceed to the coördinations between vision and hand movements. Preyer and Tournay observed that during the seventeenth week the child looked at his hands for the first time in a systematic way. Wallon[13] who quotes this, seems to envisage it as an indication of a general truth.

Observation of our children unfortunately does not corroborate these dates; rather it seems to show that coördination between vision and hand movements is a continuous process depending on functional use more than on acquisitions which can definitely be placed in time. The only date which is easy to determine is that of the appearance of the following behavior pattern: at a given moment the child grasps the objects which he has perceived in the same visual field as his hand and before grasping them he alternately looks at his hand and the objects. Now this occurrence (fixed by Preyer at the 17th week) took place in Jacqueline's case at 0;6 (1), in Lucienne's at 0;4 (15) and in Laurent's at 0;3 (6). It characterizes what we shall call the fourth stage of prehension. But earlier all sorts of coördinations between vision and hand movements may be observed, coördinations which begin at the present stage and continue through the third. Here are those which we have observed during the second stage:

*Observation 60.*—Lucienne, at 0;2 (3), that is to say, the day after the day she began systematically to suck her thumb, twice looked at her fingers as they came out of her mouth (see above, Obs. 23). This glance was fleeting, but with accommodation of the eye to distance. At 0;2 (12), on the other hand, and the next day she looked at her hand more attentively. At 0;2 (15) I watch her while she lies on her right side and sucks her bib. Her hands move in front of her (the fingers constantly moving), grasping and letting go the sheets, scratching the cover, and the right hand or both hands momentarily enter her mouth. Lucienne's eyes seem to follow the movements of her hands (her glance rises and falls correctly, etc.) but her hands do not yield to the exigencies of the visual field. Vision, therefore, adapts itself to the movements of the hand but the converse is not yet true.—At 0;2 (16) Lucienne is lying on

[13] *Op. cit.*, pp. 97–98.

her left side, her right hand pulling at the pillow; she attentively looks at this hand. At 0;2 (17) Lucienne is on her back, her right hand stretched out and the fingers moving slightly; she looks at this hand most attentively and smiles. A moment later she loses sight of it (her hand having lowered); her glance obviously seeks it and, when her hand is raised again, it immediately follows it. At 0;2 (20) Lucienne continues to look at her hands, including the left one. For instance the hands open and close alternately; they do so simultaneously and frequently outside the visual field which surely reveals that an entirely motor circular reaction independent of vision is involved. But as soon as the phenomenon is produced opposite her face, Lucienne directs her glance to her hand and watches it for a long time. She also inspects her right hand which scratches a piece of material. At 0;2 (27) she looks at her right hand which is holding a doll but is unable to keep this spectacle in her visual field. She also looks at her empty hands, the left almost as much as the right, but also without keeping them in the visual field; the glance searches for the hands but they are not subordinated to the glance. At 0;3 (3) she looks attentively at her right hand which scratches a quilt, then relinquishes it, then grasps it again, etc. While her hand loses contact with the quilt she looks at the latter but without coördination with the hand movements. Her hand rediscovers the quilt through tactile accommodation and not through accommodation with sight. That evening, she watches her hand open and close. There is as yet no precise coördination between these movements and sight except that the fingers seem to move more when Lucienne looks at them. At 0;3 (8 and 9) she looks attentively at her clasped hands while sucking the index finger and the back of her right hand.—We shall stop with this observation for, from this date on, Lucienne begins to carry to her mouth the objects she has grasped which marks the beginning of the third stage.

*Observation 61.*—Jacqueline seems not to have looked at her hands before 0;2 (30). But on this date and the following days she frequently notices her moving fingers and looks at them attentively. At 0;3 (13) she rumples her quilt with both hands. When her hands move into her visual field she looks fixedly at them just as she looks at the folds of the quilt when they appear before her but, if her eyes attempt to see the hands, the hand movements do not yet depend on vision at all. At 0;3 (21) likewise, her eyes follow her hands. At 0;3 (22) her glance follows her hands which turn aside and she seems very much surprised to see them reappear.

*Observation 62.*—At 0;2 (4) Laurent by chance discovers his right index finger and looks at it briefly. At 0;2 (11) he inspects for a moment his open right hand, perceived by chance. At 0;2 (14), on the other hand, he looks three times in succession at his left hand and chiefly at his

raised index finger. At 0;2 (17) he follows its spontaneous movement for a moment, then examines it several times while it searches for his nose or rubs his eye. Next day, same observation. At 0;2 (19) he smiles at the same hand after having contemplated it eleven times in succession (when it has been untied); I then put this hand in a bandage again; as soon as I detach it (half an hour later) it returns to the visual field and Laurent again smiles at it. The same day he looks very attentively at his two clasped hands. At 0;2 (21) he holds his two fists in the air and looks at the left one, after which he slowly brings it toward his face and rubs his nose with it, then his eye. A moment later the left hand again approaches his face; he looks at it and touches his nose. He recommences and laughs five or six times in succession while moving the left hand to his face. He seems to laugh before the hand moves, but looking has no influence on its movement. He laughs beforehand but begins to smile again on seeing the hand. Then he rubs his nose. At a given moment he turns his head to the left but looking has no effect on the direction. The next day, same reactions. At 0;2 (23) he looks at his right hand, then at his clasped hands (at length). At 0;2 (24) at last it may be stated that looking acts on the orientation of the hands which tend to remain in the visual field. Thus we reach the third stage.

It may thus be seen of what the coördinations between vision and the first circular reactions of the hand and fingers consist. We can say that the visual schemata tend to assimilate the manual schemata without the converse being yet true. In other words, the glance tries to follow what the hand does but the hand does not tend in any way to realize what the glance sees; it does not even succeed in remaining in the visual field! Later, on the contrary, the hand will be regulated by vision, and vice versa; this will enable the child to grasp the objects seen. But, for the time being, the hand moves independently of the glance, the few vague circular reactions to which it gives rise being only directed by touch, kinesthetic sensations, or sucking. The relations between sight and hand movements are therefore different from those which exist between sucking and these movements; in the case of sucking, the schemata external to the hand movements control them and incorporate them (sucking entails circular reaction of the arms and hands) while in the case of vision hand movements are autonomous and the glance is limited to assimilating without controlling them. It is therefore clear that from this point of view sucking is ahead of vision. Thus at the third

stage we shall see the hands grasp objects to carry them to the mouth and not yet in order to look at them.

In a general way we may conclude the following with regard to the second stage. During this stage the hand movements are no longer controlled only by reflex and impulsive mechanisms but give rise to some acquired circular reactions. The reactions certainly remain indefinite and it seems as though with respect to the most primitive of them (opening and closing the hands, scratching with the finger tips, grasping and letting go, etc.) that a simple impulsive automatism were always involved. But the question is to know if these behavior patterns are indeterminate because they are still entirely "impulsive" or because as yet they only constitute empty circular reactions without interest in the object grasped. The case of prehension is, in effect, exactly analogous to that of sucking, vision, or hearing. Just as there exists empty sucking, tongue sucking, etc., so also the nursling can wave his arms, open and close his hands, clench them, move the fingers, etc., for weeks without an object and without true contact with a reality which resists. And just as vision passes through a stage during which objects are aliments for the glance without assuming interest as external images, so also the first contacts of the hand with the things it grasps, touches and scratches by chance, bear witness to a purely functional phase of assimilation (grasping for the sake of grasping) by repetition and not yet by combined generalization and recognition. It is to this phase that Observations 50–52 apply. On the other hand, Observation 53 and Observations 55–58 are evidence of a generalizing assimilation and a beginning of tactile recognition in addition to this primitive functioning. On the one hand, as soon as the child learns to scratch and pull at objects (Obs. 53) he extends this behavior to everything, including his face and his own hands (Obs. 55–58). On the other hand, through the very fact of the extension of the schema it becomes differentiated and gives rise to a recognitory assimilation. This is why the child recnizes his nose, eyes and hands by touch, when he is searching for them. In correlation with the progress of assimilation there is gradual accommodation to objects. The hand takes the form of a thing, the thumb gradually is opposed to the other fingers,

beginning the third month (or even shortly before) it is enough to touch the back of the hand to make the hand attempt to grasp, etc. With regard to coördinating organizations there is, as we have seen, a beginning of coördination with sucking and with vision but without reciprocal assimilation of the schemata. The mouth sucks the hands but the hands do not try to carry to the mouth everything they grasp nor to grasp everything that the mouth sucks, and the eye looks at the hands but the hands do not try to feel or to grasp everything the eyes see. These two essential coördinations will develop during the three succeeding stages. The coördination between sucking and grasping is more precocious and thus characterizes the third stage. But there is no logical necessity for this order of succession and it is possible to conceive of the existence of a partial reversal in the case of certain exceptional subjects.

During a *third stage* notable progress is revealed: henceforth there is *coördination between prehension and sucking.* In other words, the hand grasps objects which it carries to the mouth and reciprocally it takes hold of objects which the mouth sucks.

Let us first describe the facts in order later to analyze their various aspects.

*Observation 63.*—At 0;3 (8) Lucienne grasps her coverlet in her right hand and sucks it. I then place a pencil in her hand; her hand moves slightly toward her mouth and stops. As yet it is impossible to decide between chance and coördination. But that evening three times in succession I place a soft collar in her right hand which is stretched out on the coverlet and each time she carries it to her mouth. No attempt at seeing. At 0;3 (9) I place a wooden object in her hand; she brings it toward her mouth, then lets it go. At 0;3 (13), same experiment: she holds the object, carries it to her mouth and alternately licks the object and her hand without appearing to dissociate these two bodies from one another. At 0;3 (24) she grasps bib, quilt, covers by herself and carries them to her mouth. At 0;4 (4) she grasps a toy by chance (of course without seeing it) and holds it firmly for a few moments. Then a sudden movement to put it in her mouth without trying to look. Same reaction with a part of the coverlet. She does not yet direct the object; she sucks that which comes first. There exists therefore in some way a conjunction of two schemata (grasping and holding) and (putting the hand to the mouth) and not yet the single act of putting the object to the mouth.

*Observation 64.*—At 0;4 (9) I put a rattle before her eyes: no reaction. Then I place it in her hands: she immediately puts it in her mouth, sucks it, then moves it at random while looking at it. It seems as though this time the act of grasping a substance in order to suck it forms a single organized whole. This is confirmed by the following reaction. That evening I show Lucienne her usual rattle: she looks at it fixedly, opens her mouth, makes sucking movements, opens her mouth again, etc., but she does not grasp it. The sight of the toy, therefore, set in motion movements of sucking and not of prehension. But barely touching her stretched-out hand with the handle of the rattle suffices to produce movements of prehension: successive attempts with the fingers until the opposition of the thumb leads to success. The rattle, as soon as it is grasped, is carried to the mouth. At 0;4 (10) same reactions: the object, as soon as it is grasped, regardless of the visual field, is carried to the mouth. If it falls to the side groping ensues until success is attained.

*Observation 65.*—At 0;4 (10) Lucienne is lying on her back. I put a doll in front of her mouth. She manages to suck it while moving her head, but with difficulty. She then moves her hands but without bringing them together appreciably. A moment later, on the other hand, I place the rattle in her mouth, the handle lying on her chest; she immediately brings her hand to it and grasps it. The experiment is repeated three times: same reactions. At 0;4 (15) as soon as the rattle is in contact with the mouth, the hand moves in this direction. But Lucienne does not persevere. That evening, however, she grasps it immediately. This behavior seems to be definitely acquired and coördinated. To accomplish this Lucienne does not look at her hands at all and as soon as she touches the rattle she succeeds in grasping it. She does this with her left as well as with her right hand, but less often. From this observation on, Lucienne begins to coördinate her grasping movements with vision, and thus enters the fourth stage.

*Observation 66.*—At 0;3 (2) Lucienne carries to her mouth what she has grasped at random, opposing her thumb to the other fingers. At 0;4 (8) too, she carries to her mouth ribbons, corners of pieces of material, her bib, etc.

*Observation 66 repeated.*—Already at 0;2 (17) Laurent, after grasping his sheet, sucks it at the same time as his hand. There is therefore a chance connection between the schema of prehension and that of finger sucking. The next day he sucks the bandage on his left hand while holding it with his right. The following days the relations between prehension and sucking remain at random. On the other hand, at 0;2 (28) it is enough for me to place his rattle in his left hand (outside the visual field and the extended arm) to cause Laurent to introduce this object

into his mouth and suck it. The experiment succeeds a series of times, with the right hand as well as the left, and the systematization of the reaction shows that this new schema was constituted several days before. Same result on the following days. At 0;3 (4) he carries to his mouth ribbons, fringes of covers, cloth dolls, etc., and, at 0;3 (5) he does the same with unfamiliar objects (package of tobacco, cigarette lighter, tobacco pouch, etc.) which I put in front of his face and which he grasps after having touched them while putting his hands together. So also it suffices that I place in his outstretched hand, outside the visual field, an object which is unfamiliar (visually and tactilely) such as a clothes pin, for Laurent to carry it immediately to his mouth and not to his eyes.

Thus it may be seen that from the second half of the third month there exists, in Laurent, coördination between sucking and grasping but, as we shall see later, this third stage was shortened in his case by a certain precocity in coördination between vision and prehension. Moreover the sequential order of the acquisition of coördinations was almost reversed in the case of this child.

Such observations are interesting inasmuch as they indicate how systematic prehension is acquired. Following the circular reactions of the second stage (pure, generalizing, and recognitory assimilations) the child begins to interest himself in the objects themselves which his hand has touched. Here the same phenomenon is produced as with respect to vision or hearing. After having looked for the sake of looking the child becomes interested in the objects he looks at, because the assimilation of reality to vision is completed through coördination between vision and the other schemata. So also, after having practiced in space various hand movements and having grasped for the sake of grasping, after having used his prehension with respect to all the solid objects he encounters and having thus acquired an increasingly precise accommodation to objects concomitant to generalizing assimilation, after having even developed a sort of tactile-motor recognition of things, the child finally becomes interested in the objects he grasps inasmuch as prehension, which has thus become systematic, is coördinated with an already completed schema, such as that of sucking. How can this coördination be explained? In the beginning (Obs. 63) it seems that there is only partial coördination—that is to say, simple conjunction of two partially independent schemata. The hand takes hold of the

objects and the mouth attracts the hand. Thus at 0;4 (4) Lucienne still indifferently sucks hand or object when the hand brings the objects to the mouth. At a given moment, on the other hand, coördination becomes total. But here as with regard to sight and hearing, it clearly appears that this coördination results from a reciprocal assimilation of the schemata under consideration. The mouth seeks to suck what the hand grasps just as the hand seeks to grasp what the mouth sucks. In effect, in Observation 64, the mouth is ready to suck before the hand has discovered the object and then what the child grasps is at once brought to the mouth. Inversely, at 0;4 (10) (Obs. 65) Lucienne seeks to grasp the object which her mouth sucks when this object has not previously passed through manual prehension. Thus it may once more be seen of what the progressive organization of schemata consists: a mutual adaptation with reciprocal accommodation and assimilation.

This leads us to the coördination between vision and prehension. We recall that during the second stage the glance already follows the hand movements but the latter are not governed by the former. During the fourth stage we shall see that prehension itself is controlled by vision. With regard to the third stage which concerns us at the moment, it may be said that vision, without yet controlling prehension (which still only depends on touch and sucking) already exerts an influence on hand movements. The act of looking at the hand seems to augment the hand's activity or on the contrary to limit its displacements to the interior of the visual field.

*Observation 67.*—Lucienne, at 0;3 (13) looks at her right hand for a long time (her arm is outstretched) and opens and closes it. Then her hand moves quite suddenly toward her left cheek. Her eyes follow this movement with exactitude, her head moving continuously as though there were prevision. The hand then resumes its position. Lucienne looks at it again and smiles broadly while shaking herself, then the same game begins again. The following days her visual interest in hand movements or the hand holding an object remains constant, but vision does not seem to have any effect other than a vague dynamogenization of these movements.

*Observation 68.*—At 0;4 (9) Lucienne makes no motion to grasp a rattle she is looking at. But when she subsequently brings to her mouth the rattle she has grasped independently of sight and sees the hand which holds this object, her visual attention results in immobilizing

the movement of her hands; however, her mouth was already open to receive the rattle which is 1 cm. away from her. Then Lucienne sucks the rattle, takes it out of her mouth, looks at it, sucks again, and so on.—The same day, a new experiment: I place a case in her left hand. Lucienne carries it directly to her mouth, but, as she is about to put it in (her lips already open) she perceives it, moves it away and holds it before her eyes at a distance of about 10 cm. She looks at it most attentively while holding it almost motionless for more than a minute. Her lips move and she carries the object to her mouth and sucks it for several seconds but she removes it to look at it.—The same day, Lucienne engages in the same play with her coverlet, but as yet there is no coördination between the sight of an object or of the hand and prehension as such.

*Observation 69.*—At 0;4 (10) Lucienne looks at her rattle with the same reactions of buccal desire. She opens her mouth, makes sucking-like movements, raises her head slightly, etc. But she does not stretch out her hands although they make grasping-like movements. A moment later, her right hand being outstretched, I place a rattle next to her. Lucienne looks alternately at her hand and the rattle, her fingers constantly moving, but she does not move her hand closer. However, when the rattle touches her hand, she grasps it immediately.

*Observation 70.*—Jacqueline, at 0;4 (1) looks attentively at her right hand which she seems to maintain within the visual field. At 0;4 (8) she sometimes looks at the objects which she carries to her mouth and holds them before her eyes, forgetting to suck them. But there does not yet exist prehension directed by sight nor coördinated adduction of objects in the visual field. It is when the hand passes at random before her eyes that it is immobilized by the glance.—Sometimes, too, she looks attentively at her hands which happen to be joined.—At 0;5 (12) I observe that she constantly looks at her hands and fingers but always without coördination with prehension. At 0;6 (0) she has not yet established this coördination. She watches her hand move; her hand moves toward her nose and finally hits her eye. A movement of fright and retreat: her hand still does not belong to her! Nevertheless the hand is maintained more or less successfully within the visual field.

*Observation 71.*—At 0;3 (23) Lucienne's right arm is outstretched, her hand remaining outside the visual field. I grasp this hand. She tries to free it but does not look in this direction at all. Same result at 0;4 (9), etc. It is only during the following stages that Lucienne will search with her eyes for the hand which is held.

*Observation 72.*—Jacqueline still reacts in the same way at 0;5 (12)—that is to say, during the present stage. She is on her back and I alternately hold her right and left hands which are lying flat on the mattress.

She makes vain attempts to face her hand but without looking in the right direction although she tries to see what is going on. At a given moment while wriggling about Jacqueline happens to perceive my hand which is holding her right hand. She looks attentively at this unfamiliar image but without making an effort to free herself at this exact moment. Then she resumes the struggle while looking all around her head but not in the right direction. The consciousness of effort is therefore not localized in the visual image of the hand, but in the absolute. At 0;5 (25) same reaction.

*Observation 73.*—Laurent has revealed, with regard to the coördination of vision and hand movements, remarkable precociousness which must, we feel, be attributed to his development acquired by the schema of joining the hands (see Obs. 59). In effect, through clasping of the hands which necessarily takes place in front of the face in a reclining child, Laurent eventually studied them by looking at them attentively [see Obs. 52 at 0;2 (19) and 0;2 (23)]. This regular connection, although its cause is fortuitous, results quite naturally in leading to the influence of the glance on the movement of the hand. Thus at 0;2 (24) Laurent rubs his hands, 5 to 10 cm. from his mouth, without sucking them. He separates them and then grasps them again at least twenty times in succession while looking at them. It would appear, in such an instance, that visual pleasure alone were the cause of the repetition of the phenomenon. An hour later this impression is strengthened when Laurent, having grasped his right hand with his left and having removed the bandage (placed on the right thumb) holds the bandage within the visual field and looks at it with curiosity. At 0;2 (25) Laurent looks at his motionless left hand, after having rubbed his eye. At 0;2 (26 and 28) he looks at a rattle which he has in his hands and at 0;2 (29) I observe a new combination derived from differentiation of this schema of clasping the hands. Laurent holds his hands with the tips of his fingers only, and 10 to 15 cm. away, exactly opposite his eyes. He obviously keeps them within the visual field and reveals no tendency to suck or even actually to grasp for at least a quarter of an hour. There is only involved playing with the fingers discovered tactilely and agreeable to the eye. The next day, same observation.

*Observation 74.*—The significance of the preceding behavior patterns is that, with regard to Laurent, they give rise to a very curious reaction which particularly facilitated access to the definitive coördination characteristic of the fourth and fifth stages of prehension. Beginning 0;3 (3) Laurent began to grasp my hand as soon as it appeared before his face because my hand was visually assimilated to one of his hands and so set in motion the schema of hand clasping.

At 0;3 (3), around 2 P.M. I place my motionless hand opposite his face, 10 to 15 cm. from his mouth. He looks at it and immediately makes

sucking-like movements while looking at it, as though he were assimilating it to his hand which he constantly examines before or after sucking. But he looks at my hand without trying to grasp it. Then, without changing its location, I open my hand more and manage to touch his left hand very lightly with my little finger. Laurent immediately grasps this finger without seeing it. When I withdraw it, Laurent searches for it until he rediscovers it (which is the first example of a reaction important to the development of prehension: recapturing that which escaped the hands). Finally this use of prehension passes into the visual field and Laurent looks at my finger most attentively. The same day, at 6 P.M. if I display my hand in the same position this is enough to make Laurent grasp it! I touched his hand (with my little finger) just once, then five times in succession he grasps my hand without my having previously touched his nor his having seen his hand at the same time as mine! At first I assumed this phenomenon to be a coördinated act of prehension regulated by sight of the object alone (hence an act characteristic of the fifth stage) but the rest of the observation suggested a simpler interpretation. The sight of my hand simply set in motion the habitual cycle of movements of bringing the hands together (the schema of clasping) and as my hand was in the trajectory of his hands, he met and grasped it.

The next day, at 0;3 (4) he at once grasps my hand even though I have not touched his at all. I find, moreover, confirmation of the preceding interpretation in the following three facts. In the first place, when I present Laurent with some objects instead of my hand, he does not attempt to grasp them and confines himself to looking at them. In the second place, when I present my hand at a certain distance (20 to 30 cm.) and not just in front of his face, he is content to grasp his hands without trying to reach my hand. In the third place, when I separate and clasp my hands, at a distance of about 50 cm., Laurent imitates me, as we shall see later. These three combined facts seem well to demonstrate that, if Laurent grasps my hand in front of his face, this is through assimilating my hand to the schema of clasping his own.

At 0;3 (5) Laurent imitates less well my clasping movement when I am at a distance. As soon as I bring my hand closer to his face, he joins his hands and, at the proper distance, grasps them. When I again move my hands away, he joins his. That afternoon, I present my motionless hand: he grasps it and laughs. Then I replace my hand with a package of tobacco, a cigarette lighter and finally my tobacco pouch: he grasps all three in sequence! By means of my hand and the schema of joining, Laurent thus arrives at the beginning of the fourth stage.

*Observation 75.*—At 0;3 (5)—that is to say, the third day of the preceding observation—I immobilize Laurent's hands outside the visual field: he does not look (see Obs. 71 and 72).

*Observation 76.*—Finally here is an example of the conjunction of the combined schemata of vision, prehension and sucking. At 0;4 (4) I show Lucienne my motionless hand. She looks at it attentively, then smiles, then opens her mouth wide and finally puts her own fingers into it. Same reaction many times. It appears that Lucienne assimilates my hand to hers and thus the sight of my fingers makes her put hers into her mouth. It is noteworthy that shortly afterward she looks at her own index finger, sucks it, looks at it again, etc. So also Laurent, at 0;3 (6), while looking at my hand in the same position, opens wide his mouth. Then he grasps my hand and draws it toward his open mouth while staring at my fingers.

It may thus be seen in what these coördinations between vision and hand movements consist. It is not yet possible to speak of coördination between vision and prehension, since the child knows neither how to grasp what he sees (he does not grasp what he touches or sucks) nor how to hold before his eyes that which he has grasped (he carries things to the mouth and not to the eyes), nor even how to look at his own hand when it is held by the hand of someone else (Obs. 71, 72 and 75). It can no longer be said, however, that the child is limited to looking at his hands without having them react to his looking. When the hand by chance enters the visual field it tends to remain there. It even happens that the child postpones sucking the grasped object through pure visual interest (Obs. 68 and 70). In short, there is a beginning of true coördination—that is to say, of reciprocal adaptation. The hand tends to conserve and repeat the movements that the eye looks at, just as the eye tends to look at everything the hand does. In other words the hand tends to assimilate to its schemata the visual realm just as the eye assimilates to its schemata the manual realm. Henceforth when the child perceives certain visual images (sees the fingers move, the hands hold an object, etc.) this is enough to make his hand tend to conserve them through reproductive assimilation to the extent that these images are assimilated to manual schemata.

How is this reciprocal assimilation to be explained? We understand the meaning of the assimilation of the motor realm by the visual schemata, since the hand and its movements can be seen and watched. But what does the assimilation of the visual by the manual mean? In what follows this will be tantamount to

saying that the hand tries to grasp everything that the eye sees. But this coördination will only be exactly produced later, during the fourth or fifth stage. For the time being the manual schemata only assimilate the visual realm to the extent that the hand conserves and reproduces what the eyes see of it. Now how is this possible? Associationism simply responds: the visual image of the hand, through being associated to the movements of this hand, acquires by transfer the value of a signal and governs sooner or later these very movements. Concerning the fact, as such, of this associative transfer, everyone is of course in agreement. Every accommodation involves putting into relationship known quantities imposed by experience, and the child discovers the connection between the visual image of the hands and their movements quite a while before attributing this image and the corresponding kinesthetic impressions to a unique and substantial "object." But the problem is to find out whether this relation between the visual and the motor is established by "association." On the contrary, we place in opposition to the passive concept of association the active concept of assimilation. That which is fundamental and without which no relationship between sight and hand movements could be established is that the hand's activity constitutes schemata which tend to conserve and reproduce themselves (opening and closing, grasping bodies and holding them, etc.). Through the very fact of this tendency toward conservation such activity incorporates in itself every reality capable of supporting it. This explains why the hand grasps that which it encounters, etc. Now comes the moment when the child looks at his hand which is moving. On the one hand, he is led, by visual interest, to make this spectacle last—that is to say, not to take his eyes off his hand; on the other hand, he is led, by kinesthetic and motor interest, to make this manual activity last. It is then that the coördination of the two schemata operates, not by association, but by reciprocal assimilation. The child discovers that in moving his hand in a certain way (more slowly, etc.) he conserves this interesting image for his sight. Just as he assimilates to his glance the movement of his hands, so also he assimilates to his manual activity the corresponding visual image. He moves with his hands the image he contemplates just as he looks with his eyes at the

movement he produces. Whereas until now only tactile objects served as aliments for the manual schemata, visual images now become material for exercises of the hands. It is in this sense that they may be called "assimilated" to the sensorimotor activity of the arms and hands. This assimilation is not yet an identification: the visual hand is not yet a tactile-motor hand. But the substantial identification will result from assimilation as the geometric point of the crossing of lines. The intersection of the assimilating activities will define the object, in proportion as these activities applied to the outside world will constitute causality.

An excellent illustration of this process is supplied by Observations 73 and 74. After watching his hands join for several days, at 0;3 (3) Laurent manages to grasp a privileged object as represented by my hand. How can this precocious prehension be explained, if not precisely as being due to the fact that this visual image of my hand is assimilated to the visual image of his hands, and that this latter image is already incorporated in the schema of joining the hands.[14] Here we see at work, in the most definite way, the play of assimilation in its dual reproductive and recognitory character. If the coördination of vision and prehension were a matter of pure physiological maturation of the nervous system, the differences in dates of acquisitions as revealed by three normal children such as Jacqueline, Lucienne and Laurent could not be understood. On the other hand, by following in detail Laurent's psychomotor assimilations (the use of the cycle

[14] It may be thought strange that we should assert with regard to Observation 74 that at 0;3 (3) Laurent manages to assimilate my hand to his, despite differences in size and position. But a good reason impels us to this interpretation. Beginning 0;3 (4) I have been able to establish that Laurent imitates my hand movements; he separates, then joins his hands in response to my suggestions. This imitative reaction recurred at 0;3 (5), 0;3 (6), at 0;3 (8), at 0;3 (23), etc. Now if there is imitation of such a movement, to the exclusion of so many others, it is obvious that there is assimilation. That this assimilation is entirely synthetic, without objective identification, is evident; it does not yet involve either the distinguishing of another's body and his own body or the concept of permanent and comparable objects grouped in categories, and it is doubtless even based upon a confusion rather than an actual comparison. But no more than this is needed to enable us to speak of assimilation. Assimilation, which is the source of imitation as it is of recognition, is an earlier mechanism than objective comparison and, in this sense, there is no obstacle to asserting that a 3-month-old child can assimilate another's hand to his own.

to the joining of the hands, the assimilation to this schema of the visual image of the hands and finally the assimilation of my hand to his hands) the reason for his precocity may be understood.

The same applies to the still more complex example of assimilation of the visual to the manual furnished by observation 76. At 0;4 (4) Lucienne sucks her hand while looking at mine. Before that time Lucienne has already coördinated the grasping of objects to sucking movements. She carries to her mouth everything that she grasps, regardless of the visual field. In addition she visually recognizes the objects she sucks or is going to suck and thus a coördination between vision and sucking is established which we have analyzed in connection with the latter. Now, among these objects the hand plays a central role, since Lucienne is visually acquainted with it at about two months, that she knows how to suck from a still earlier date and knows how to carry her hand to her mouth after having looked at it. There exists, therefore, as far as the hand is concerned, a conjunction of at least three schemata—sucking, vision, and motor activity to the exclusion of actual prehension. Lucienne looks at my hand; her reaction is to suck it immediately and perhaps to move it. But, either she confuses it at first with her own and then sucks her own, or else, what is more likely, she has the impression, due to global assimilation, of an object capable of being carried to the mouth more easily than the others, and, not knowing how to grasp what she sees, it is her own hand that she brings between her lips. In this second case, there was only semiconfusion; but in both cases the visual image of my hand is assimilated to the simultaneously visual, motor, and buccal schema of her own hand.

Regardless of these last examples, the coördination between vision and hand movements until the present only affect the latter to the exclusion of prehension. In other words, except in Observations 74 and 76, the child still only grasps objects when he touches them by chance, and, if he looks at his hands when they are holding the object, vision does not yet help at all in the actual act of grasping. During the fourth and fifth stage, the coördination between vision and hand movements will be extended until it arrives at actual prehension.

The *fourth stage* is that during which *there is prehension as*

*soon as the child simultaneously perceives his hand and the desired object.* I have been able to observe, in effect, in the most definite way with respect to my three children that the grasping of objects which have simply been looked at only begins to become systematic in cases in which the object and the hand are perceived in the same visual field.

*Observation 77.*—At 0;6 (0) Jacqueline looks at my watch which is 10 cm. from her eyes. She reveals a lively interest and her hands flutter as though she were about to grasp, without however discovering the right direction. I place the watch in her right hand without her being able to see how (the arm being outstretched). Then I again put the watch before her eyes. Her hands, apparently excited by the contact just experienced, then proceed to move through space and meet violently, subsequently to separate. The right hand happens to strike the watch: Jacqueline immediately tries to adjust her hand to the watch and thus manages to grasp it. The experiment is repeated three times: it is always when the hand is perceived at the same time as the watch that the attempts become systematic.—The next day, at 0;6 (1) I resume the experiment. When the watch is before her eyes Jacqueline does not attempt to grasp it although she reveals a lively interest in this object. When the watch is near her hand and she happens to touch it, or it is seen at the same time as her hand, then there is searching, and searching directed by the glance. Near the eyes and far from the hands, the watch is again simply contemplated. The hands move a little but do not approach each other. I again place the object near her hand: immediate searching and again, success. I put the watch a third time a few centimeters from her eyes and far from her hands: these move in all directions but without approaching each other. In short, there are still two worlds for Jacqueline, one kinesthetic and the other visual. It is only when the object is seen next to the hand that the latter is directed toward it and manages to grasp it.—That evening, the same experiments with various solid objects. Again and very regularly, when Jacqueline sees the object facing her without perceiving her hands, nothing happens, whereas the simultaneous sight of object and of hand (right or left) sets prehension in motion. Finally it is to be noted that, that day, Jacqueline again watched with great interest her empty hand crossing the visual field: The hand is still not felt to belong to her.

*Observation 78.*—Lucienne, at 0;4 (12) looks attentively at her mother's hand while taking the breast. She then moves her own hand while looking at her mother's. Then she perceives her own hand. Her glance then oscillates between the aforementioned hands. Finally she grasps her mother's hand.—The same day, in the same situation, Lucienne again perceives her mother's hands. She then lets go the breast to stare at

this hand while moving her lips and tongue. Then she puts forward her own hand in the direction of the maternal hand and suddenly, she puts her own between her lips, sucks it a moment and takes it out, the while looking at her mother's hand. There ensues a reaction analogous to that of Observation 65. As she did a week earlier, Lucienne sucks her own hand out of confusion with the hand she perceives. But this time the confusion does not last; after having removed her hand from between her lips she moves it about at random, haphazardly touches her mother's hand and immediately grasps it. Then, while watching this spectacle most attentively, she lets go the hand she was holding, looks alternately at her own hand and the other one, again puts her hand in her mouth, then removes it while contemplating the whole time her mother's hand and finally grasping it and not letting it go for a long moment.

*Observation 79.*—Lucienne, at 0;4 (15) looks at a rattle with desire, but without extending her hand. I place the rattle near her right hand. As soon as Lucienne sees rattle and hand together, she moves her hand closer to the rattle and finally grasps it. A moment later she is engaged in looking at her hand. I then put the rattle aside; Lucienne looks at it, then directs her eyes to her hand, then to the rattle again, after which she slowly moves her hand toward the rattle. As soon as she touches it, there is an attempt to grasp it and finally, success.—After this I remove the rattle. Lucienne then looks at her hand. I put the rattle aside. She looks alternately at her hand and at the rattle, then moves her hand. The latter happens to leave the visual field. Lucienne then grasps a coverlet which she moves toward her mouth. After this her hand goes away haphazardly. As soon as it reappears in the visual field, Lucienne stares at it and then immediately looks at the rattle which has remained motionless. She then looks alternately at hand and rattle after which her hand approaches and grasps it.

*Observation 80.*—The same day progress is revealed after the facts related in Observation 65 (taking the rattle placed against the mouth). I put the rattle above Lucienne's face. The immediate reaction consists in trying to suck it; she opens her mouth, makes sucking-like movements, pulls her tongue, pants with desire. Thereupon her hands approach her mouth and seem to stretch toward the object. As soon as the right hand is seen, it directs itself toward the rattle and grasps it. It is therefore the desire to suck the object which set the hand in motion; therein is progress toward the fifth stage.—I then place the rattle higher up. Same expression of buccal drive. The hand tries to grasp, in space. As soon as Lucienne perceives her hand she looks alternately at the rattle and at her hand, then tries to grasp, which she achieves after some groping.—At 0;4 (19) same reactions with my finger: she makes sucking-

like movements while looking at it, then moves her hand toward her mouth and when she sees the hand, grasps.

*Observation 81.*—At 0;3 (6)—that is to say following Observation 73 and 74—Laurent looks at my watch which I hold not in front of his face, but to his right. This spectacle sets in motion activity of both hands, but not a joining movement. The right hand remains in the zone of the watch, as though he were searching for it. As soon as Laurent sees watch and hand together, he grasps! The hand was well oriented, open, with the thumb opposite.—A moment later, I present a cloth doll on the left side. The reaction is the same: Laurent looks at the doll, then perceives his left hand, looks at it, then his eyes return to the doll. He then grasps it, carries it to his mouth and sucks it.

That evening, an essential observation. Laurent's hands are outstretched and he looks ahead of him, wide awake. I present to him the customary objects (rattle, cloth doll, package of tobacco, etc.): he grasps nothing and looks at them as though he knew nothing whatever about grasping. Thereupon I place my motionless hand in front of his face, in the same place as the objects: he grasps it immediately; my hand is barely in position when his hands move and with one motion seize my hand.—It seems that, through lack of seeing a hand, Laurent did not have the idea of grasping the objects presented at first, and that the sight of my hand (in its role of hand and not of object) immediately stimulated his schema of prehension.

A little later I present to Laurent a cloth doll (on the left side): he looks at it attentively without moving his hand (except for a few vague movements). But, as soon as he sees his hand (I observe his glance through the hood of the bassinet), he grasps. Same experiment with the customary series of objects and same reactions.

*Observation 82.*—At 0;3 (7), the following day, Laurent is motionless, his hands outstretched, engaged in babbling, when I begin the day's first experiment. I present to him on the left side (without showing myself) a roll of tinfoil (to him an unfamiliar object). Three definite reactions succeed one another. In the first place, his hands immediately move, open and tend to approach one another. Meanwhile Laurent looks at the object without looking at his hands. His left hand passes near the paper, very slowly, but instead of bifurcating in the direction of the object, it pursues its trajectory toward the other hand which comes to meet it. The hands then meet while Laurent continues to look at the object. The sight of the object has therefore set in motion the cycle of the functioning of the hands, without modifications. In the second place, while Laurent's hands are joined, I put the tinfoil opposite him. He looks at it but does not react at all. In the third place, I put the paper in the same visual field as his hands. He then looks at his hands, losing sight of the object for a moment, then again looks at

the object; then he separates his hands and directs them toward the object which he manages to grasp. The simultaneous viewing of hands and object is therefore still necessary for prehension.

The next day, same observations, in the morning. In the afternoon I present one of his rattles to Laurent; when the rattle is in the trajectory of his hands, he grasps it immediately. Otherwise he looks alternately at his hand and the object. In particular, when I place the rattle on his quilt, before his face at a distance of about 10 cm., he looks at length at his hand and the rattle before attempting to grasp; his hand remains 5 cm. from the rattle. Then at last he makes an attempt and succeeds.

Same reaction for two more days, then Laurent enters the fifth stage.

*Observation 83.*—During this fourth stage I was able to observe in Laurent a beginning of the reciprocal relation between vision and actual prehension. But it is only a beginning. At 0;3 (7) when he has succeeded in grasping the tinfoil, Laurent lets it go soon after (with his left hand). He then turns his head to look at his empty hand. Same observation a moment later. I then hold each of his hands in succession, outside the visual field, to see whether he takes notice of the position. In seven attempts Laurent succeeds twice on the left side, but not at all on the right. Then I place an object in his right hand (tinfoil). He at once carries it to his mouth. But, before introducing it between his lips, he perceived it and then maintains it in his visual field.

At 0;3 (8) after the experiment with the rattle (Obs. 82) he loses it on the right side (but he has let it go with his left hand while he was shaking it from side to side). Laurent then looks four or five times in succession at his empty left hand. He even shakes his hand very markedly, at a certain moment, as though this shaking would start the sound of the rattle! Regardless of this last point, in any case he marks with his glance the position of his hand.

The importance of this fourth stage is evident. Henceforth the child grasps the objects which he sees, and not only those he touches or those he sucks. It is thus the beginning of the essential coördination which will aid prehension. The only limitation which still exists and which thus makes the fourth stage stand in the way of the fifth is that the child only tries to grasp the objects seen to the extent that he perceives, in the same visual field, his own hand. It is even, as is clearly apparent from examining the facts, the simultaneous sight of hand and object which induces the child to grasp; neither the sight of the object nor of the hand alone leads to this result. It might appear that an ex-

ception should be made of Observation 80. Lucienne tries to grasp the rattle or the finger that she wants to suck. But the exception is only ostensible. Either Lucienne simply brings her hand to her mouth and it is on seeing it that she tends to grasp the object, or else from the outset it is in order to grasp that she simply prolongs the behavior patterns recorded in Observation 65 (grasping objects placed against her mouth) which was revealed several minutes before those of Observation 80 which is in question.

How can we explain this tendency to grasp objects when they are perceived in the same visual field as the hand? It is possible to waver between two extreme solutions: the associative transfer and the Gestalt. Concerning associationism, as the sight of the hand holding the object has been associated to the act of grasping a certain number of times, it suffices, at a given moment, for the hand and the object to be visually perceived separately but simultaneously, in order that this perception may set prehension in motion. But, as we have already seen with regard to the third stage, such an explanation overlooks the element of activity peculiar to such establishing of relationships. The visual image of the hand is not only a signal which sets prehension in motion; it constitutes, with the grasping movements a total schema, in the same way that, during the third stage, the visual schemata of the hand are coördinated with the motor schemata other than that of prehension. Is it then necessary to speak of Gestalt and to say that the simultaneous sight of the hands and of the object causes a "structure" to appear which neither the sight of the hands nor of the object was sufficient to give rise to? About the fact itself we are certainly in agreement, and Observations 77–83 may be compared to those of W. Köhler according to which the monkey uses the cane when he perceives it at the same time as the objects to be drawn to him and not when the cane has been seen outside of the same visual field. Only it must be noted that this "structure" did not appear suddenly, but in close relation to a whole series of earlier searchings and of coördinations between sight and hand movements. Once the child has learned, during the third stage, to conserve and reproduce by means of movements of the hand that which the eye has been able to see

of these same movements, he becomes able to grasp under the influence of the glance. In other words, it is not so much the new "structure" which is of importance here as the process leading to this structure. This is why we speak of active assimilation.

Actually, once the visual schemata and the sensorimotor schemata of the hand have been mutually assimilated during the third stage (the eye looks at the hand just as the hand reproduces those of its movements which the eye sees), this kind of coördination is applied sooner or later to the very act of grasping. Looking at the hand which grasps an object, the child tries, with the hand, to maintain the spectacle which the eye contemplates as well as continuing, with his eye, to look at what his hand is doing. Once this double schema has been constituted, it is self-evident that the child will try to grasp an object while he looks at the same time at his hand when he is not yet capable of this behavior when he does not see his hand. Grasping the object when he sees the object and the hand at the same time is therefore, for the child, simply assimilating the sight of the hand to the visual and motor schema of the act consisting in "looking at grasping."

Proof that this act of "looking at grasping" simply constitutes a double schema of assimilation and not a "structure" independent of the effort and progressive activity of the subject is that this act was revealed at 0;3 (6) by Laurent, at 0;4 (12 to 15) by Lucienne and at 0;6 (0–1) by Jacqueline—that is to say, with a distance of nearly three months between the extremes. Now this difference between one child and another is explained by the whole history of their ocular-manual coördinations. Lucienne looked at her fingers since 0;2 (3), Laurent since 0;2 (4) whereas Jacqueline waited until 0;2 (30) and 0;3 (0), etc. However, nothing justifies us in considering Jacqueline retarded in relation to Lucienne. The explanation is very simple: Jacqueline, born January 9th and spending her days outdoors on a balcony, was much less active in the beginning than Lucienne and Laurent, born in June and in May. Furthermore and by virtue of this very fact, I made many fewer experiments on her during the first months, whereas I was constantly busy with Laurent. Regarding the latter, his precocity can be explained, as we have seen, first by

the fact that he sucked his fingers much earlier than the others (partly because of my experiments), and chiefly by the fact that this finger sucking gave rise to a very firm schema, that of joining the hands (Obs. 59). Constantly joining his hands, he applied himself to watching them function (Obs. 73). Once accustomed to this spectacle, he precociously grasped my hands through assimilation with his own (Obs. 74), and he thus arrived quite naturally at grasping objects (see Obs. 81: he only grasps objects, at a given moment, after having seen and grasped my hand). It seems therefore that the appearance of the essential coördinations between sight and grasping depends upon the whole psychological history of the subject and not on structures determined by an inevitable physiological unfolding. It is therefore the history, the assimilating process itself, which is of the essence and not the isolated "structure" of this history. It even appears as though a certain chance intervenes in the child's discoveries and the assimilating activity which utilizes these discoveries is thus more or less slowed down or accelerated, as the case may be.

During the *fifth stage,* at last, *the child grasps that which he sees* without limitations relating to the position of the hand.[15] First here are the facts:

*Observation 84.*—At 0;6 (3)—that is to say, three days after the beginning of the fourth stage—Jacqueline, at the outset, grasped pencils, fingers, neckties, watches, etc., which I present to her at a distance of about 10 cm. from her eyes, regardless of whether or not her hands are visible.

*Observation 85.*—The same day Jacqueline brings before her eyes the objects I put into her hand outside the visual field (pencils, etc.). This reaction is new and did not appear on the previous days.

*Observation 86.*—Finally, the same day, Jacqueline instantaneously looks in the right direction when I hold her hand outside the visual field. This too is new (see Obs. 72). These three behavior patterns which appeared simultaneously (grasping what one sees, carrying the objects to the eyes, and looking at the hand which is being held) were maintained and established the following days.

*Observation 87.*—Lucienne, at 0;4 (20), looks at my finger and opens her mouth in order to suck. Meanwhile her right hand touches mine,

[15] See in this connection H. Hetzer, with H. H. Beaumont and E. Wiehemeyer, Das Schauen und Greifen des Kindes, *Zeitschr. f. Psychol.,* 113, 1929, p. 239 (see in particular pp. 257 and 262–263).

feels it and climbs little by little toward the finger, while her glance is lowered and looks for the hand. This coördination of the direction of the glance with a gesture made by the hand outside the visual field is new in connection with the fourth stage and presages the fifth stage. —So also, a moment later, Lucienne looks at a rattle located above her face. Without seeing her hand, she raises it toward the rattle. As soon as she perceives the hand, prehension follows (left hand). When the rattle is higher, Lucienne wavers between putting her hands in her mouth and trying to grasp. The sight of the hand stimulates prehension. At 0;4 (21) in the same situation, Lucienne brings her hand at once into the visual field, looks alternately at this hand and at the rattle and grasps. When I place the rattle higher up, on the other hand, she gesticulates without bringing the hand nearer and it is necessary for her to have perceived the hand for her to attempt to grasp the object. When the rattle is lower down, the hand is then brought into the visual field and then the simultaneous sight of hand and object induces grasping. Likewise, when the rattle is high up, but Lucienne has just touched it (without seeing it), she tries to grasp while steering her hand in the right direction.—All these facts, therefore, indicate an intermediary behavior pattern between the fourth and the fifth stage. Sight of the hand remains adjuvant to prehension, but sight of the object suffices to bring the hand into the visual field.

*Observation 88.*—Beginning at 0;4 (26), on the other hand, it seems that sight of the object at once sets prehension in motion in the case of Lucienne; all the day's attempts are positive. At 0;4 (28) she seems at first to have regressed; simultaneous sight of hand and object is necessary, at the beginning of the day. But that evening she immediately tries to grasp what she sees. For instance, I place my slide rule above her eyes. She looks for a moment at this unfamiliar object, then both her hands simultaneously direct themselves toward it. From 0;5′(1) there is no longer any hesitation; Lucienne attempts to grasp everything she sees.

*Observation 89.*—At 0;5 (1) also, Lucienne immediately brings to her eyes the object she grasps or which is placed in her hands, regardless of the visual field. Then she sometimes sucks the object, but not always. It is only on an average of 3 out of 10 times that she sucked before looking. Furthermore, at the moment that she brings the object toward the visual field she expects to see something and searches with her glance even before seeing.

*Observation 90.*—At 0;5 (1) Lucienne looks in the direction of her hand which is being held. For example, I clasp her right hand while she looks to the left; she immediately turns in the right direction. Until now such an experiment yielded negative results.—A moment later, I place in her left hand (outside the visual field) a bulky object (a

gourd), which she immediately tries to grasp but which I retain. She then definitely looks for this hand, even though her arm is outstretched beside her body and thus her hand is hard to see.

At 0;5 (18), Lucienne corroborates these last acquisitions: taking what she sees, bringing the object before her eyes when she has grasped it outside the visual field, and looking in the direction of the hand which is being held.

*Observation 91.*—At 0;3 (11) Laurent is pulling toward himself sheets, covers, etc., to suck them (he does this a part of each day since he has learned how to grasp). When I hold out directly in front of him a package of tobacco, he grasps it immediately, without looking at his hand. Same reaction with an eraser. At 0;3 (12) under the same conditions he grasps my watch chain which is on his left and outside the trajectory of the joining of his hands. That evening, same reaction with this chain and with a roll of cardboard. At 0;3 (13) he immediately grasps a case which I hold out to him. He does not look at his hands or attempt to join them but at once directs the right hand toward the case. When he has grasped it, he does not suck, but examines it.

*Observation 92.*—At 0;3 (12) when I put a key in his hand, outside the visual field, he carries it to his mouth and not to his eyes. But he is very hungry (five hours have elapsed since his last meal). That evening same reaction with the case with which he is familiar but, when I place my watch chain in his hand, he looks at it before attempting to suck it.

The next day he swings a hanging chain in order to move his rattle (see below, Obs. 98). He has grasped it without looking at it, but twice he looks at his hand while it is holding the chain. In the same way, he rolls part of his sheet into a ball before sucking it and from time to time he looks at what he is doing (with both his hands).

At 0;3 (13) likewise, while he still holds in his left hand the case which he grasped (see Obs. 91) and looks straight at me, I slip into his right hand without his noticing it, my watch chain rolled into a knot (his hand is outstretched beside him). Then I withdraw and observe through the hood of the bassinet. He immediately brings the chain before his eyes (and not to his mouth) and, as he still holds the case in his left hand, he looks alternately at case and chain.—At a certain moment he loses his case. He searches for it (without seeing and still with his left hand), then he touches it without succeeding in extricating it from the folds of the coverlet. A long attempt. As soon as he succeeds in grasping it, he brings it before his eyes!

*Observation 93.*—At 0;3 (12) Laurent's left hand is outstretched. I then hold it outside the visual field: he immediately looks at it. The experiment fails with his right hand, but he seems unnerved. That evening, when I hold his right hand, he looks at it at once.

We see what comprises the acquisitions belonging to the fifth stage which denote the definitive triumph of prehension. The coördination between vision and prehension is now sufficient for each object which meets the eye to give rise to a grasping movement even when the hand is not yet perceived in the same visual field as the object.

How can this final coördination be explained? It can be conceived as being the simple result of the effort of reciprocal assimilation which the visual and manual schemata have hitherto revealed. As early as the second stage the glance attempts to follow (hence to assimilate itself) everything that the hand performs. During the third stage the hand attempts, in return, to reproduce those of its movements which the eye sees; that is to say, as we have seen, to assimilate the visual realm to the manual schemata. During the fourth stage this assimilation of the visual to the manual is extended to prehension itself when the hand appears in the same field of observation as the object to be grasped. The hand thus takes hold of what the eye observes, just as the eye tends to look at that which the hand grasps. During the fifth stage reciprocal assimilation is finally complete. All that is to be seen is also to be grasped and all that is to be grasped is also to be seen. It is natural that the hand should seek to grasp everything that the eye looks at, since the behavior patterns characteristic of the fourth stage have taught the child that this was possible when the hand is perceived at the same time as the object. The behavior belonging to the fifth stage is in this respect only a generalization of the coördinations belonging to the fourth stage. With regard to looking at everything that is grasped, it is remarkable that this tendency appears precisely at the same time as the complementary tendency. Observations 85 and 89 show that Jacqueline at 0;6 (3) and Lucienne at 0;5 (1) bring to their eyes that which they grasp, on the very date when they begin to grasp systematically what they see. The same day they also tend to look at their hand when it is held outside the visual field (Obs. 86 and 90). Such facts adequately demonstrate how much the coördination of vision and prehension is a matter of reciprocal assimilation and not of simple and irreversible associative transfer.

In conclusion, the conquest of prehension, while much more complex than that of sucking and the other elementary acquired adaptations, confirms what we have observed of the latter. All adaptation is a putting in equilibrium of a complementary accommodation and assimilation and is itself correlative to an internal and external organization of the adaptational systems. In the realm of prehension, the *accommodation* of the hand to the object is what has chiefly held writers' attention. Pure reflex in its beginnings, it subsequently involves an apprenticeship during which the accomplishment of hand movements and the opposing of the thumb are on a par with the coördination of these movements as a function of sucking and of the tactile and visual characteristics of the object. This aspect of the question is important, particularly in regard to the elaboration of the concept of space. With respect to the *assimilation* of the real to the grasping schemata, this develops analogously to what we have seen in the other realms. The child begins by moving his hand for the sake of moving it, to grasp for the sake of grasping, and hold for the sake of holding, without any interest in the objects themselves. This purely functional or reproductive assimilation (assimilation through simple repetition) is seen in the course of the reflex stage and of the second stage. How will the subject proceed from this purely functional interest (denoting an elementary assimilation of the real to the activity) to an interest in the objects grasped? Through a dual process of complication of assimilation and coördination among the sensorimotor schemata. As far as an assimilation itself is concerned, it becomes complicated through generalization. In the beginning the nursling limits himself to grasping immobile objects of a certain consistency which come in contact with the palm of the hand or the inside of the fingers; then through very repetition of the act of prehension he applies the same schemata to objects of various consistencies, animated by different movements and which the hand approaches in different ways. There exists, therefore, "generalizing" assimilation and through that very thing, the constitution of differentiated schemata, that is to say, "recognitory" assimilation. But the manifestations of the latter are less clear in the realm of prehension than in that of sight, hearing,

etc., because prehension is too quickly subordinated to external ends, such as sucking or sight. Nevertheless tactile recognition exists as revealed by the different ways in which a child grasps a handkerchief or a pencil, for example: from the first contacts the accommodation is different. This diversification of schemata during which generalizing and recognitory assimilation are on a par with the progress of accommodation explains in part how interest in the objects grasped follows purely functional interest. But it is chiefly the coördination of prehension with sucking and vision that accounts for the progressive objectifying of the universe in its relations to the activity of the hands.

Here we reach the *organization* of the grasping schemata. These schemata become organized among themselves through the fact that they adapt themselves to the external world. Thus every act of prehension presupposes an organized totality in which tactile and kinesthetic sensations and arm, hand and finger movements intervene. Hence such schemata constitute "structures" of a whole, although they were elaborated in the course of a slow evolution and over a number of attempts, gropings, and corrections. But above all these schemata are organized in coördination with schemata of another kind, chief of which are those of sucking and of vision. We have seen in what this organization consists: it is a reciprocal adaptation of the schemata in view, naturally with mutual accommodation but also with collateral assimilation. Everything that is looked at or sucked tends to be grasped and everything that is grasped tends to be sucked and then to be looked at. This coördination which crowns the acquisition of prehension also indicates an essential progress in objectification. When an object can be simultaneously grasped and sucked or grasped, looked at and sucked, it becomes externalized in relation to the subject quite differently than if it could only be grasped. In the latter case it is only an aliment for the function itself and the subject only attempts to grasp through the need to grasp. As soon as there is coördination, on the contrary, the object tends to be assimilated to several schemata simultaneously. It thus acquires an ensemble of meanings and consequently a consistency, which endow it with interest.

§5. THE FIRST ACQUIRED ADAPTATIONS: CONCLUSIONS.—After having analyzed in detail the first adaptations which are superimposed on the reflex adaptations, it is appropriate to draw some general conclusion which can subsequently guide us in our study of actual intelligence. The behavior patterns which we have described in the foregoing chapters in effect form the transition between the organic and the intellectual. They cannot yet be described as intelligent behavior patterns because intention is lacking (the differentiation between means and ends) and also mobility, which allows adaptation to continue in new circumstances. But certain intersensorial coördinations, such as those of prehension with vision, are not far removed from intelligent connection and already presage intention. On the other hand, these adaptations can no longer be characterized as purely organic because they add to the simple reflex an element of accommodation and asimilation related to the subject's experience. It is therefore important to understand how the behavior patterns of this second stage prepare intelligence.

Expressed in ordinary language, the problem which confronts us here is that of acquired association or habit and the role of these mechanisms in the genesis of intelligence. Sucking thumb or tongue, following with the eyes moving objects, searching for where sounds come from, grasping solid objects to suck or look at them, etc., are the first habits which appear in the human being. We have described their appearance in detail but the question may be asked in a general way, what sensorimotor habit is and how it is constituted. Furthermore, and it is with this sole aim that we have studied the first acquired adaptations, it may be asked in what way habitual association prepares the intelligence and what the relationships are between these two types of behavior patterns. Let us begin with this last point.

In psychology there has always been a tendency to trace back the active operations of intelligence to the passive mechanisms arising from association or habit. To reduce the causal link to a matter of habit, to reduce the generalization characteristic of the concept to the progressive application of habitual systems, to reduce judgment to an association, etc.—such are the common positions of a certain psychology dating from Hume

and Bain. The idea of the conditioned reflex, which is perhaps misused today, undoubtedly revives the terms of the problem, but its application to psychology certainly remains in the prolongation of this tradition.—Habit, too, has always seemed to some people to be the antithesis of intelligence. Where the second is active invention, the first remains passive repetition; where the second is awareness of the problem and an attempt at comprehension, the first remains tainted with lack of awareness and inertia, etc. The solution we shall give to the question of intelligence thus partly depends on that which we shall choose in the realm of habit.

Now at the risk of sacrificing precision to a taste for symmetry, we believe that the solutions between which one may waver with regard to the relationships between habit and intelligence are five principal ones and that they are parallel to the five solutions (delineated in our Introduction) regarding the genesis of the morphological reflex structures and their relationship to intelligence. Let us examine these various solutions.

The first consists in asserting that habit is a primary fact from which, through progressive complication, intelligence is derived. This is the associationist solution and the doctrine of conditioned reflexes insomuch as the latter attempts to be an instrument of general explanation in physiology. We have seen (Introduction, §3) to which Lamarckian attitude this first solution in biology proper corresponds.

The second solution which is on a par with vitalism in biology and the doctrine of "intelligence faculty" in psychology, consists in considering habit as being derived, through automatization, from higher operations involving intelligence itself. Thus to Buytendijk, the formation of habits, in animal psychology, presupposes something quite different from association: "Not only are the phenomena much more complicated, but here we see appear, in the sensitive-motor realm, phenomena which bear considerable analogy to the higher process of thought."[16] This analogy, according to this writer, rests upon the fact that "the center whence emanate all the functions of the soul . . . is an

[16] Buytendijk, *Psychologie des animaux*, Paris, Payot, p. 205.

immaterial cause, as much of sensorial activities as of the motor (activities) of the animal's psychic function."[17]

A third and a fourth solution, which are on a par with the theories of preformation and mutation in biology and with apriority and pragmatism would be tantamount to saying that habit is absolutely or relatively independent of intelligence and in some ways even its opposite. Although this point of view has not yet been systematically supported with regard to the theory of habit itself, many indications of it may be found in connection with intelligence in the work of writers whose chief common preoccupation is to emphasize the originality of the intellectual act. Thus it is that the Gestalt theory (third solution) radically opposes the putting into structures fitted to understanding and the simple automatism due to habit. Among the French psychologists H. Delacroix is also very decisive: "Far from necessarily depending on habit, it seems on the contrary as though it (generalization) were connected with the power to free oneself of it. . . . Thus, even while asserting the importance of habit as a means of grouping, all generalization remains irreducible to it."[18] So also when Claparède (fourth solution) describes intelligence to us as a searching arising on the occasion of the defeats of instinct and of habit, he partially puts the latter in opposition to the former.[19]

A fifth solution is, finally, conceivable: this is to consider the formation of habits as being due to an activity whose analogies with intelligence are purely functional, but which will be found again at the point of departure of intellectual operations when the suitable structures will permit it to go beyond its initial structure. As we understand the very important work of J. M. Baldwin, it seems to us that the concept of "circular reaction" is precisely destined to express the existence of this active factor, the principle of habit, and at the same time the source of an adaptational activity which intelligence will prolong by means of new techniques. It is through being inspired by this kind of tradition that we have, for our part, interpreted the genesis of

[17] *Ibid.*, pp. 290–291.

[18] Delacroix, in Duman, *Traité*, 1st edition, II, p. 135.

[19] Claparède, *op. cit.*, pp. 137–161.

the first habits of the nursling in terms of active assimilation and accommodation. This is not to say that this adaptational activity of which habit is only an automatization is already intelligence. For this it lacks the structural characteristics (intention, systems of mobility, etc.) whose advent we shall describe in connection with the following stage. But it does present all the functional characteristics of intelligence which will arise from it through reflexive progress and a differentiation of the relationships between subject and object more than by simply being placed in opposition to acquired habits.

Having distinguished these five solutions, let us try to discuss them in the light of the facts previously established. This will provide us with an opportunity to state precisely the meaning of the general concepts of conditioned reflex, associative transfer, habit, and circular reaction to which we have alluded without sufficient criticism, and finally to elaborate upon the concepts of accommodation, assimilation and organization which will subsequently aid us in analyzing intelligence itself.

The *first solution* is tantamount to explaining the formation of habits by the hypothesis of training or of passive association. Do the facts which we have analyzed in the course of §§1–4 lend themselves to such an interpretation? We do not think so. Neither the physiological concept of the "conditioned reflex" transposed into psychology without addenda, nor the concept of "associative transfer" seems adequate to account for the formation of the first habits which we have described.

As far as the conditioned reflex is concerned it is certain that this concept corresponds to facts which have been well established in physiology. But are these facts of sufficient importance in that field to support on themselves alone the whole weight of psychology, as some people demand of them today? In the second place, assuming that they are utilized in psychology, should they then be translated into the language of association as desired by the new associationism born of reflexology, or else have they a quite different meaning? To the first of these two questions we shall reply that the conditioned reflex is essentially fragile and unstable if it is not constantly "confirmed" by the external environment. And to the second, we shall reply that,

to the extent that the conditioned reflex is "confirmed" it ceases to be a simple association and is inserted in the more complex schema of the relationships between need and satisfaction, consequently the relationships of assimilation. That the conditioned reflex is fragile—that is to say, that the results of training are soon lost if not constantly confirmed by new training—has been brought to light by the physiologists. Moreover, they have remained much more prudent than psychologists in employing this concept. In order that a conditioned reflex may become stabilized it is, in effect, necessary either that it cease to be conditioned and become hereditarily fixed, or else that it be "confirmed" by experience itself. But the hereditary fixing of conditioned reflexes, maintained at first by Pavlov who subsequently withdrew his assertion, and then by MacDougal, seems improbable and we have seen why in our Introduction. Therefore there only remains stabilization by the environment itself and this brings us back to psychology.

A conditioned reflex can be stabilized by experience when the signal which sets the reflex in motion is followed by a confirmation—that is to say, a situation in which the reflex has the opportunity to function effectively. Hence in order to confirm the association between a sound and the salivary reflex, the animal is periodically given real food which gives the signal back its first meaning. So also could many of our observations be interpreted in terms of conditioned reflexes confirmed by experience. When the nursling makes ready to nurse when in his mother's arms and then actually finds the breast; when he turns his head to follow with his eyes a moving object and indeed finds it; when he searches with his eyes for a person whose voice he has heard and succeeds in discovering his face; when the sight of an object excites his movements of prehension and he subsequently manages to grasp it, etc.; it might be said that the reflexes of sucking, of visual and auditory accommodation and of prehension have been conditioned by signals of a postural, visual, etc., nature and that these conditioned reflexes have been stabilized because of being constantly confirmed by experience itself. But such a manner of speaking would elude the main question: How does experience confirm an association; in other

words, what are the psychological conditions necessary in order that success may consolidate a behavior pattern? It is in order to answer this question that we have invoked combined assimilation and accommodation and this is why the term pure conditioned reflex seems to us inadequate.

In effect, when a conditioned reflex is confirmed by experience, through this very fact it enters a whole schema—that is to say, it ceases to be isolated and becomes an integral part of a real totality. It is no longer a simple term in the series of acts leading to satisfaction and it is this satisfaction which becomes the essential. In effect a series of movements resulting in gratifying a need cannot be interpreted as a juxtaposition of associated elements: it constitutes a whole—that is to say, the terms which compose it only have meaning in relation to the act which regulates them and to the success of this act. It is because the objects perceived by the child are thus assimilated to the act of grasping —that is to say, because they have set in motion the need to grasp and allow it to be gratified—that the hand reaches for them, and not because an association has been established between a visual image and the reflex of prehension. The latter association, in the capacity of a conditioned reflex, is only an abstraction, only a movement artificially cut in the series itself which also presupposes an initial need and a final satisfaction. Judgment has long been explained by the association of images or of sensations. Today we know that the simplest association already presupposes some activity which participates in the judgment. In the same way the act of grasping objects visually perceived may be explained by a chain of conditioned reflexes. But the links will never be coördinated except insomuch as a single act of assimilation will confer on the object seen the meaning of an object to be grasped.

What we say about the conditioned reflexes is all the more acceptable in that this is already true of the simple reflexes themselves. It is known to what extent the story of reflexes has been renovated by Sherrington's fine works. We became aware that the classic reflex arc is an abstraction rather than a reality. In the living organism, the reflexes form organized totalities and not juxtaposed mechanisms. According to Graham Brown, a

rhythm of the whole always precedes the differentiation in reflexes: "The reflex does not explain the rhythm. In order to understand the reflex, the rhythm itself must first be invoked." And Herrick and Coghill, in studying the embryological development of the locomotor reflexes in the batrachians, speak of a "total" locomotor reaction which is subsequently dissociated into a particular reflex.[20] If all this is true of the reflexes themselves there is all the more reason to accept it as true of the conditioned reflexes. Let us be careful not to make of the conditioned reflex a new psychological element through whose combinations we shall reconstruct complex acts, and let us wait until the biologists have defined its real meaning rather than use it inordinately to explain that which is most elementary and consequently most obscure in mental phenomena.

In short, wherever we may speak of conditioned reflexes being stabilized as the result of experience, we always perceive that a schema of the whole organizes the parts of the associations. If the nursling seeks the breast when he is in position to nurse, follows moving objects with his eyes, tends to look at the people whose voice he hears, grasps objects he perceives, etc., it is because the schemata of sucking, vision, and prehension have assimilated increasingly numerous realities, through this very fact endowing them with meaning. Accommodation and assimilation combined, peculiar to each schema, insure its usefulness and coördinate it to the others, and it is the global act of complementary assimilation and accommodation which explains why the relationships of the parts which presuppose the schema are confirmed by experience.[21]

But is that not a completely verbal explanation and would not things be clarified if, for the concepts of assimilation and accommodation, the apparently clearer concept of "associative transfer" were substituted? The concept of the associative trans-

[20] Concerning all these points see Larguier, *Introduction à la psychologie*, pp. 126–138.

[21] This continuous subordination of the conditioned reflexes to organized totalities or global schemata of assimilation is demonstrated experimentally in the realm of the conditioned motor behavior patterns by a series of studies which M. André Rey, Chief of the work at our Institute, is pursuing at the present time and which will soon be published.

fer is more general than that of the conditioned reflex: association is involved, no longer only between a signal and some movement. Thus the sight of stairs suffices to set in motion appropriate movements of the legs and feet in the subject accustomed to climbing a staircase, etc. The associative transfer is thus regarded as the basis of the habit, by the first of the five above-mentioned solutions. According to this hypothesis our schemata of assimilation would not be anything other than the totalities of associative transfers whereas, according to us, every associative transfer presupposes a schema of assimilation in order to be constituted. It is therefore fitting to discuss this point closely; only this discussion is able to elucidate the true nature of sensorimotor assimilation and accommodation.

Let us first distinguish the two principal cases in which the associative transfer seems to intervene—the associations which are constituted within the same schema and the associations between heterogeneous schemata. The criterion of this distinction is the following. When the sensorial movements and elements are associated which do not yet present themselves, by another road, in the isolated state, we shall say that there is a single schema. We shall say, on the contrary, that coördination between schemata exists when they are able to function separately in other situations. For example, putting the thumb in the mouth constitutes a single schema and not a coördination between the sucking schema and the manual schemata because, at the age at which the child learns to suck his thumb he knows, it is true, how to suck something other than his thumb, but he does not know how to accomplish in other circumstances, by means of his hand, the action which he performs in putting it into his mouth (the few spontaneous movements of the hands which we have noted at about 1 to 2 months cannot even yet be definitely considered as independent schemata for it is not certain that they already constitute circular reactions distinct from impulsive movements). On the other hand, the behavior pattern consisting in grasping objects seen (4 to 5 months) may be cited as an example of coordination between heterogeneous schemata, for grasping objects independently of sight constitutes, as early as the fourth month, an autonomous schema and looking at objects independently of